ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

Associated with document Ref. Ares(2025)3876292 - 14/05/2025

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101181056
Project name:	Demonstrating circular value chains for complex plastics via collection, reuse, repair, remanufacture, recycling and ecodesign of PVC tarpaulins
Project acronym:	RETAIN
Call:	HORIZON-CL6-2024-CIRCBIO-02
Topic:	HORIZON-CL6-2024-CircBio-02-2-two-stage
Type of action:	HORIZON-IA
Service:	REA/B/03
Project starting date:	fixed date: 1 May 2025
Project duration:	42 months

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PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

Tarpaulins are complex composite-like materials, typically composed of polyethylene terephthalate (PET) as reinforcement material and plasticized/soft polyvinyl chloride (PVC) as coating material. These durable plastics have a wide range of applications going from trailer covers to professional tents and from advertisement banners to biogas membranes. Each year 200 kton of tarpaulins enter the EU market, 65% of this volume , corresponding to ca. 200 Mio m2, being produced in the EU. After their functional lifetime, which ranges from a few months up to 30 years, over 99% of tarpaulins currently end up as landfill or are being incinerated as today's recycling technologies are insufficient. The target of RETAIN is to contribute to the final ambition of having by 2036 at least 90% of circular tarpaulins use (re-used, remanufactured or recycled). To reach our ambition, we will further develop various recycling technologies (i.e. mechanical, physical, and selective dissolution) and demonstrate circular business practices (advanced collection, monitoring, sorting, repair/reuse/recycling and conversion of post-production and end-of-life tarpaulins) at TRL8 scale at three DEMO locations in the EU. Additionally, we will develop and implement an efficient decision-making mechanism for process and lifecycle optimization such as the RETAIN gate control procedure for post-industrial and end-of-life tarpaulins. Finally, we will implement a digital product passport (DPP) for tarpaulins to ensure traceability, support certification and quality validation, quantify environmental, social, and economic impacts, incorporate learning and training activities, contribute to relevant standards, and connect with other recycling initiatives in the EU and abroad.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	SIOEN	SIOEN INDUSTRIES NV	BE	985990257
2	BEN	CTB	CENTRE SCIENTIFIQUE & TECHNIQUE DEL'INDUSTRIE TEXTILE BELGE ASBL	BE	998868171
3	BEN	SATTLER	SATTLER PRO-TEX GMBH	AT	879347584
4	BEN	RENOLIT	Renolit SE	DE	873202052
5	BEN	EKOTREND	EKOTREND Ludky s.r.o.	CZ	879510641
6	BEN	SF	Serge Ferrari SAS	FR	925657130
7	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
8	BEN	AIMPLAS	AIMPLAS - ASOCIACION DE INVESTIGACION DE MATERIALES PLASTICOS Y CONEXAS	ES	999513415
9	BEN	VEOLIA	VEOLIA NV-SA	BE	940521895
10	BEN	VWERVEN	KUNSTSTOF RECYCLING VAN WERVEN BV	NL	919114092
11	BEN	VINYLPLUS	VINYLPLUS	BE	941446402
12	BEN	EUPC	EUROPEAN PLASTICS CONVERTERS EUPC AISBL	BE	998424978

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC

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13	BEN	BIG	BEAULIEU INTERNATIONAL GROUP NV	BE	940353018
14	BEN	KKF	KKF REVINYL GMBH	DE	878787894
15	BEN	RINA-C	RINA CONSULTING SPA	IT	999951467
16	BEN	MEHLER	MEHLER TEXNOLOGIES GMBH	DE	878725329

LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
WP1	Logistics, traceability, monitoring and sorting	7 - IRIS		1	24	D1.1 – Logistics Value Chain and Gate Control Protocols strategy report D1.2 – Interoperable Data Platform and DPP Prototype D1.3 – Sorting and Monitoring strategies for circularity in Logistics Value Chain
WP2	Repair/reuse/remanufacturing, and deconfectioning	10 - VWERVEN		1	30	D2.1 – Repair kits for tarp, based on mapping of common damage D2.2 – Reuse and remanufacture cases of post-production and EoL tarps D2.3 – Protocol for semi-automated deconfectioning of EOL tarps
WP3	Cascade recycling, separation and purification	2 - CTB		1	30	D3.1 – Strategy for cascaded tarp recycling – intermediate report for demo D3.2 – Optimisation of mechanical recycling options for tarps D3.3 – Processes for solvent-based recycling of tarps and related extraction of valuable additives D3.4 – Optimization strategy for cascaded tarp recycling and implementation in public Decision Support Tool

WP4	Ecodesign using recycled and biobased materials to produce high value products, and related standardization aspects	1 - SIOEN		1	39	D4.1 – Ecodesign framework for circular RETAIN value chains D4.2 – Development of circular and

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						sustainable high value soft PVC products from secondary materials from tarp D4.3 – Overview of standardization input
WP5	Demonstration of circularity for complex PVC based value chains in three EU regions	9 - VEOLIA		15	42	D5.1 – Design & planning of pilots, integrating RETAIN preparatory results D5.2 – Demo RETAIN gate control process for screening and sorting D5.3 – Demo RETAIN cascaded recycling D5.4 – Demo repair, remanufacturing & uptake of recycled products D5.5 – Assessment of RETAIN recycling chain and 3 pilots
WP6	Sustainability assessment	15 - RINA-C		19	42	D6.1 – LCA & LCC Sustainability report preliminary version (T6.1& 6.2) D6.2 – LCA & LCC & SLCA Sustainability report - final version

WP7	Market acceptance, uptake and deployment, incl. dissemination, communication, exploitation, and training & learning	12 - EUPC		1	42	D7.1 – RETAIN communication basics: logo, website, brochure & templates D7.2 – DEC plan - first preliminary version for regular updating D7.3 – Preliminary policy brief D7.4 – DEC plan – review and detailed plan for final year DEC actions D7.5 – Business models for RETAIN technologies, incl. circular tarp hubs D7.6 – Final DEC materials (project video, final event) & training & learning materials D7.7 – Final exploitation & IPR management plan for RETAIN KERs

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D7.8 – Final policy brief

WP8	Coordination	1 - SIOEN		1	42	D8.1 – Quality Manual: definition for monitoring project progress and for internal self-evaluation D8.2 – Risk Management Plan, first preliminary version for regular updating D8.3 – Data & Ethics Management Plan, first preliminary version for regular updating D8.4 – Data & Ethics Management Plan updated version D8.5 – Risk Management Plan - final version D8.6 – Data & Ethics Management Plan final version

Work package WP1 – Logistics, traceability, monitoring and sorting

Work Package Number	WP1	**Lead Beneficiary**	7 - IRIS
Work Package Name	Logistics, traceability, monitoring and sorting		
Start Month	1	**End Month**	24

Objectives

- Provide platform giving all actors along the value chain (incl. consumers, economic operators, and competent publicauthorities) access and ability to securely exchange relevant product information
- Offer traceability of complex, flexible PVC products along logistics value chain, thereby implementing a digital productpassport for tarps
- Facilitate verification of product compliance by public authorities using innovative monitoring
- Enable tracking of all substances of concern (incl. legacy additives) throughout the lifecycle of complex PVC productsusing a combination of spectroscopic techniques, like Raman and LIBS.

Description

Task 1.1: Definition of logistic value chain and gate control for complex, flexible PVC waste (M1-M9) VEOLIA, SIOEN, SATTLER, EKOTREND, IRIS, VINYLPLUS, BIG, MEHLER, SF
Within this task the industrial partners in the consortium will define the logistics value chain and gate control mechanisms required for effective DPP implementation. This will include mapping the entire product lifecycle from production to end-of-life, identifying key checkpoints, and establishing gate control protocols to ensure accurate data collection and monitoring at each stage for the three pilot demonstrators. The outputs from this task will provide the foundational logistics framework for the subsequent development and integration of the monitoring technologies and will inform the Digital Product Passport development. Specific metrics will be established to quantify the efficiency of data collection and gate control accuracy. The results of this task will be reported in deliverable D1.1.

Task 1.2: Development of interoperable data platform (IDP) and Digital Product Passport (DPP) prototypes (M4-M24) IRIS, SIOEN, VEOLIA (TRL5->6)
The development of the DPP will involve a participatory co-creation process engaging relevant actors from the logistic value chain, including raw materials suppliers, polymer designers, manufacturers, wholesalers/retailers, end-users, waste managers, environmental experts, and regulatory bodies. A detailed requirements analysis performed by VEOLIA and SIOEN will identify the data attributes and processes necessary for the DPP and interoperable data platform (IDP) to implement the gate control process. The technical specifications will ensure compliance with NGSI, European Blockchain Service Infrastructure, and Gaia-X concepts. The DPP will be designed by IRIS to integrate seamlessly with the spectroscopic monitoring technologies (LIBS and Raman) developed in Task 1.3, ensuring that data on hazardous substances, pigments, polymer types, and legacy additives are accurately captured and accessible. The project will engage with standardization bodies to ensure the data ecosystem is interoperable, scalable, and aligned with evolving industry standards and legal requirements. The results of this task will be showcased in deliverable D1.2.

Task 1.3: Validation of spectroscopic photonics-based monitoring technologies for the identification of legacy additives (M4-M24) IRIS, CTB, AIMPLAS, VWERVEN, SIOEN, EKOTREND (TRL5->6)
Building on the outputs of Task 1.1, IRIS assisted by AIMPLAS and CTB will adapt, calibrate, and validate within existing benchtop prototypes an optical LIBS and Raman spectroscopy-based monitoring system for real-time capture and analysis of data. These technologies will be integrated with the VISUM Cloud platform (IRIS Trademark for such monitoring technologies) to process information on hazardous substances, pigments, polymer types, and legacy additives (such as DEHP, lead, tin, and bromine derivatives) within the tarpaulin value chain. The system will ensure accurate, real-time data collection that will enhance traceability and sorting processes. Calibration and validation metrics will include detection limits, specificity, and accuracy for each legacy additive, ensuring that the system meets regulatory and operational requirements for compliance verification and can be installed at the facilities of VWERVEN, SIOEN and EKOTEND for the demonstration activities. The results of this task will be reported in deliverable D1.3.

Task 1.4: Definition of best sorting strategies for circularity in logistic value chain (M4-M24) VWERVEN, SF, VEOLIA, SIOEN
Sorting strategies will be developed by VWERVEN assisted by SF, VEOLIA and SIOEN for tarpaulins from different logistic routes, utilizing data from the DPP and spectroscopic analyses when available. When DPP data is unavailable, information will be gathered using LIBS and Raman technologies from Task 1.3 to assess composition, purity, and other

critical factors such as the presence of legacy additives. The sorting strategies will optimize the reuse and recycling of materials in accordance with the waste hierarchy, while minimizing risks to workers and the environment. The task will include specific quantifiable outcomes, such as the reduction of hazardous waste, improved sorting accuracy, and the increase in material recovery rates. The results of this task will be reported in deliverable D1.3.

Work package WP2 – Repair/reuse/remanufacturing, and deconfectioning

Work Package Number	WP2	**Lead Beneficiary**	10 - VWERVEN
Work Package Name	Repair/reuse/remanufacturing, and deconfectioning		
Start Month	1	**End Month**	30

Objectives

- Work out repair solutions for confectioned tarps
- Work out remanufacturing solutions for deconfectioned tarps
- Develop semi-automated workstation for high throughput cost-efficient deconfectioning of tarps

Description

Task 2.1: Repair solutions for confectioned tarps (M1-M9) SIOEN, SF, MEHLER
The 3 tarp manufacturers namely SIOEN, SF and MEHLER will start with making an overview of common damages. Typical damages range from cuts to damaged eyelets and from broken zippers to torn out welds. Next to type, this overview will include a gradation of damages as depending on the severity of the damages, repairs can be executed on location or at the confectioning company.
Small scale repair cases at location (at least 3) will be executed and documented as basis to spread the (economic) opportunities among confectioning companies. Based on the gained experiences, SIOEN will prepare 'tarp repair kits' that customers can apply to perform simple repairs themselves.
Via this task we want to stimulate repairing of confectioned tarps. Partner SIOEN, also being a confectioning company, will coordinate this repair task. This task prepares the repair pilot (part of Task 5.3) and will provide input for WP7 where we will create awareness for these repair solutions and will define economically viable business models for it (Task 7.3). The results of this task will be reported in deliverable D2.1.

Task 2.2: Remanufacture of post-production and EOL tarps (M10-M24) SIOEN, EKOTREND, SF, MEHLER, AIMPLAS
One of the main advantages of PVC is its excellent thermal and UV resistance and thus its long lifetime under indoor and outdoor conditions. For that reason, PVC coated textiles (i.e. tarps) have been applied for numerous applications over the past 65 years. As many of those applications do not require a 20+ years lifetime (e.g. publicity banners, typical lifetime of 6-12 months; trainer curtains, typical lifetime of 5-10 years), the corresponding materials are still in (very) good shape at their 'end of life'. Within this task SIOEN, EKOTREND , SF and MEHLER will bring assisted by AIMPLAS circular design strategies into practice by developing at least 3 remanufacture cases based on post-production and EOL tarps. A potential case study could be the reuse of 'EOL' publicity banner materials as transport packaging materials for parts of (offshore) wind turbines. The results of this task will be reported in deliverable D2.2.

Task 2.3: (Semi)automized deconfectioning of EOL tarps (M10-M24) VWERVEN, VEOLIA, IRIS (TRL5->6)
The objective of this task is to apply circular design rules on the deconfectioning of end-of-life (EOL) tarps, thus preparing these materials for repair/reuse/remanufacture, or recycling. As a typical tarp contains all kinds of functionalities (e.g. zippers, eyelets, buckles, etc.), these must be removed before further processing can take place. In the first phase of this task the deconfectioning of EOL tarps will be studied. Starting from manual deconfectioning, VWERVEN will study and develop together with IRIS a semi-automated workstation to mechanize this process, thereby minimizing manpower and maximizing materials output and value. In the second phase at least 3 deconfectioning cases defined by VEOLIA (e.g. trailer curtain, publicity banner, etc.) will be executed to further optimize the deconfectioning process. The knowledge and know-how as, generated in this task, will be demonstrated at scale in WP5. The results of this task will be reported in deliverable D2.3.

Work package WP3 – Cascade recycling, separation and purification

Work Package Number	WP3	**Lead Beneficiary**	2 - CTB
Work Package Name	Cascade recycling, separation and purification		
Start Month	1	**End Month**	30

Objectives

Objectives:
• Develop and evaluate different Mechanical recycling options
• Develop and evaluate different solvent-based recycling options
• Method for Selective extraction of valuable additives from PVC fraction
• Definition of strategies and implementation in decision support tool for cascade tarp recycling

Description

Task 3.1: Mechanical recycling of PVC tarps (M1-M24) KKF, EKOTREND, VWERVEN, SIOEN, SATTLER, VEOLIA, MEHLER (TRL5->6)
Shredding, milling and purification of PVC tarps (KKF). To optimize the shredding and milling process and to be sure that enough material is available at the beginning of the project, trials will be performed with post industrial waste tarps supplied by the PVC coating companies SIOEN, MEHLER and SATTLER. Firstly, a first shredding step will be performed by VWERVEN to reduce the volume and make the materials easier to handle. Shredding parameters will be optimized to make it possible to do the shredding at a 1000 kg/hr scale. Afterwards KKF will mill the pre-shredded materials into small particles (0-4 mm). The milling step is the slowest step, limiting the processing of tarps to 1000 kg/8hrs. These milled particles can then be used as secondary raw materials for film production or injection moulding or will be used for further extraction of valuable components. In case the PET still present in the particles cause to many problems for film production the PET will be separated from the PVC by means of air treatment, vibration and sifting. Later on, VEOLIA will provide post consumer EOL tarps.
Shaving separation (EKOTREND). PVC tarps will be cut into smaller strips or pieces. This makes it easier to handle and process the material. The next step is then to pass the cut PVC tarp through a shaving machine which shaves off the PVC from the polyester resulting in a thin layer of PVC and a polyester PVC fraction. Parameters such as the sharpness of the razor, the number of razors, the pressure and speed will be varied to obtain the highest yield of the pure PVC fraction. A speed of 600m2/hour of tarp is targeted. The soft PVC fraction will be molten, extruded into thin strands and then cut into small pellets. These pellets are the raw material for new PVC products. The enriched PET fraction will be sent to solvent based recycling. The results of this task will be reported in deliverable D3.2.

Task 3.2: Solvent based recycling (M4-M30). SF, CTB
Texyloop process (SF). Shredded PVC tarps derived from task 3.1 will be put in a solvent tank. The solvent will dissolve the PVC into a liquid form. This process can take several hours, depending on the amount and type of PVC and this will be optimized during this task by varying the temperature and size of the shredded PVC pieces also considering that the fibres should have a minimum length (>50 mm) to be able to be reused. After dissolving the PVC, fibres and impurities will be filtered off to ensure that the final product is of high quality. The dissolved PVC fraction will be then precipitated by adding a non-solvent. Finally, the solid PVC particles will be collected through filtration or centrifugation. The targeted yield is 300kg/day. (TRL5->6)
Soak-and-separate process (CTB). In this task it will be investigated if the PVC can be separated from the PET fibres in an economical feasible way. Shredded material from task 3.1 is pre-separated by a cyclone system in order to sort the non-bounded fibres. Then the material is soaked in a solvent. This causes the PVC fraction to swell. Next, steam is injected, triggering a rapid evaporation of the solvent (popcorn effect). As a result, the PVC coating separates from the PET substrate. To obtain different individual streams, the materials will be separated using conventional methods like zig zag air separation. The solvent is recovered through a distillation process and reused in a closed loop system. Process parameters such as input material size, soaking time, material/solvent ratio, steam injection time, temperature and pressure will be optimized to be able to recycle 50kg of tarp/day. The influence of material features, such as thickness and PVC/PET ratio, on the efficiency of this process will also be investigated. The results of this task will be reported in deliverable D3.3. (TRL4->5)

Task 3.3: Selective extraction of valuable additives from tarps (M10-M30) AIMPLAS, CTB, SF, EKOTREND, BIG (TRL4->5)

The recycled tarpaulins will undergo a two-step extraction process in order to remove additives and other materials to obtain a high-quality PVC and recover the extracted additives that could be used in similar applications.
Purification of recycled tarpaulins (AIMPLAS). The extraction of different additives contained in the tarpaulins (e.g., plasticizers, pigments, fibres and FRs) will be done by physical recycling. In this process different solvents, dissolution conditions and extraction techniques will be tested on the shredded tarpaulins from task 3.1.1 and 3.1.2, separately, in order to extract the different additives without degrading the PVC. For this physical recycling process, green solvents will be preferred for the dissolution, together with precipitation, distillation and filtration techniques to isolate the individual additives. First, a preliminary characterization of the tarps will be performed to determine which additives are present and to which extent, this will be done in collaboration with manufactures of the tarps. For this purpose, multiple identification techniques could be used, such as FT-IR, TGA and chromatographic methods. The composition analysis will lead the extraction strategy for each additive fraction, by the implementation of a design for experiment at pilot plant scale (few kilograms), to determine the best conditions to achieve the highest recovery efficiency and purity, followed by a scale-up to treat the recycled tarpaulin. Target 10-50kg/day of treated tarp, with > 99% efficiency or purity per extracted fraction (PVC and additives).
Devolatilization and re-compounding (AIMPLAS). Part of the purified PVC from subtask T3.1.2 will be stabilized with a suitable stabilizer pack in order to complement the loss of these additives during the extraction process. The dry blend obtained will be devolatilized in an extruder to remove the residual solvents from the physical recycling process. For the devolatilization, different options could be used, such high vacuum, with the aid of stripping agents, such as ScCO2, water, or low molecular weight solvents.
Characterization of recycled materials (CTB). The properties of the recycled materials will be investigated by assessing their physical, thermal, and chemical properties. The density and melt flow index (for the recovered PVC and PET fibres) will be measured to identify material types and processing behaviours. Chemical composition and contaminant analysis will be performed to identify the presence of impurities that could affect material performance. In addition, the additives extracted at high extent (e.g., plasticizer), will be recycled and tested to verify the potential use in a new plastisol by SF, EKOTREND and BIG, thus the possibility to include in new tarps. The results of this task will be reported in deliverable D3.3.

Task 3.4 Strategies definition and development of decision support tool for recycling processes (M10-M30) RINA-C, VINYLPLUS, AIMPLAS, CTB, SF, KKF, VWERVEN, EKOTREND.
This task will be divided into two parts. In the first one a preliminary analysis of the process will be performed in order to develop the first version of the optimization problem, while the second one includes the coding activities and the tool's testing phase.
Definition of decision support tool optimization problem (RINA-C). Here we will make a detailed definition of optimization strategies and the optimization problem that will be the base for the decision support tool (see below). The optimization strategy will be defined among two approaches: The first approach involves formulating a Mixed Integer Linear Problem (MILP) or a Mixed Integer Nonlinear Problem (MINLP), which is then solved using a derivativebased solver. The second one is based on the use of evolutionary algorithms that iteratively interrogate a data driven model representing the process. To perform this activity, an analysis of the recycling processes will be done to identify the variables (representing the controllable parameters), the inputs, and the constraints. This analysis will be carried out considering the inputs available through installed devices (e.g., through sensors or manually) or the DPP and the processes investigation already carried out in Tasks 3.1, 3.2 and 3.3. Moreover, the clarification of the optimization criteria will allow to set properly the parameters of the objective functions embedded in the optimization algorithms. At the end of this preliminary analysis, the most suitable recycling processes for which to develop the tool will be defined. In parallel, to have the datasets for the forecasting models of the required inputs, a Data Acquisition and Preprocessing activity, which entails gathering data from sensors, manual inputs, historical records, and industry databases, will be carried out. This phase includes data cleaning to ensure data quality, data integration to compile a comprehensive dataset, and data transformation to normalize and standardize the data. Subsequently, an Exploratory Data Analysis (EDA) will be conducted to comprehend data characteristics and uncover patterns. This involves calculating descriptive statistics, visualizing data distributions and relationships using graphs, and performing correlation analysis. RINA will lead this activity coordinating the work, the technical partners involved in Tasks 3.1, 3.2 and 3.3 will support the recycling processes analysis and IRIS will support the navigation into the DPP available data in WP2.
Development of decision support tool (RINA-C). This task will develop optimization strategies to provide a decision support tool able to evaluate the optimal setting of the controllable parameters related to the specific recycle process. This decision support tool will be tested in Task 5.3 selecting a set of relevant scenarios. All the algorithms will be developed with Python programming language. Finally, Implementation and Monitoring concentrate on deploying datadriven models for inputs forecasting, real-time monitoring using sensors and actuators, and iteratively refining datadriven models via a feedback loop. After these stages will start the last phase of Reporting and Stakeholder Engagement convey findings through detailed reports, presenting outcomes to stakeholders, and integrating feedback

to foster ongoing model improvement. This approach ensures robust, data-driven optimization strategies aimed at enhancing recycling

process efficiency and effectiveness. RINA-C will lead this task and develop the code, and the technical partners will technically assist with the input definition for the test scenarios setting. The results of this task will be reported in deliverable D3.1 and D3.4.

Work package WP4 – Ecodesign using recycled and biobased materials to produce high value products, and related standardization aspects

Work Package Number	WP4	**Lead Beneficiary**	1 - SIOEN
Work Package Name	Ecodesign using recycled and biobased materials to produce high value products, and related standardization aspects		
Start Month	1	**End Month**	39

Objectives

- Setting the ecodesign framework
- Investigation of potential markets and applications for tarp based recyclates
- Uptake of recycled and biobased materials to produce high value products
- Standardization scenario development by evaluating existing standards, identify gaps and propose input to existing orwriting new standards for the secondary raw materials, processes and products that contain recycled content

Description

Task 4.1: Setting the ecodesign framework for circular RETAIN value chains (M1-M9) SIOEN, SF, SATTLER, MEHLER, VINYLPLUSP, RENO, BIG
In this task, SIOEN, SF, SATTLER, MEHLER, VINYLPLUS, SF, RENO and BIG will set the ecodesign framework for the circular RETAIN value chains. It comprises 2 main aspects: (i) circular tarp design to cover both the high R-strategies and to ease recycling at EoL and (ii) optimal use of the recycled fractions into tarp and other soft PVC products. Both aspects are further elaborated on in the next paragraphs.
Circular tarp design is the practice of creating durable, reusable, repairable and recyclable products that generate zero waste to support a circular economy. To support one of the key objectives of the RETAIN project, being the demonstration of circularity for tarpaulin materials, we will develop and apply circular design strategies in this task. Early in the project, we will organize workshops on circular design in close collaboration with external circular designers to rethink tarp product design. Focus will be on the one side be on the design and production of the actual tarp, and on the other side on circular design strategies to facilitate repair/reuse/remanufacturing, and deconfectioning of post-production and EOL tarps. A further element will be to introduce biobased and/or bio-attributed materials. The knowledge and know-how as generated within this task will subsequently be applied in the following tasks.
Optimal use of recycled fractions. As tarp recycling is nearly non-existent today there is little to no knowledge on the properties and use of anticipated recycled streams, i.e. the various streams obtained from tarp: PVC, PET, additives, each of them potentially in different typologies. VINYLPLUS, representing the plastics industry and as such well positioned, will lead the investigation of potential markets and applications for tarp recyclates. The focus will not only be on the applications of the RETAIN partners but also on injection moulding, rotation extrusion and other technologies and their corresponding products. The results of this task will be reported in deliverable D4.1.

Task 4.2: Introducing secondary raw materials from tarp into high value products (M10-M30) RENO, BIG, SIOEN, CTB, EKOTREND, KKF (TRL4-6)
In this task the (new) high value products will be developed using the recycled materials from WP3. RENO will focus on the development of roofing films using (plasticized) PVC recyclate. Such films will be produced by means of calendaring. BIG will apply the recovered plasticizers to develop cushion vinyl flooring by means of plastisol coating technology. They will also test if the quality of the recovered PVC is comparable to virgin PVC (porosity, particle size, colour, K-value, REACH compliant). SIOEN will study the use of recycled PET yarns of at least 50 mm for producing PET nonwovens. The latter can, depending on the weight and mechanical properties, be applied for sewer renovation, geomembranes or acoustic isolation. CTB will leverage recycled PET fibres from task 3.2 to produce highquality high-tenacity (HT) polyester yarns. Initially, rPET pellets will be prepared using the Vacurema compounder, which prevents viscosity reduction in rPET during compounding. If necessary, a viscosity booster will be added, or solidstate polymerization (SSP) will be applied to ensure the rPET materials have the correct molecular weight and properties to be converted into high performance fibres. In the next phase, rPET yarns will be produced using the multifilament

extrusion line. The mechanical properties and shrinkage of these yarns will then be assessed to ensure they meet the required standards for quality and performance.
Finally, EKOTREND and KKF will also search for potential applicators outside the RETAIN consortium. For example, rPVC will be supplied to injection moulding companies. The results of this task will be reported in deliverable D4.2.

Task 4.3: Development of circular and sustainable PVC tarps (M10-M30) SIOEN, MEHLER, SATTLER, SF (TRL4->5) Partners SIOEN, MEHLER, SATTLER and SF will explore the potential for producing more sustainable and circular tarpaulins through several innovative approaches and following the ecodesign aspects from Task 4.1. First, by utilizing multifilament yarns from Task 4.2 to create rPET fabrics, we aim to enhance the eco-friendliness of our products. Additionally, we will incorporate biobased PVC to further increase sustainability. Recovered components, such as flame retardants, plasticizers, stabilizers, and pigments from Task 3.3, will be used to develop advanced PVC coating formulations. Moreover, we will investigate the feasibility of replacing current fossil-based plasticizers with biobased alternatives, aiming to reduce the environmental footprint and contribute to a circular economy. The ultimate target for partners SIOEN and MEHLER will be to produce circular tarp by means of calendering using rPVC obtained from EoL tarps. The results of this task will be reported in deliverable D4.2.

Task 4.4: Standardization scenario development (M10-M39) CTB, VINYLPLUS, AIMPLAS
The task is subdivided into two parts, with the goal of providing valuable results for developing new standards or for using them as input for already existing standardization activities.

Objectives			
• Make concrete demo plan for the 3 pilots integrating on latest RETAIN developments • Demonstration of gate control process and deconfectioning • Demonstration of Reuse/Repair/Remanufacturing/Recycling processes • Demonstration of uptake of recycled material in RETAIN value chain products • Evaluation of the pilots & production of public ‘lessons learned’			
Work Package Number	WP5	**Lead Beneficiary**	9 - VEOLIA

Work Package Name	Demonstration of circularity for complex PVC based value chains in three EU regions		
Start Month	15	**End Month**	42

Description

Task 5.1: Preparation of demonstration phase (M15–M30). VEOLIA, SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VWERVEN, BIG, KKF, RINA-C, MEHLER
Investigation of actual standardization scenario by CTB, VINYLPLUS and AIMPLAS. The purpose of this task is to evaluate the currently available standards, perform a gap analysis and identify needs for improving existing standards or for writing new standards to pave the way for market uptake of the materials, processes and prototypes developed in the RETAIN project. A first pre-evaluation has shown that in particular standardization for the recycling, reuse, repair and remanufacture processes are still high level and not the product and production process specific. In a first step relevant existing standards and ongoing standardization activities will be identified;. As a second step a gap analysis will be made to identify the specific needs for the RETAIN materials, processes and prototypes which are not covered by existing standards or standards under development. The task foresees a combination of desk studies, including a full screening of the existing standards data bases (e.g. CEN-CENELEC, ISO-IEC, ASTM), and in-depth interviews with partners (internal stakeholders) and experts in the field of RETAIN project (external stakeholders). Standardization strategy development. Based on the results of the first parts, a specific standardization strategy will be developed by the task 4.4 partners on how to address the identified standardization gaps, taking into account the inputs from the industrial players. Additionally, a standardization workshop will be held in Month 24 to gather input from a broader audience. The strategy will include concrete proposals for providing input to ongoing standardization development and/ or proposals for new work. If new work is identified, the option to write a normative document in the form of a CEN workshop agreement (CEN CWA) will be evaluated. Partner CTB, together with the Belgian national standards body (NBN) and its other partners, can provide the necessary secretariat support for this. The results of this task will be reported in deliverable D4.3

Work package WP5 – Demonstration of circularity for complex PVC based value chains in three EU regions

The RETAIN solutions as developed within WP1-4 will be demonstrated at three distinct EU regions. This demonstration will be performed at 3 levels: 1) technical demonstration in an industrially relevant environment; 2) showcasing symbiosis in a circular economy model of matching local partners as part of the technical demonstration; and 3) assessing the business cases (technical, economical, regulatory), and generating public lessons learned from the various demos. To have a successful demonstration phase, a clear and detailed demonstration plan needs to be developed for these elements. Led by partner VEOLIA, necessary documentation for the various demonstrations will be prepared. Interaction and collaboration of all partners is key to allow effective realisation of the demonstrations. WP1 till WP4 will provide the necessary specifications and production details such that the demos can be successfully executed at scale in the three EU regions. The results of this task will be reported in deliverable D5.1.

Task 5.2: Demonstration of gate control process and deconfectioning (M24-M36). IRIS, VEOLIA, RINA-C, VWERVEN, SIOEN
The in WP1 prepared collection schemes and developed quality monitoring systems and DPP as well as the developed semi-automated deconfectioning device from WP2 will be demonstrated in an industrial environment (TRL6->8). This demonstration will take in the Benelux region at an existing waste collection hub of VEOLIA. This demonstration shows the integration of the AI based spectroscopy monitoring on the sorting line which is capable to sort the waste tarps according to the RETAIN gate control process into 3 streams (1) repair/reuse/remanufacturing, 2) recycling, 3) waste) with a throughput condition of 10 tons/day. Secondly it will also be demonstrated how the presence of a DPP in the EOL tarp will make the sorting procedure less complex.
An important part of this demo task is focusing on the semi-automated workstation for the deconfectioning step, which will remove all the non-sheet parts (e.g. eyelets) from the tarps (TRL6->8). This will make these tarps ready for remanufacturing or recycling (see Task 5.3). The results of this task will be showcased in deliverable D5.2.

Task 5.3: Demonstration at scale of reuse/repair/remanufacturing and cascaded recycling (M28-40). VEOLIA, EKOTREND, SF, RINA, VWERVEN, SIOEN, KKF, CTB, AIMPLAS
At least 6 processes and technologies, as developed in WP2 and WP3, will be demonstrated in an industrial environment. Although the exact pilot implementation will depend on the RETAIN developments and thus still need to be decided (see Task 5.1), we foresee at this moment the demo of these elements as follows distributed over the 3 pilots:

- Benelux: 1) Demonstration of at least 4 cases of repair, reuse and/or remanufacture of post-production and EOL tarps(Sioen and VWERVEN - TRL6->8), 2) Physical recycling which separates PVC from the fibres at 50 kg/day (CTBTRL5->6).
- France-Spain-Italy: 3) Selective solvent based recycling to dissolve the PVC and filter the fibres (>50 mm length) ata speed of 300kg/day (SF - TRL6->7), 4) Selective extraction of plasticizers and other valuable tarp components (e.g. stabilizers, flame retardants and/or fillers) at a capacity 10-50kg/day treated tarp with >99% efficiency for high purity extracted fractions (Aimplas - TRL5->6).
- Germany-Austria-Czech Republic: 5) Mechanical recycling to obtain grinded PVC/PET material with a capacity of10 tons per day followed by PET/PVC separation (KKF - TRL6->8); 6) Mechanical recycling by shaving off the PVC from the PET reinforcement at a speed of 600m2/hour of tarp (EKOTREND - TRL6->7).

Part of the piloting of these processes, will be to demonstrate how the RETAIN decision support tool helps to optimize the value obtained from tarp recycling (RINA-C). The results of this task will be showcased in deliverable D5.3.

Task 5.4: Demonstration of products with recycled content (M29-M40). RENOLIT, SIOEN, MEHLER, BIG, SATTLER
As developed in WP4, different recyclates resulting from the cascade recycling processes of Task 5.3 will be converted into recycled (similar value) and upcycled (higher value) products. At least 5 recycled/upcycled products will be produced in a production environment to demonstrate these (potential) applications for tarp recyclates at the facilities of RENOLIT, SIOEN, MEHLER, BIG and SATTLER (TRL6->8). These products can be related to tarps, flooring, roofing, profiles, foils, injected molded parts, PET nonwovens, and high tenacity multifilament PET yarns. The results of this task will be showcased in deliverable D5.4.

Task 5.5: Assessing the RETAIN recycling chain, and demos (M37-M42). VEOLIA, SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VWERVEN, BIG, KKF, RINA-C, MEHLER
An important part of the work will be to evaluate and define lessons learned based on the different demos as executed in tasks 5.2, 5.3, and 5.4. VEOLIA will evaluate at three levels: (i) very specific for the partner(s) involved, confidential, with the goal to use internally, e.g. the (quantification of the) benefits of the DPP; (ii) from a financial and (local) regulatory perspective (can these pilots be made profitable in a business context), and (iii) from a wider perspective with the partners involved to determine what pilot information can be made public as 'lessons learned' so that WP5 input can be used for dissemination and communication activities. The results of this task will be reported in deliverable D5.5.

Description

Task 7.1: Development of dissemination, exploitation and communication (DEC) plan (M1-M42) EUPC, SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, VINYLPLUS, BIG, KKF, RINAC, MEHLER
The DEC plan will start with desk study research, which continues with input from consortium members from the beginning to the end of the project. At M6 the initial version of this plan will be submitted, containing an initial plan

Work package WP6 – Sustainability assessment

Work Package Number	WP6	**Lead Beneficiary**	15 - RINA-C
Work Package Name	Sustainability assessment		
Start Month	19	**End Month**	42

Objectives

- sustainability assessment by applying LCA, LCC and SLCA methods to the solutions proposed within the RETAINproject
- provide insights in energy and resources consumption, CO2 emissions, in addition to social and socio-economicanalysis as input for steering the pilot demonstrations

Description

Task 6.1: Experimental LCC and LCA of proposed processes (M19-M42) RINA-C, SIOEN, CTB, SATTLER, EKOTREND, SF, AIMPLAS, VEOLIA, VWERVEN, VINYLPLUS, BIG, KKF, RINA-C, MEHLER
The outcome of the task is an LCA and LCC that will provide a valuable set of information about the energy and resources used by the processes and CO2 production. The required inputs will be gathered by RINA-C with the support of the technical partners involved in the definition and realization of the processes. Evaluation of the developed solutions using LCA to quantify environmental impacts, comparing them with benchmark technologies, adhering to ISO 14040 and 14044. LCC will be conducted per SETAC's Code of Practice, estimating total costs (OPEX and CAPEX) throughout the materials' life cycle. Compare proposed alternatives with benchmarks to quantify economic savings, considering stakeholders' specific costs and gains within the supply chain. The results of this task will be reported in deliverable D6.1 (preliminary report) and D6.2 (final report).

Task 6.2: Experimental SLCA of proposed processes (M19-M42) RINA-C, SIOEN, CTB, SATTLER, EKOTREND, SF, AIMPLAS, VEOLIA, VWERVEN, VINYLPLUS, BIG, KKF, RINA-C, MEHLER
The SLCA will be performed under the same hypothesis, boundaries and framework of the LCA. SLCA will assess social and socio-economic aspects of developed materials and components following the UNEP/SETAC guideline (2020). Stakeholder perspectives will be considered to identify and manage both positive and negative impacts related to human rights, working conditions, and similar areas of interest. The results of this task will be reported in deliverable D6.1 (preliminary report) and D6.2 (final report).

Work package WP7 – Market acceptance, uptake and deployment, incl. dissemination, communication, exploitation, and training & learning

Objectives

- Development of DEC plan and its implementation, including IPR management
- Development of business models for RETAIN exploitation
- Compilation of training & learning materials and their promotion

Work Package Number	WP7	**Lead Beneficiary**	12 - EUPC
Work Package Name	Market acceptance, uptake and deployment, incl. dissemination, communication, exploitation, and training & learning		

Start Month	1	**End Month**	42

of activities for the entire project duration. It will also consider the Exploitation plan and align activities to efficiently complement exploitation and scaling of the project with dissemination. Periodic updates will be submitted at M24 and M42. The results of this task will be reported in deliverable D7.2. and D7.4.

Task 7.2: Exploitation strategies and intellectual property (IPR) management (M1-M42). EUPC, SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, VINYLPLUS, BIG, KKF, RINAC, MEHLER
The IPR exploitation activities will be developed based on preliminary KERs identified in the proposal. They will be updated with the partners according to the progress and emerging results of the project. Later, EUPC will explore different exploitation routes with a 5-year roadmap and make iterations of the IP landscape and seminars to develop an extensive exploitation plan to disseminate and exploit the project results found during the project execution and at the end of the project. This plan will be updated and aligned with the partners' interests to create specific roadmaps for the exploitation results (reports, materials developments, software creation, etc.). These activities will also be focused on finding stakeholders to set the bases for future market and business deployments (market analysis, PESTLE, CANVAS, SWOT, etc.). The results of this task will be reported in deliverable D7.7.

Task 7.3: Circular business model supporting RETAIN exploitation (M10-M36) RINA-C, VINYLPLUS, CTB, SIOEN, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, BIG, KKF, MEHLER
Within this task, a preliminary analysis of the overall tarp sector will be performed with the aim of evaluating legislative framework, specific practices of the sector (e.g. for collection) and supply chains currently in place (along with involved business actors, their economic drivers, etc.). RINA-C, VINYLPLUS and CTB will analyse at which points of the existing supply chain organizational model modifications need to be put in place to enable the new practices (repair/reuse/ remanufacturing/recycling) and the related supply chains. For this activity, the point of view of the different stakeholders and business actors intending to commercialize the solutions will be analysed (including via surveys), evaluating the main technical and non-technical barriers that may pose limitations to market entry, including legislative/regulatory restrictions, if any. This will be done for each of the regions where RETAIN plans pilot activities. Once the most promising RETAIN results in terms of readiness to the market and innovativeness are identified, dedicated business models will be investigated and developed by RINA for the new technologies (products/process/services) deployment.
The preliminary analysis developed will be the basis to perform the final analysis aiming at identifying potential business models, which will especially include a description of the new practices and of their capability to create value so that each actor involved in the business model will have direct economic benefits, along with descriptions of the value proposition, of aspects related to customers/end-users, of required partnerships, key recourses and infrastructures, and of the mechanisms for revenue generation. To this aim, consolidated methodologies (e.g. Business Model CANVAS, Flourishing CANVAS, New Framework CANVAS, etc.) will be put in place and related outputs will orient subsequent activities aimed at the evaluation of financial and economic sustainability of the project results. The results of this task will be reported in deliverable D7.5.

Task 7.4: Communication and dissemination activities (M1-M42) EUPC, VINYLPLUS, SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, BIG, KKF, RINA-C, MEHLER
This task aims to create an effective and attractive visual identity of the RETAIN project that will help deliver its message to key stakeholders and the wider public. The material to be created includes 3 main categories. i) Project visual identity and branding; ii) Website & social media channels and iii) C&D materials (banners, posters, leaflets, and flyers). As delineated and monitored in the C&D plan, C&D activities will ensure targeted communication of the project and its progress at the local/national and EU levels. In addition, partners will use the channels developed during the project and their own, ensuring both wider reach and one-to-one communication. Dissemination actions will be strategically planned to make results public to maximize the project's impact and exploitability with actions such as social media campaigns, online and offline events, scientific and non-scientific publications. In addition, the final project event (RETAIN Day) will be organized in Brussels (Belgium) with an aim to showcase the project's results to the industry representatives, general public, policymakers and other stakeholders. Also 2 policy briefs will be prepared with recommendations of tarp recycling in the EU. The results of this task will be reported in deliverable D7.1, D7.3, D7.6 and D7.8.

Task 7.5 Training & learning material creation (M25-M42) EUPC, VINYLPLUS, SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, BIG, KKF, RINA-C, MEHLER
The task goal is to create a comprehensive suite of learning and training materials (appropriate documents and videos) that not only convey the necessary knowledge but also inspire and motivate further development of the proposed solutions. The material will serve as the foundation for educating team members and stakeholders about the project's objectives, methodologies, and technologies. The creation process involves several key steps: i) identifying the learning objectives (clearly define what the learners should know or be able to do after the training), ii) content development (gather information from subject matter experts to create accurate and relevant content), iii) designing the material

(use engaging formats that cater to different learning styles, ensuring the content is accessible and comprehensible), iv) iterative feedback (collaborate with users to refine the materials based on feedback), v) pilot testing (conduct a trial run with a small group to identify any issues or areas for improvement), vi) finalizing the content (incorporate feedback from the pilot test and finalize the materials for distribution), evaluation (after the training, evaluate its effectiveness). The material will be offered to the partners and other external actors via an e-learning platform integrated in the RETAIN website. The results of this task will be reported in deliverable D7.6.

Work package WP8 – Coordination

Work Package Number	WP8	**Lead Beneficiary**	1 - SIOEN
Work Package Name	Coordination		
Start Month	1	**End Month**	42
Objectives			
• Coordinating RETAIN successfully and cost-effectively, promoting multi-stakeholder synergies & impact management • Overseeing technical strategic decision-making, and project quality management • Preparing project meetings and reports, ensuring sound financial administration, interaction with EC • Implement a pro-active risk management to ensure implementation of circular tarp pilots• Compile and implement Data and Ethics Management Plan			
Description			

Task 8.1. Project coordination and impact management (M1-M42), SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, VINYLPLUS, EUPC, BIG, KKF, RINA-C, MEHLER
This task covers the general leadership and coordination of the RETAIN project, providing cohesion and focus with a view on the final results and success. The proper coordination of the project will also require working with all the WP leaders and the technical responsibilities of each task for the execution and to refine and refocus the project activities as necessary. With this aim, a robust management structure has been designed and includes the Project Management Board (PMB), the Project Technical Committee (PTC), the Exploitation Committee and any other ad-hoc Committees that will be set up as needed. To closely monitor the project progress, frequent communication will take place (at least once per month by teleconference and, if necessary, in person) within the project consortium, helping to enforce timely proceedings. In M02, the Coordinator (Bert Groenendaal from SIOEN) will produce a project management handbook aimed at supporting the partners in efficient project management and quality assurance including guidelines and templates for financial reporting, definition of standards for deliverables and reports to the EC, measures to ensure timely reporting, payment procedures plan and calendar, etc. Internal communication procedures will also be established. As part of the impact management, we will also build connections that can strengthen the impact of RETAIN by collaborating with other Horizon Europe and EU-funded projects as well as national, local and industry initiatives. This initiative involves sharing and exchanging information on best practices, and project partners will build these relationships, which could take the form of participation in relevant events, co-organizing events with other projects and initiatives, supporting each other's online presence, or communication activities. We will also seek to cluster with the other projects selected in the call and similar calls at identifying synergies, exchanging good practices and providing evidence for EU policy developments. The results of this task will be reported in deliverable D8.1.

Task 8.2. Technical coordination and project quality assessment (M1-M42) SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, VINYLPLUS, EUPC, BIG, KKF, RINA-C, MEHLER
Task 8.2 covers technical and scientific coordination and leadership, as well as project quality assessment. It will involve a continuous monitoring of the state of the art, providing awareness of scientific and technical progress in the field and any relevant issues that may emerge. High quality standards will be applied to all the work undertaken, good performance being a priority in this project and stressing the importance of quality procedures in the Consortium. For this purpose, a quality management process and plan will be implemented to ensure the quality of all outputs of the project. This task also entails the continuing assessment of the project in the fulfilment of its objectives and validation of its scope. The results of this task will be reported in deliverable D8.1.

Task 8.3. Day-to-day management, reporting and finances (M1-M42) SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, VINYLPLUS, EUPC, BIG, KKF, RINA-C, MEHLER
Day-to-day assistance will be provided to all partners, assuming the financial control, formal revision, and submission of

the progress reports to the EC, related costs statements and any other document or communication about the project. All the partners will also be informed about the project progress through interim reports, meeting minutes, etc. The liaison between the EC and the consortium will also be the responsibility of the project coordinator. During the project execution 2 technical progress and financial reports will be submitted (M24 and M42). Linked to these reporting periods the project coordinator will organize 2 review meetings, and any additional meeting that the EC might request. Additionally, the EC manager will be invited to the project events and the PC will gather the feedback from the commission services for any strategic issue relevant for this project. To ensure smooth and efficient daily Project Management, the coordinator's team will include a highly experienced Project Coordinator, Mr. Bert Groenendaal. He will guarantee that the project management will follow the quality standards defined for this project and will ensure coordination with all the partners and stakeholders. Six-monthly PMB and PTC meetings will be held to track the progress of the project.

Task 8.4: Risk management (M1-M42) CTB, SIOEN, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, VINYLPLUS, EUPC, BIG, KKF, RINA-C, MEHLER
During proposal preparation the main risks have been identified and contingency plans established (Table 3.1e). However, unexpected risks can arise during project implementation and CTB, together with the coordinator and WP leaders, will oversee the overall risk management plan. The proactive management of the risks includes the identification of any risk that may arise, the evaluation of causes and effects and their relevance and the definition of alternative tasks or paths when applicable to ensure the demonstration via circular tarp pilots. The results of this task will be reported in deliverable D8.2 and D8.5.

Task 8.5: Data management and ethics (M1-M42) EUPC, SIOEN, CTB, SATTLER, RENOLIT, EKOTREND, SF, IRIS, AIMPLAS, VEOLIA, VWERVEN, VINYLPLUS, BIG, KKF, RINA-C, MEHLER
RETAIN activities involve data collection, generation of new data, and ethical aspects, which will be monitored by EUPC. A Data & Ethics Management Plan will be compiled and updated. The task provides an analysis of the main elements of the data management policy that will be used throughout RETAIN, regarding all the datasets that will be generated by the project. As a result, a Data Management Plan (DMP) will be elaborated, first version by M6. The DMP will consider the "Guidelines on FAIR Data Management". The DMP will include information on: i) the handling of research data during and after the end of the project; ii) what data will be collected, processed, and/or generated; iii) which methodology and standards will be applied; iv) whether data will be shared/made open access and, v) How data will be curated and preserved (including after the end of the project. It will be a live document, continuously evolving the final version of M42.
Ethics. Ensures the safety, rights, dignity and well-being of human participants in research activities. First guidelines already provided in 'ethics' section, included in Part A. EUPC will ensure compliance with the Horizon Europe's ethics standards throughout the project. Activities relevant to human participation, personal data collection/processing and project-related health and safety concerns will be prioritized. First version at M6, an updated version at M24 and a final version at M42. The results of this task will be reported in deliverable D8.3, D8.4 and D8.6.

STAFF EFFORT

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Logistics Value Chain and Gate Control Protocols strategy report	WP1	9 - VEOLIA	R — Document, report	SEN - Sensitive	9
D1.2	Interoperable Data Platform and DPP Prototype	WP1	7 - IRIS	DEM — Demonstrator, pilot, prototype	PU - Public	24
D1.3	Sorting and Monitoring strategies for circularity in Logistics Value Chain	WP1	10 - VWERVEN	R — Document, report	PU - Public	24
D2.1	Repair kits for tarp, based on mapping of common damage	WP2	1 - SIOEN	R — Document, report	PU - Public	9
D2.2	Reuse and remanufacture cases of postproduction and EoL tarps	WP2	1 - SIOEN	R — Document, report	SEN - Sensitive	24

D2.3	Protocol for semi-automated deconfectioning of EOL tarps	WP2	10 - VWERVEN	R — Document, report	SEN - Sensitive	24
D3.1	Strategy for cascaded tarp recycling – intermediate report for demo	WP3	15 - RINA-C	R — Document, report	SEN - Sensitive	18
D3.2	Optimisation of mechanical recycling options for tarps	WP3	14 - KKF	R — Document, report	SEN - Sensitive	24
D3.3	Processes for solvent-based recycling of tarps and related extraction of valuable additives	WP3	6 - SF	R — Document, report	SEN - Sensitive	30
D3.4	Optimization strategy for cascaded tarp recycling and implementation in public Decision Support Tool	WP3	15 - RINA-C	R — Document, report	PU - Public	30
D4.1	Ecodesign framework for circular RETAIN value chains	WP4	1 - SIOEN	R — Document, report	SEN - Sensitive	9
D4.2	Development of circular and sustainable high value soft PVC products from secondary materials from tarp	WP4	4 - RENOLIT	R — Document, report	SEN - Sensitive	30

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D4.3	Overview of standardization input	WP4	2 - CTB	R — Document, report	PU - Public	39
D5.1	Design & planning of pilots, integrating RETAIN preparatory results	WP5	9 - VEOLIA	R — Document, report	SEN - Sensitive	30
D5.2	Demo RETAIN gate control process for screening and sorting	WP5	7 - IRIS	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	36
D5.3	Demo RETAIN cascaded recycling	WP5	9 - VEOLIA	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	40
D5.4	Demo repair, remanufacturing & uptake of recycled products	WP5	4 - RENOLIT	DEM — Demonstrator, pilot, prototype	PU - Public	40
D5.5	Assessment of RETAIN recycling chain and 3 pilots	WP5	11 - VINYLPLUS	R — Document, report	PU - Public	42

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D6.1	LCA & LCC Sustainability report - preliminary version (T6.1& 6.2)	WP6	15 - RINA-C	R — Document, report	SEN - Sensitive	24
D6.2	LCA & LCC & SLCA Sustainability report - final version	WP6	15 - RINA-C	R — Document, report	SEN - Sensitive	40

D7.1	RETAIN communication basics: logo, website, brochure & templates	WP7	12 - EUPC	DEC —Websites, patent filings, videos, etc	PU - Public	3
D7.2	DEC plan - first preliminary version for regular updating	WP7	12 - EUPC	R — Document, report	SEN - Sensitive	6
D7.3	Preliminary policy brief	WP7	12 - EUPC	R — Document, report	PU - Public	20
D7.4	DEC plan – review and detailed plan for final year DEC actions	WP7	12 - EUPC	R — Document, report	SEN - Sensitive	24
D7.5	Business models for RETAIN technologies, incl. circular tarp hubs	WP7	15 - RINA-C	R — Document, report	SEN - Sensitive	36
D7.6	Final DEC materials (project video, final event) & training & learning materials	WP7	12 - EUPC	DEC —Websites, patent filings, videos, etc	PU - Public	42
D7.7	Final exploitation & IPR management plan for RETAIN KERs	WP7	12 - EUPC	R — Document, report	SEN - Sensitive	42
D7.8	Final policy brief	WP7	12 - EUPC	R — Document, report	PU - Public	42
D8.1	Quality Manual: definition for monitoring	WP8	1 - SIOEN	R — Document, report	SEN - Sensitive	3

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (⚠ automatically posted online)
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EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)

	project progress and for internal selfevaluation					
D8.2	Risk Management Plan, first preliminary version for regular updating	WP8	2 - CTB	R — Document, report	SEN - Sensitive	6
D8.3	Data & Ethics Management Plan, first preliminary version for regular updating	WP8	12 - EUPC	DMP — Data Management Plan	SEN - Sensitive	6
D8.4	Data & Ethics Management Plan updated version	WP8	12 - EUPC	DMP — Data Management Plan	SEN - Sensitive	24
D8.5	Risk Management Plan - final version	WP8	2 - CTB	R — Document, report	SEN - Sensitive	40
D8.6	Data & Ethics Management Plan - final version	WP8	12 - EUPC	DMP — Data Management Plan	SEN - Sensitive	40

Deliverable D1.1 – Logistics Value Chain and Gate Control Protocols strategy report

Deliverable Number	D1.1	**Lead Beneficiary**	9 - VEOLIA
Deliverable Name	Logistics Value Chain and Gate Control Protocols strategy report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	9	**Work Package No**	WP1
Description			
The report outlines how companies streamline supply chain processes to optimize the movement of PVC waste from customers to recyclers. Secondly it will provide gate protocols for effective DPP implementation (T1.1).			

Deliverable D1.2 – Interoperable Data Platform and DPP Prototype

Deliverable Number	D1.2	**Lead Beneficiary**	7 - IRIS
Deliverable Name	Interoperable Data Platform and DPP Prototype		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP1
Description			
This deliverable demonstrates how the DPP will look like and the working of the data platform allowing seamless data exchange between different systems and stakeholders in the supply chain, ensuring compatibility across various platforms by using standardized formats and protocols, enabling real-time access to accurate product information (T1.2).			

Deliverable D1.3 – Sorting and Monitoring strategies for circularity in Logistics Value Chain

Deliverable Number	D1.3	**Lead Beneficiary**	10 - VWERVEN
Deliverable Name	Sorting and Monitoring strategies for circularity in Logistics Value Chain		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP1
Description			
It reports on how the DPP and spectroscopic photonics-based monitoring technologies tools were used to evaluate the quality and composition of the EOL tarps and how these tarps were accordingly sorted for recycling, reusing/ remanufacturing or disposing (T1.3 &T1.4).			

Deliverable D2.1 – Repair kits for tarp, based on mapping of common damage

Deliverable Number	D2.1	**Lead Beneficiary**	1 - SIOEN
Deliverable Name	Repair kits for tarp, based on mapping of common damage		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP2
Description			

This deliverable reports on how damaged tarps can be efficiently repaired either by the customer or by sending them back to the confectioner (T2.1).

Deliverable D2.2 – Reuse and remanufacture cases of post-production and EoL tarps

Deliverable Number	D2.2	**Lead Beneficiary**	1 - SIOEN
Deliverable Name	Reuse and remanufacture cases of post-production and EoL tarps		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP2
Description			
This deliverable is focusing on strategies for the reuse and remanufacture of post-production and EoL tarps. It will include a detailed description of how these materials can be repurposed, outlining specific processes, techniques, and potential applications for extending their lifecycle. The document will also explore practical methods for transforming used tarps into valuable products, reducing waste and promoting sustainable resource management (T2.2).			

Deliverable D2.3 – Protocol for semi-automated deconfectioning of EOL tarps

Deliverable Number	D2.3	**Lead Beneficiary**	10 - VWERVEN
Deliverable Name	Protocol for semi-automated deconfectioning of EOL tarps		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP2
Description			
This deliverable will outline a detailed protocol for transitioning from manual to semi-automate deconfectioning processes for EoL tarps. It will provide a comprehensive description of the steps involved in reducing manual intervention and improving efficiency through the development of a semi-automated deconfectioning unit. The deliverable will also report on the design, functionality, and operational parameters of the semi-automated unit, highlighting the advancements made in streamlining the deconfectioning process and reducing labor-intensive tasks (T2.3).			

Deliverable D3.1 – Strategy for cascaded tarp recycling – intermediate report for demo

Deliverable Number	D3.1	**Lead Beneficiary**	15 - RINA-C
Deliverable Name	Strategy for cascaded tarp recycling – intermediate report for demo		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			
This deliverable outlines the initial phases of developing an optimization problem to support the decision-making process for recycling technologies (T3.4).			

Deliverable D3.2 – Optimisation of mechanical recycling options for tarps

Deliverable Number	D3.2	**Lead Beneficiary**	14 - KKF
Deliverable Name	Optimisation of mechanical recycling options for tarps		

Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP3
Description			
A detailed report will be provided on the optimisation of mechanical recycling processes for End-of-Life (EOL) tarpaulins at scale, focusing on practical and efficient solution (T3.1).			

Deliverable D3.3 – Processes for solvent-based recycling of tarps and related extraction of valuable additives

Deliverable Number	D3.3	**Lead Beneficiary**	6 - SF
Deliverable Name	Processes for solvent-based recycling of tarps and related extraction of valuable additives		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP3
Description			
The deliverable reports the solvent-based recycling processes for End-of-Life (EOL) tarpaulins, focusing on the recovery of pure PVC, PET fibres and valuable additives at scale (Task3.2 & 3.3).			

Deliverable D3.4 – Optimization strategy for cascaded tarp recycling and implementation in public Decision Support Tool

Deliverable Number	D3.4	**Lead Beneficiary**	15 - RINA-C
Deliverable Name	Optimization strategy for cascaded tarp recycling and implementation in public Decision Support Tool		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	30	**Work Package No**	WP3
Description			
This final deliverable presents a comprehensive definition and development of the optimization problem for the decision support tool tailored to recycling technologies. The task included a thorough analysis of the recycling processes to identify variables, inputs, constraints, and optimization criteria, forming the foundation for the tool (T3.4).			

Deliverable D4.1 – Ecodesign framework for circular RETAIN value chains

Deliverable Number	D4.1	**Lead Beneficiary**	1 - SIOEN
Deliverable Name	Ecodesign framework for circular RETAIN value chains		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	9	**Work Package No**	WP4
Description			

Ecodesign framework for circular RETAIN value chains (T4.1). A report will be provided on an ecodesign framework to support the development of circular RETAIN value chains, focusing on durability, reuse, and recyclability. The framework will outline strategies for designing products that retain value throughout their lifecycle, minimizing waste and maximizing material recovery. It will include guidelines on repair/reuse/remanufacturing, and deconfectioning of post-production and EOL tarps and end-of-life recovery processes to support closed-loop systems (T4.1).

Deliverable D4.2 – Development of circular and sustainable high value soft PVC products from secondary materials from tarp

Deliverable Number	D4.2	**Lead Beneficiary**	4 - RENOLIT
Deliverable Name	Development of circular and sustainable high value soft PVC products from secondary materials from tarp		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP4
Description			
A report will be provided on the development of circular and sustainable high-value soft PVC products using secondary materials recovered from tarpaulins. The focus will be on integrating recycled PVC into new products without compromising performance, quality, or durability. Key considerations will include material processing, additive optimization, and product design for longevity and recyclability. The report will also address challenges related to contamination, mechanical properties, and market acceptance, proposing scalable solutions for industrial applications (T4.2 & T4.3).			

Deliverable D4.3 – Overview of standardization input

Deliverable Number	D4.3	**Lead Beneficiary**	2 - CTB
Deliverable Name	Overview of standardization input		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	39	**Work Package No**	WP4
Description			
A report will be provided outlining standardization input for circular processes related to tarpaulin recycling and highvalue product development. The report will focus on aligning recycling practices with existing standards and identifying gaps for new standards in material recovery, product quality, and safety. It will provide recommendations on best practices for classification, testing, and certification of secondary materials. Additionally, the report will highlight opportunities to influence ongoing standardization efforts at both national and international levels (T4.4).			

Deliverable D5.1 – Design & planning of pilots, integrating RETAIN preparatory results

Deliverable Number	D5.1	**Lead Beneficiary**	9 - VEOLIA
Deliverable Name	Design & planning of pilots, integrating RETAIN preparatory results		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP5
Description			

A report will be provided on the design and planning of pilot projects, integrating preparatory results from the RETAIN initiative. The pilots will focus on testing circular processes for recycling tarpaulins and producing high-value products from secondary materials. Key aspects will include pilot objectives, timelines, required technologies, and stakeholder involvement. The report will also outline how insights from initial research will be applied to optimize scalability, efficiency, and real-world impact (T5.1).

Deliverable D5.2 – Demo RETAIN gate control process for screening and sorting

Deliverable Number	D5.2	**Lead Beneficiary**	7 - IRIS
Deliverable Name	Demo RETAIN gate control process for screening and sorting		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5
Description			
This deliverable demonstrates of the RETAIN gate control process for screening and sorting incoming End-of-Life (EOL) tarpaulins. The process will focus on identifying materials based on type, condition, and recyclability to optimize recycling efficiency. It will show the use of automated and manual inspection techniques for separating reusable, recyclable, and non-recyclable materials (T5.2).			

Deliverable D5.3 – Demo RETAIN cascaded recycling

Deliverable Number	D5.3	**Lead Beneficiary**	9 - VEOLIA
Deliverable Name	Demo RETAIN cascaded recycling		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	40	**Work Package No**	WP5
Description			
The deliverable demonstrates the RETAIN cascaded recycling process for End-of-Life (EOL) tarpaulins, focusing on sequential recycling methods to maximize material recovery and quality. The process will integrate mechanical and chemical recycling techniques to retain the highest possible value of materials throughout multiple recycling stages at scale (T5.3).			

Deliverable D5.4 – Demo repair, remanufacturing & uptake of recycled products

Deliverable Number	D5.4	**Lead Beneficiary**	4 - RENOLIT
Deliverable Name	Demo repair, remanufacturing & uptake of recycled products		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	40	**Work Package No**	WP5
Description			
The deliverable will demonstrate the repair, remanufacturing, and the uptake of recycled products derived from End-ofLife (EOL) tarpaulins. The focus will be on restoring tarpaulins to their original functionality through repair processes, remanufacturing them into new products, and promoting the adoption of these recycled items in various applications at scale (T5.4)			

Deliverable D5.5 – Assessment of RETAIN recycling chain and 3 pilots

Deliverable Number	D5.5	**Lead Beneficiary**	11 - VINYLPLUS
Deliverable Name	Assessment of RETAIN recycling chain and 3 pilots		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	42	**Work Package No**	WP5
Description			
A comprehensive assessment of the RETAIN recycling chain and its three pilot projects will be provided, focusing on the evaluation of circular processes for recycling tarpaulins and producing high-value products from secondary materials. The assessment will include an analysis of the pilot objectives, methodologies, outcomes, and scalability potential. Key considerations will encompass material recovery rates, product quality, environmental impact, and economic feasibility. The report will also identify challenges encountered during the pilots, such as contamination, process optimization, and market acceptance, and propose strategies for overcoming these obstacles to enhance the effectiveness of the recycling chain (T5.5).			

Deliverable D6.1 – LCA & LCC Sustainability report - preliminary version (T6.1& 6.2)

Deliverable Number	D6.1	**Lead Beneficiary**	15 - RINA-C
Deliverable Name	LCA & LCC Sustainability report - preliminary version (T6.1& 6.2)		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP6
Description			
This preliminary report provides an initial assessment of the Life Cycle Assessment (LCA) and Life Cycle Costing (LCC) for circular tarps, focusing on their environmental and economic impacts throughout their lifecycle. The analysis covers key stages, including material extraction, manufacturing, usage, and end-of-life management, with emphasis on durability, recyclability, and waste minimization (T6.1 & T6.2).			

Deliverable D6.2 – LCA & LCC & SLCA Sustainability report - final version

Deliverable Number	D6.2	**Lead Beneficiary**	15 - RINA-C
Deliverable Name	LCA & LCC & SLCA Sustainability report - final version		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	40	**Work Package No**	WP6
Description			
This comprehensive Life Cycle Assessment (LCA) and Life Cycle Costing (LCC) report provides a detailed evaluation of the environmental and economic impacts of circular tarps throughout their lifecycle. The assessment covers all key phases, including material extraction, production, use, and end-of-life management, focusing on metrics such as greenhouse gas emissions, energy consumption, and cost-efficiency (T6.1 & T6.2).			

Deliverable D7.1 – RETAIN communication basics: logo, website, brochure & templates

Deliverable Number	D7.1	**Lead Beneficiary**	12 - EUPC

Deliverable Name	RETAIN communication basics: logo, website, brochure & templates		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	3	**Work Package No**	WP7
Description			
This deliverable provides the foundational communication tools for RETAIN, including a modern logo, a user-friendly website, and an informative brochure. It also includes branded templates for PowerPoint presentations and deliverables to ensure consistent and professional communication. (T7.4).			

Deliverable D7.2 – DEC plan - first preliminary version for regular updating

Deliverable Number	D7.2	**Lead Beneficiary**	12 - EUPC
Deliverable Name	DEC plan - first preliminary version for regular updating		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP7
Description			
The first preliminary version of the Dissemination, Exploitation, and Communication Plan provides an initial framework for promoting the project's results to key stakeholders and ensuring their practical use. It outlines early strategies for raising awareness, sharing outcomes, and maximizing impact through tailored communication channels. This version establishes baseline targets and activities, which will be regularly updated as the project progresses. The document will evolve over time to reflect new insights, stakeholders, and opportunities for enhanced dissemination and exploitation (T7.1).			

Deliverable D7.3 – Preliminary policy brief

Deliverable Number	D7.3	**Lead Beneficiary**	12 - EUPC
Deliverable Name	Preliminary policy brief		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	20	**Work Package No**	WP7
Description			
This deliverable will present an initial policy brief outlining key findings, challenges, and recommendations identified during the early stages of the project. The brief aims to inform policymakers, industry stakeholders, and research partners about the project's progress, the potential impact of circular PVC innovations, and the regulatory considerations required to facilitate their adoption (T7.4).			

Deliverable D7.4 – DEC plan – review and detailed plan for final year DEC actions

Deliverable Number	D7.4	**Lead Beneficiary**	12 - EUPC
Deliverable Name	DEC plan – review and detailed plan for final year DEC actions		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP7

Description
This deliverable outlines the key activities for maximizing the visibility and impact of the project's results in its final year. This detailed plan includes strategies for disseminating findings to relevant stakeholders, ensuring their engagement through targeted communication channels, and identifying opportunities for exploitation. It specifies the final communication campaigns, events, and publications planned to ensure the project's outcomes reach a wide audience. The plan also covers the final steps for ensuring sustainability, such as collaboration with external partners, policy recommendations, and commercializing research outcomes (T7.1).

Deliverable D7.5 – Business models for RETAIN technologies, incl. circular tarp hubs

Deliverable Number	D7.5	**Lead Beneficiary**	15 - RINA-C
Deliverable Name	Business models for RETAIN technologies, incl. circular tarp hubs		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP7
Description			
This report outlines innovative business models for RETAIN technologies, with a focus on the integration of circular tarp hubs to promote sustainability and resource efficiency. The models explore value creation through product-as-aservice frameworks, repair and refurbishment networks, and (semi) closed-loop recycling systems for tarps. Emphasis is placed on collaborative partnerships, scalability, and financial viability, ensuring alignment with circular economy principles. The report provides actionable strategies for implementing and scaling circular tarp hubs, enabling costeffective operations, extended product lifecycles, and enhanced environmental benefits (T7.3).			

Deliverable D7.6 – Final DEC materials (project video, final event) & training & learning materials

Deliverable Number	D7.6	**Lead Beneficiary**	12 - EUPC
Deliverable Name	Final DEC materials (project video, final event) & training & learning materials		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	42	**Work Package No**	WP7
Description			
This deliverable includes a comprehensive set of resources to effectively disseminate the project's results and ensure knowledge transfer. It features a project video that highlights the key outcomes, impact, and lessons learned, designed to engage a broad audience. The deliverable also includes the final event, where the project's findings will be presented to stakeholders, fostering discussions on future exploitation and collaboration opportunities. Additionally, training and learning materials will be provided, offering educational content and resources for capacity building, aimed at ensuring the continued application and sustainability of the project's results (T7.4 & T7.5).			

Deliverable D7.7 – Final exploitation & IPR management plan for RETAIN KERs

Deliverable Number	D7.7	**Lead Beneficiary**	12 - EUPC
Deliverable Name	Final exploitation & IPR management plan for RETAIN KERs		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	42	**Work Package No**	WP7

Description
The final version of the Dissemination, Exploitation, and Communication Plan consolidates the project's complete strategy for ensuring effective outreach, impact, and sustainability of results. It includes a detailed account of the communication activities carried out, key audiences reached, and exploitation measures implemented throughout the project lifecycle. The plan highlights lessons learned and provides actionable steps to ensure continued use and visibility of the project outcomes beyond its duration. This final version serves as a comprehensive reference for maximizing the project's long-term impact across relevant sectors (T7.2).

Deliverable D7.8 – Final policy brief

Deliverable Number	D7.8	**Lead Beneficiary**	12 - EUPC
Deliverable Name	Final policy brief		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	42	**Work Package No**	WP7
Description			
The final policy brief will offer a detailed summary of the project's findings, focusing on the challenges and opportunities related to tarp recycling. This comprehensive document is designed to inform policymakers, industry stakeholders, and environmental experts, offering evidence-based recommendations for advancing tarp recycling technologies and practices (T7.4).			

Deliverable D8.1 – Quality Manual: definition for monitoring project progress and for internal self-evaluation

Deliverable Number	D8.1	**Lead Beneficiary**	1 - SIOEN
Deliverable Name	Quality Manual: definition for monitoring project progress and for internal selfevaluation		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	3	**Work Package No**	WP8
Description			
This document outlines the standards and procedures for tracking project progress and conducting internal selfevaluations. It serves as a comprehensive guide to ensure consistency, quality control, and continuous improvement across projects by defining key performance indicators and self-assessment processes (T8.1 & T8.2).			

Deliverable D8.2 – Risk Management Plan, first preliminary version for regular updating

Deliverable Name	Risk Management Plan, first preliminary version for regular updating		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP8
Description			
This plan provides an initial framework for identifying, assessing, and mitigating potential risks that could impact the project's success. It outlines key risk categories, probability levels, potential impacts, and proposed mitigation measures. This version establishes a dynamic risk-monitoring process that will be regularly updated to reflect new risks and changing project conditions (T8.4).			

Deliverable D8.3 – Data & Ethics Management Plan, first preliminary version for regular updating

Deliverable Number	D8.3	**Lead Beneficiary**	12 - EUPC
Deliverable Name	Data & Ethics Management Plan, first preliminary version for regular updating		

Deliverable Number	D8.2	**Lead Beneficiary**	2 - CTB

Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP8
Description			
The first version of the Data & Ethics Management Plan establishes the foundational framework for ethical data handling throughout the project. It outlines the principles and protocols for collecting, storing, and processing data, ensuring compliance with relevant ethical standards and regulations, such as GDPR. This version also identifies potential ethical challenges and presents strategies for mitigating risks related to privacy, security, and consent. The plan serves as an initial guide to ensure that all data-related activities are conducted responsibly and transparently (T8.5).			

Deliverable D8.4 – Data & Ethics Management Plan updated version

Deliverable Number	D8.4	**Lead Beneficiary**	12 - EUPC
Deliverable Name	Data & Ethics Management Plan updated version		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP8
Description			
The updated version of the Data & Ethics Management Plan provides a refined framework for managing the ethical aspects of data collection, storage, and usage throughout the project. It includes detailed updates on data protection measures, compliance with relevant regulations (e.g., GDPR), and procedures for handling sensitive or personal data. This version also incorporates any ethical concerns or challenges that arose during the project, alongside strategies to address them. The updated plan ensures that data management practices remain in line with ethical standards, promoting transparency, accountability, and responsible use of data (T8.5).			

Deliverable D8.5 – Risk Management Plan - final version

Deliverable Name	Risk Management Plan - final version		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	40	**Work Package No**	WP8
Description			
The final version of the Risk Management Plan presents a comprehensive and updated assessment of all identified risks and their corresponding mitigation measures throughout the project. It includes a detailed analysis of risks that materialized, the effectiveness of the mitigation actions taken, and lessons learned from risk management practices. The plan ensures that all residual risks are clearly documented and addressed to support the project's long-term sustainability. This final version serves as a reference document to guide future risk management processes and improve resilience in similar projects (T8.4).			

Deliverable D8.6 – Data & Ethics Management Plan - final version

Deliverable Number	D8.6	**Lead Beneficiary**	12 - EUPC
Deliverable Name	Data & Ethics Management Plan - final version		

Deliverable Number	D8.5	**Lead Beneficiary**	2 - CTB

Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	40	**Work Package No**	WP8
Description			
The final version of the Data & Ethics Management Plan provides a comprehensive framework for ensuring the ethical handling of data throughout the project. It includes updated protocols for data collection, storage, sharing, and processing, ensuring full compliance with relevant regulations such as GDPR. The plan addresses any ethical challenges encountered during the project, outlining strategies for ongoing risk mitigation and data protection. This final version serves as the definitive guide for ethical data management, ensuring transparency, accountability, and the responsible use of data beyond the project's conclusion (T8.5).			

LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	GO for digital platform development and start of solvent based recycling; RETAIN online presence (WP7)	WP1, WP3, WP7	7 - IRIS	Samples available for solvent based recycling (from Task3.1). Communication basics available & website online	3
2	GO for remanufacturing & deconfectioning trials and GO for development of ecodesigned products	WP1, WP4, WP2, WP3	1 - SIOEN	Logistics Value Chain and Gate Control Protocols (developed & Ecodesign framework defined	9
3	Start of planning for demo & of assessment studies	WP8, WP6, WP5, WP3	15 - RINA-C	Cascade recycling strategy document and assessment of potential of tarp circularisation technologies available and accepted by the partners.	24
4	GO for screening demo activities	WP1, WP6, WP5	15 - RINA-C	Successful development and initial testing of DPP prototype and monitoring technologies specifications and preliminary assessment performed	24
5	GO for recycling demo activities, relying on the developed options for remanufacturing, deconfectioning, recycling and ecodesign	WP4, WP2, WP5, WP3	1 - SIOEN	At least 4 cases for repair, reuse and remanufacture, at least 3 recycling routes and at least 3 novel ecodesigned products are available for demonstration	30
6	End of experimental demo activities, switch focus on evaluation and finalisation of documentation	WP6, WP5, WP7	9 - VEOLIA	Experimental work on demonstrators successfully finished. Raw data available for WP6 & 7 and for finalising T5.5	40

7	Ready for 'Handover', i.e. to start the beyond RETAIN phase	WP6, WP5, WP7	15 - RINA-C	Final demo evaluation executed, exploitation plans, final report and training material available, final event organized	42

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	Concerns over data security undermines confidence in the DPP development among test users and hinders engagement with industry (likelyhood medium / impact high = M/H)	WP1	IRIS will design a comprehensive security framework, including: robust encryption, access control, blockchain enhancements, data anonymization, employee training, effective incident response plan, as well as audit of the security protocols. In case there is still disagreement we will target agreement on a minimum info level of content within a DPP
2	Inability to access relevant data from industry partners can severely impede the development and implementation of the DPP (M/M)	WP1	Use of widely accepted industry standards and data formats to facilitate easier data exchange (this will be facilitated thanks to the cooperation with standardization bodies and plastic operators); Ensure security. Establish strong relationships with key industry stakeholders (also beyond the consortium) to foster cocreation and trust and collaboration.
3	Discarded tarp materials processing is too expensive to offer economically viable remanufactured products (L/H)	WP2	RETAIN semi-automated workstation for cost efficient processing of EoL tarps; re-focus remanufactured tarp applications to high added value products requiring a high amount of confection, e.g. dedicated packaging for nacelles (part of wind turbines).
4	Recycled PVC has inferior properties/performance compared to virgin PVC (H/M)	WP3	Inferior/deviating properties of recycled matter can be expected but ecodesign is included (WP4) to find solutions for this (adapt manufacturing techniques and product design).
5	Recycled streams not directly fit for use as secondary material because of insufficient purity or degradation during recycling processes (H/L)	WP4	RETAIN decision support tool exactly made to deal with such difficulties, i.e. to provide best strategy for a recycled stream. For example: to maximise the value of the secondary raw materials from EoL tarps, additives and pretreatment steps might be employed or alternative manufacturing techniques might be used.

6	Completion of (part of) pilot not possible, due to higher efforts and resources requirements than planned. (M/H)	WP5	Prioritize & focus on most promising developments, both technically and budget-wise. In work plan, an explicit task for planning the pilots (T5.1).
7	Data missing for LCA and LCC analyses (M/L)	WP6	WP6 leader will share tailored data collection templates with involved partners. In case of missing data, secondary (literature) data will be used.
8	Difficulties in management of fairly large project, e.g. with administrative aspects or partner loss (leaving, bankruptcy) (M/M)	WP8	Shared coordination effort foreseen, via the project steering committee (see 3.2.1) and relying on the WP leaders. In case of loss of a partner, rely on wide partner network for replacement.

PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

Review No	**Timing** (month)	**Location**	**Comments**
RV1	26	To be checked and confirmed	Review of RP1 (M1 - M24)
RV2	42	To be checked and confirmed	Review of RP2 (M25 - M42)

RETAIN - Demonstrating circular value chains for complex plastics via collection, reuse, repair, remanufacture, recycling and ecodesign of PVC tarpaulins

History of changes

Date	Nature of change and reason
Part A	
17/01/2025	• **All task descriptions** – for clarity, following adaptation: ○ References to specific deliverables were included ○ Leading and contributing partners are mentioned ○ The specific start and end TRLs are provided for each technological development • **Table 3.1c**: a deliverable description is included to comply with the guidelines • **Deliverables** – following updates as discussed and agreed with the Project Officer: ○ D2.3 & D7.3: due dates have been adapted to M24 (instead of M30) to align with the midterm review ○ D6.2, D8.5 & D8.6: due dates have been adapted to M40 (instead of M42) to avoid too many deliverables at project end ○ Two policy brief deliverables are added: D7.7 preliminary policy brief (M20) and D7.8 final policy brief (M42) • **Task duration and naming** – for consistency following adaptations: ○ *Task 2.3*: duration adapted to M10-M24 instead of M10-M30 ○ *Task 3.2*: correction of process name ('Polysep' to 'soak-and-separate process') ○ *Task 5.1*: starting time of adapted to M15 instead M18 ○ *Task 5.3*: duration adapted to M28-40 instead of M31-M40 ○ *Task 5.4*: duration adapted to M29-40 instead of M31-M40

07/02/2025	• **Short name of the beneficiaries**: to be consistent with the short names on the portal, following changes: VWERV to VWERVEN, RINA to RINA-C and VP to VINYLPLUS in part A. • **Task descriptions** – following changes as discussed and agreed with the Project Officer: ○ *Task 3.4*: added explanation for MILP (=Mixed Integer Linear Problem) and MINLP (=Mixed Integer Nonlinear Problem). ○ *Task 4.4*: added explanation for NBN (=Belgian national standardisation body) ○ *Task 5.5*: task leader typo corrected ○ *Task 8.1*: task title was missing on the portal, added 'clustering' has been added ○ *Task 8.3*: adapted the task description as follows: '*During the project execution 2 technical progress and financial reports will be submitted (M24 and M42). Linked to these reporting periods the project coordinator will organize 2 review meetings ….*' This to adapt the change from the anticipated 3 reporting periods at proposal stage to the actual 2 reporting periods. • **Milestones table**: means of verification rewritten to stress that the milestones are indeed control points in the project as discussed and agreed with the Project Officer
26/03/2025	• **Critical Risks table**: the likelihood and impact of the risks have been explicitly specified as discussed and agreed with the Project Officer
03/04/2025	• **WP3 & WP5:** to align with Table 2, the TRL levels and scale of operation have been adjusted in the task descriptions
Part B	
17/01/2025	• **Part B entirely**: figure numbers have been added for clarity • **Table 3.1a**: total person months for WP8-Coordination corrected to 45PM (instead of 43)
	• **Gantt chart**: adapted according to the adapted task timings (see Part A) • **Ethics self-assessment**: included to ensure compliance with the guidelines
7/02/2025	• **Part B entirely**: ○ *Short name of beneficiaries*: to be consistent with the short names on the portal, following changes: VWERV to VWERVEN, RINA to RINA-C and VP to VINYLPLUS in part B to make all partner naming uniform. ○ *Tables*: have been numbered for clarity • **Section 1.2.2 - overall approach**: the table with the overview of the demo's has been updated as discussed and agreed with the Project Officer • **GANTT chart**: the deliverables are added as discussed and agreed with the Project Officer • **Table 3.1h**: the tasks related to the costs were added as discussed and agreed with the Project Officer
26/03/2025	• **List of abbreviations and partners:** has been added for clarity • **Section 1.2.2 - overall approach**: the table with the overview of the demo's has been further updated as discussed and agreed with the Project Officer

	• **GANTT chart**: updated for readability as discussed and agreed with the Project Officer
03/04/2025	• **Section 1.1. KPI's:** to align with Table 2, the scale of operations has been adjusted • **Table 2:** has been adjusted and updated to reflect the progress targeted in WP5, as discussed and agreed with the Project Officer

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List of abbreviations

AI	artificial intelligence
ALTAI	Assessment List for Trustworthy AI
ANNs	Artificial Neural Networks
API	Application Programming Interface
appr	approximately
AU	Austria
BC	business case
BE	Belgium
BENELUX	Belgium, The Netherlands, Luxembourg
CA	Consortium Agreement
ca	Circa
CAGR	compound annual growth rate
CAPEX	capital expenditure
CDTI	Centre for the Development of Industrial Technology
CEAP	circular economy action plan
CEN TC	European Committee for Standardization Technical Committees
CEN-CENELEC	European Committee for Standardization &European Committee for Electrotechnical Standardization
COMM	Commercialization
CZ	Czech Republic
D	deliverable
DE	Germany
DEC	dissemination, exploitation and communication
DEHP	di(2-ethylhexyl) phthalate
DID	Distributed Identity
DLT	decentralized ledger technology
DMP	Data Management Plan
DNSH	Do No Significant Harm
DPP	digital product passport
DST	decision support tool
e.g.	for example
EC	European Commission
ECHA	European Chemicals Agency

EIB	European Investment Bank
EIC	European Innovation Council
EIT	European Institute of Innovation & Technology
EOL	end of life
EP	Exploitation Plan
eq	equivalent
ES	Spain
ESPR	Eco-design for Sustainable Products Regulation
ESWA	European Waterproofing Association
etc.	et cetera
EU	European Union
FAIR	Findability, Accessibility, Interoperability and Reusability
FR	flame retardant
FR	France
FTE	Full Time Equivalent
GA	General Assembly
GRC	Governance, Risk, and Compliance
GS	Global Standards
HCl	hydrogen chloride
HEUR	Horizon Europe
i.e.	that is
IDP	interoperable data platform
IoT	internet of things
IP	Intellectual property
IPR	Intellectual property rights
ISO	International Organization for Standardization
IT	Italy
IVK	Industrieverband Kunststoffbahnen e.V.
JSON	JavaScript Object Notation
KER	Key Exploitable Result
KG	Knowledge Generation
KIC	Knowledge and Innovation Community
KO	key objective

KPI	key performance indicator
kt	kiloton
LCA	Life cycle assessment
LCC	Life-cycle costing
LE	large enterprises
LIBS	Laser Induced Breakdown Spectroscopy
Ltd	limited
M	month
Mio	million
MSE	Mean Squared Error
NBN	Bureau for Standardization
NGO	Non-governmental organisation
NGSI	Next Generation Service Interfaces
NL	The Netherlands
PAT	Process Analytical Technologies
PET	polyethylene terephthalate
PL	Poland
PSC	Project Steering Committee
Pty	Proprietary Limited
PVC	polyvinyl chloride
QR	Quick-Response
R&D	Research and development
R&I	Research and Innovation
REACH	registration, evaluation, authorisation and restriction of chemicals
RFID	Radio-Frequency Identification
RMSE	Root Mean Squared Error
rPET	recycled PET
rPVC	recycled PVC
RTD	Research and Technological Development
SDG	Sustainable Development Goal
SE	Sweden
SETAC	Society of Environmental Toxicology and Chemistry
SLCA	Social Life Cycle Assessment
SMART	Specific, Measurable, Achievable, Relevant, and Time-Bound

SME	Small and Medium-sized Enterprises
SOTA	State-of-the-art
SSH	Social Sciences and Humanities
SWOT	Strengths, Weaknesses, Opptunities, Threats
Tarp	Tarpaulin
TCH	Tarp Circularisation Hubs
TRL	Technology readiness level
TSG	transition support group
U	end user
UK	United Kingdom
UNE	Spanish Association for Standardization
UNEP	United Nations Environment Programme
USD	United States Dollar
VC	Verifiable Credentials
VisumHSI	Hyperspectral imaging system
WP	work package
y	year

List of partners

Participant no.	Short name	Participant organization name	Country	Organization Type
1	SIOEN	SIOEN Industries NV	BE	LE
2	CTB	CENTRE SCIENTIFIQUE & TECHNIQUE DEL'INDUSTRIE TEXTILE BELGE ASBL	BE	RTO
3	SATTLER	Sattler PRO-TEX GMBH	AU	LE
4	RENOLIT	Renolit SE	DE	LE
5	EKOTREND	Ekotrend Ludky sro	CZ	LE
6	SF	Serge Ferrar SAS	FR	LE
7	IRIS	Iris Technology Solutions Sociedad LIMITADA	ES	SME
8	AIMPLAS	Asociación de Investigación de Materiales Plásticos y Conexas	ES	RTO
9	VEOLIA	Veolia NV SA	BE	LE
10	VWERVEN	Kunststof recycling Van Werven BV	NL	SME
11	VINYLPLUS	VinylPlus	BE	Assoc.
12	EUPC	European Plastics Converters EUPC AISBL	BE	Assoc.
13	BIG	Beaulieu Intl. Group NV	BE	LE
14	KKF	KKF reVinyl GMBH	DE	SME
15	RINA-C	Rina Consulting SPA	IT	LE

16	MEHLER	Mehler Texnologies GMBH	DE	LE

1. Excellence

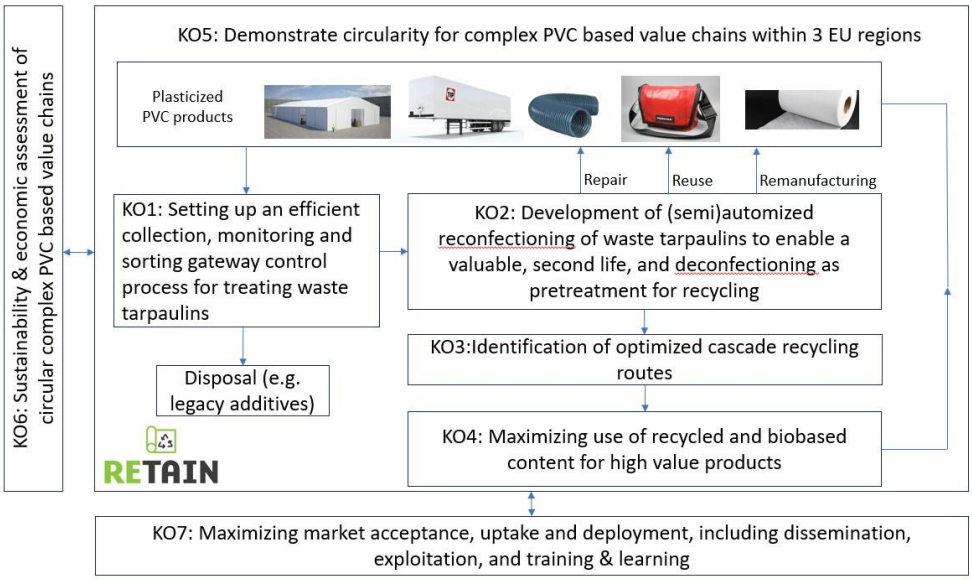
Tarpaulins are complex composite-like materials, typically composed of polyethylene terephthalate (PET) as reinforcement material and plasticized/soft polyvinyl chloride (PVC) as coating material. These durable plastics have a wide range of applications going from trailer curtains to professional tents, and from advertisement banners to biogas membranes. Each year 200 kton of tarpaulins enter the EU market, 65% of this volume, corresponding to ca. 200 Mio m2, being produced in the EU. After their functional lifetime, which ranges from a few months up to 30 years, over 99% of tarpaulins currently end up as landfill or are being incinerated as today's recycling technologies are insufficient. The target of RETAIN is to contribute to the final ambition of having, by 2036, at least 90% of circular use (repaired, reused, remanufactured or recycled) of the tarpaulins produced in EU. To reach our ambition, we will further develop various recycling technologies (i.e. mechanical, physical, and selective dissolution) and demonstrate circular business practices (advanced collection, monitoring, sorting, repair/reuse/recycling and conversion of postproduction and end-of-life tarpaulins) at TRL8 scale at three DEMO locations in the EU. Additionally, we will develop and implement an efficient decision-making tool for process and lifecycle optimization such as the RETAIN gate control procedure for post-industrial and end-of-life tarpaulins. Finally, we will implement a digital product passport (DPP) for tarpaulins to ensure traceability, support certification and quality validation, quantify environmental, social, and economic impacts, incorporate learning and training activities, contribute to relevant standards, and connect with other recycling initiatives in the EU and abroad.

1.1 Objectives and ambition

1.1.1. Objectives

Tarpaulins (short: tarps), composed of a soft polyvinyl chloride (PVC) matrix reinforced with polyethylene terephthalate (PET) fibres, are flexible, waterproof plastic sheets that are widely used for numerous outdoor applications such as trailer covers, professional tents, advertisement banners and biogas membranes. At their end of life (EOL), 99+% of these PVC tarpaulins end up in landfills or are incinerated, by lack of circular solutions at scale. Increased recovery and efficient recycling technologies at scale are urgently needed to increase the sustainability of these complex plastic materials. Thus, the overall objective of RETAIN is to increase the circularity of PVC tarpaulins by deploying 3 EU demonstration pilots for collection, monitoring, sorting, repairing/ reusing/remanufacturing, and recycling. We aim to install these demonstration pilots in the Benelux, the southwest of Europe covering France, Spain and Italy, and central Europe involving Germany, Austria and Czech Republic because they are home to the largest European tarp producers and the companies that will be recycling these materials.

To ensure that these materials are kept in circulation at the highest level of quality, economic and environmental value, our circularity model is based on: 1) a strict gate control procedure; 2) repairing/ reusing/ remanufacturing tarpaulins that are still in good condition; 3) advanced mechanical, physical or dissolution recycling of post-production and EOL tarps that are free of legacy additives; 4) reprocess/ecodesign the recycled products into new, high-value products; and 5) safe disposal of tarpaulins that still contain legacy additives in order to avoid toxic components re-entering the market (figure 1). An additional target is to make tarpaulins more sustainable by adding biobased PVC and biobased and/or recycled



plasticizers, and to investigate if high tenacity recycled PET fibres can be produced and applied in tarps. As such RETAIN is addressing the priorities set in the circular economy action plan (CEAP)[1] for achieving a sustainable, circular economy.

Figure 1: retain circularity model

To reach this ambitious overall target and to maximize the RETAIN impact, we define the following SMART key objectives (KOs) and key performance indicators (KPIs), and show their link with the call requirements and CEAP:

KO1: Setting up an efficient collection, monitoring and sorting gate control process for post-production and EOL tarpaulins => WP1

KPI(s): i) Installation of an efficient collection, monitoring and sorting process with a project throughput capacity of 10 tons/day, including an AI based spectroscopy inline monitoring quality check for condition i.e. physical and colour integrity and presence of legacy additives by M30. This will generate 3 main streams, namely: 1) repair/reuse/remanufacturing: legacy additives free and almost undamaged tarps, 2) recycling: legacy additives free, but non reusable tarps and 3) disposing: legacy additives containing tarps. ii) Interoperable data platform integrated with circular tarpaulins value chain processes, DPP and software services for storage, processing and visualization of data originating from, and related to, the collection, monitoring, sorting, repair/reuse/remanufacturing/recycling, and conversion of post-production and EOL processes will also be developed. Initial DPP Prototype by Month 18 and optimized DPP with TRL 8 by end of the project.

Link to call: i) the project should consider the use of digital solutions (including technologies such as AI, robotics, IoT and blockchain) and demonstrate their benefits for increased circularity. ii) Proposals should aim to implement traceability solutions in support of recent policy developments, e.g. regarding the digital product passport. Link to CEAP: solutions for high-quality sorting and removing contaminants from waste, including those resulting from incidental contamination

KO2: Reconfectioning of EOL tarpaulins to enable a valuable, second life, and (semi)automized deconfectioning as pretreatment for recycling => WP2

KPI(s): i) EOL tarpaulins that still have a sufficient quality, e.g. visually (colour, appearance, a rating of at least 4 on the grey change scale), mechanically (tensile strength, adhesion, etc.; >80% remaining), will be reconfectioned and prepared for a second life as tarpaulin. During the project, several options will be investigated to promote at least 3 novel “recovery and reuse routes” not excluding the possibility of cross-fertilization among different sectors (e.g. repairing of a trailer curtain or remanufacturing of an advertisement banner as a packaging material for wind turbine parts). ii) For tarps that must go to recycling, a semi-automated deconfectioning device will be developed to remove all kinds of functionalities such as eyelets, buckles, zippers, and seams, at a speed of at least 250 m²/h/person.

Link to call: Proposals should also demonstrate circular business practices, in particular in the uptake of repair and reuse, remanufacture, product-service-systems, and in the full lifetime of products or services. Link to CEAP: improving product durability, reusability, upgradability and repairability.

KO3: Identification of optimized cascade recycling routes => WP3

KPI(s): Recycling technologies will be optimized and brought at scale via the deployment of an AI based optimization digital tool, for tarpaulin recycling: i) mechanical recycling to obtain grinded PVC/PET material with a project capacity of 10 tons per day; ii) mechanical recycling process with air treatment, vibration and sifting to separate the PVC from the fibres and special exhaust system to prevent release of

[1] European Commission: Directorate-General for Communication, Circular economy action plan – For a cleaner and more competitive Europe, Publications Office of the European Union, 2020, https://data.europa.eu/doi/10.2779/05068

microplastics; iii) mechanical recycling by shaving off the PVC from the PET reinforcement at a speed of 600m^2/hour; iv) physical recycling process which separates PVC from the fibres at 50 kg/day; v) selective solvent based recycling to dissolve the PVC and filter the PET fibres (>50 mm length) at a speed of 300kg/day; vi) selective extraction of plasticizers and other valuable tarp components (e.g. stabilizers, flame retardants and/or fillers) at a capacity of 10-50kg/day treated tarp with >99% efficiency for high purity extracted fractions.

Link to call: The projects should consider the use of digital solutions (including technologies such as AI, robotics, IoT and blockchain) and demonstrate their benefits for increased circularity. ii) Proposals should demonstrate increased recycling and upcycling rates. Link to CEAP: high-quality recycling, avoiding unintentionally released microplastics during recycling

KO4: Maximizing use of recycled and biobased content for high value products => WP4

KPI(s): i) at least 5 recycled/upcycled products will be prepared to demonstrate potential applications for the recycled materials.
These products are related to profiles, foils, injected moulded parts, PET nonwovens and high tenacity multifilament PET yarns. Another expected output is the development of more sustainable tarpaulins which contain bio-attributed PVC, recycled and/or biobased plasticizers, and recycled PET fibres. The tarps will contain at least 50% recycled/biobased content. ii) A standardization scenario will be developed with the goal of providing valuable results for developing new standards or for using them as input for already existing standardization activities to make sure that there is information about the quality of the recycled materials.

Link to the call: i) Proposals should demonstrate a higher uptake of secondary materials for high value applications, ii) The projects should provide contributions to relevant standards or best practices. Link to CEAP: i) call for mandatory recycled content and the controlled use of bio-based materials; ii) enhance the role of standardisation based on the on-going assessment of existing standardisation work at national, European and international levels.

KO5: Demonstrate circularity for complex PVC based value chains within 3 EU regions => WP5

KPI(s): setting up of the complete circularity value chain that covers collectors, sorters, repairers, re-users, recyclers, logistics and end users of the recycled products in each of the 3 EU demo sites and show that PVC tarpaulins can be recirculated at large scale (>10 kt/year). Deployment and monitoring of the RETAIN technologies.

Link to the call: i) Proposals should demonstrate and deploy at large scale innovative solutions and designs for increased quality, non-toxicity and durability of secondary materials and increased share of secondary materials in new products. ii) Proposals should demonstrate increased recovery, recycling and upcycling rates and a higher uptake of secondary materials for high value applications. Link to CEAP: scaling up the separate collection of plastic waste and ensuring better infrastructure for sorting and recycling.

KO6: Sustainability & economic assessment of circular complex PVC based value chains => WP6

KPI(s): Assessment of environmental, economic, and social impacts via LCA/LCC and SLCA developed according to established robust methodologies aligned with existing standards and guidelines (e.g. ISO 14040 and 14044, UNEP/SETAC guideline (2020)) to proving positive RETAIN impact assessment at Demonstrators level.

Link to call: Environmental, social and economic impacts should be assessed from a lifecycle perspective as product, organization and consumption environmental footprints, using the respective methods developed by the European Commission and through costing methods. ii) demonstrate circular business practices, in particular in the uptake of repair and reuse, remanufacture, product-service-systems, and in

the full lifetime of products or services. To achieve this, targeted market size, economic feasibility, cost efficiency and social acceptance need to be addressed. Link to CEAP: analyse how the impact of circularity on climate change mitigation and adaptation can be measured in a systematic way; assess the feasibility of establishing a market observatory for key secondary materials

KO7: Maximizing market acceptance, uptake and deployment, including dissemination, exploitation, and training & learning => WP7

KPI(s): we will contribute to the European PVC industry's commitment for sustainable development. A 5-year roadmap will be prepared to explore different exploitation routes. Training materials such as implementation manuals and tutorial videos will be available to endow workers in this occupational group with the right skillset to deploy the innovative technologies developed. Three training workshops will be organized at the identified "replicability" sites, engaging (200+) professionals involved in process development, to pave the way for faster RETAIN replicability.

Link to the call: i) develop training material to endow workers in this occupational group with the right skillset in order to deploy the new technologies developed. ii) consider the development of learning resources for the current and future generations of employees. Link to CEAP: EU-wide market for secondary raw materials.

1.1.2 Ambition & Beyond State-of-the-art (SOTA)

Current situation: PVC is a thermoplastic polymer made from ethylene (mainly petrol based) and chlorine (derived from sea salt) and is one of the most used plastics in the world. In the EU-27 + UK, Norway, and Switzerland, about 6.5 million tons of PVC products are manufactured every year[2]. PVC is known for its durability and resistance to chemicals, weathering, and abrasion. It can be rigid or flexible, clear, or coloured, and it can be modified to have different levels of hardness. These versatile properties make that PVC has a wide range of applications such as pipes and fittings, cable insulation, profiles, flooring, toys, medical devices, banners, packaging, protective clothing, automotive components, furniture and covers. When the PVC product is not complex (i.e. homopolymer) it can be readily recycled through established and advanced mechanical recycling systems. For example, PVC window profiles can already be made of 100% recycled PVC or at least contain a high percentage of recycled PVC[3]. On the other hand, the recycling of more complex PVC based materials such as PVC tarpaulins is much more challenging, and no large-scale solutions are existing[4].

This is further illustrated by the table on the next page for which the numbers were taken from the VINYLPLUS-Progress-Report2024[5]. When looking at the tonnage of coated fabrics, of which tarps are part, it can be seen that **only 83 ton was recycled in 2023,** and this was all pre-consumer (=post-production) waste. Post-consumer (=EOL) waste was not recycled.

[2] https://pvc.org/pvc-applications/
[3] https://vinylplus.eu/wp-content/uploads/2021/05/Recycling-PVC-2021-web-Final-1.pdf
[4] https://op.europa.eu/en/publication-detail/-/publication/e9e7684a-906b-11ec-b4e4-01aa75ed71a1
[5] https://www.vinylplus.eu/wp-content/uploads/2024/05/VinylPlus-Progress-Report-2024-web.pdf

Table 1: TONNAGES OF PVC RECYCLED IN THE EU-27 PLUS NORWAY, SWITZERLAND AND THE UK

Type of PVC (waste origin)	**Tonnage recycled in 2023**		
	Total ton	Post-consumer	Pre-consumer
Profiles	391093	152585	238508
Pipes	30471	6983	23488

Cables	97586	88345	9241
Flexibles	67720	25107	42613
Rigid film	20399	5140	15259
Coated fabrics (incl. tarps)	83	0	83
Flooring	118379	3248	115131
Other rigid	11913	1250	10663
Total	**737644**	**282658**	**454986**

Although tarpaulins are not the largest PVC application, it is still an important market in the EU because about 200 Mio m² or ca.120kton (average 600 g/m²) is annually produced in the EU representing €700 Mio Euro. This volume represents ca. 2/3 of the tarps that enter the EU market. Considering that the average functional lifetime of most of these materials is between 5 and 10 years, a significant amount of waste is generated every year that currently goes to landfill or incineration as today's recycling technologies are insufficient.

Overall Ambition: **The overall ambition of RETAIN is to demonstrate circularity chains that will enable us to reuse/remanufacture/recycle over 90% of these tarps by 2036. As technologies are still at relatively low TRL's (5-6) today, our target is to bring these technologies to TRL 8, which will enable us to successfully demonstrate increased recovery, recycling, and upcycling rates (at ton scale) and a higher uptake of secondary materials for high value applications during the project. After a successful project, the consortium is committed to further expand these technologies to achieve the 2036 objective.**

To reach this general ambition, the overall objectives of RETAIN are to collect, sort, repair/reuse/recycle and convert postproduction and post-consumer (EOL) tarpaulins at large scale in 3 different EU regions and to achieve this objective, several steps beyond the state of the art in the circularity chain are required and this will be described below. Furthermore, another challenge in recycling soft PVC materials is the potential presence of legacy additives. Legacy additives in PVC products refer to substances that were historically used in the production of these products but are now recognized as problematic due to environmental and/or health concerns. These additives were commonly incorporated into PVC formulations to enhance certain properties or processing characteristics but have since been phased out or restricted due to regulations or voluntary industry commitments such as VINYLPLUS[2]. The most common legacy additives used in tarpaulins are heavy metal-based stabilizers (e.g. lead), specific short-chain plasticizers such as di(2-ethylhexyl) phthalate (DEHP) and heavy metal-based pigments (e.g. cadmium, chromium). The presence of DEHP has led to the closure of the specialised PVC recycling business at Ferrara, Italy (Vinyloop, focus on cable recycling) in 2018, because it was not economically viable to extract the plasticizer.[3] Also compared to the situation in 2018, the current in use tarpaulins contain almost no legacy additives anymore considering 80+% of these tarps have a lifetime between 5-10 years and knowing that EU tarpaulin manufacturers are not using DEHP anymore as plasticizer.

Expected steps beyond SOTA (State-of-the-Art) and increase in TRL levels for the core RETAIN developments.

The project is positioned at an advanced stage of Research and Innovation (R&I) maturity. It builds on established technologies and methodologies while integrating innovative approaches to address specific challenges in the recycling of PVC tarps. This positioning reflects a transition from research and development to practical application and market integration. By implementing demonstration pilots in regions with significant industry presence, the project aims to validate and optimize these innovations in real-world conditions, ensuring readiness for wider adoption and scalability. To achieve this, the main technology elements that need to be forwarded beyond SOTA and the related TRL progress are summarized below.

Collecting, monitoring, and sorting of the tarpaulins. TRL 5 → 8.

[2] https://www.vinylplus.eu/

[3] https://www.plastech.biz/en/news/VinyLoop-closed-PVC-recycling-project-in-Italy-12649

State of the art: Sector-specific take-back systems and separate collection are successfully applied to post-consumer waste from the construction and building sector (e.g. windows, pipes) in most of the EU Member States. However, the (limited) available information for complex flexible PVC materials suggests that separate collection remains a challenge in practice for these waste streams. Regarding efficient sorting, spectroscopic methods are frequently used, being rapid, reliable and often non-destructive analytical techniques. Process Analytical Technologies (PAT) used for real-time monitoring of critical quality parameters, are the next step for the competitive operation of many industries.[4] This technology is available at non-industrial scale at IRIS (TRL5).

Beyond SOTA: Setting up a smart collection and gate control procedure. Sensing technologies will be used to identify the presence of legacy additives as: Raman Spectroscopy for identification of **DEHP**, a plasticizer, and certain **bromo derivatives** (from fire retardants) combined with Laser Induced Breakdown Spectroscopy (LIBS) for detection of other tarp components **lead derivatives** (from pigments) and **tin derivatives** (from thermal stabilizers). Customized algorithms will allow process control for the optimization of processes and final product classification. RETAIN will demonstrate the use at pilot scale (TRL8).

A standard-based approach to Digital Product Passports as an enabler for industry circularity. TRL 5 → 8.

State of the art: Today very little is known about end-of-life aspects of tarpaulins apart from the fact that 36% ends up in landfills and 63% is being incinerated. If we want to efficiently recycle tarpaulins at their end-of-life, we must introduce lifetime traceability of these materials. Moreover, the databases on material classifications are held by big enterprise resource planning (ERP) system providers. Those systems are centrally managed by industry associations or private businesses and provide a static source of public data which is prone to future data leaks due to the nature of centrally stored data.[5]

The concept of a Digital Product Passport (DPP) can be a key enabler of the shift towards circularity as a business model by creating transparency over raw materials sourcing, circular design principles, and waste stream trajectories, and opening up new opportunities for repairing existing products, giving them a second life, as well as creating trust in the secondary market by providing customers with a reliable tool to verify a product's authenticity and history. By prioritising circular design principles and ensuring end-of-life product recovery, companies can significantly decrease their dependence on virgin materials, bolster customer loyalty, enhance brand reputation, and seamlessly adhere to increasingly stringent sustainability regulations.

The technologies upon which a DPP are built are not new, and as Europe moves forward in leading the way in DPP developments and deployments, a key focus needs to be on architecting DPPs that are designed based on **standards, interoperability, scalability, security and trustworthiness** and taking into consideration currently available **data frameworks** and how the DPP can access relevant data that already exists in industry. This is important as DPPs will not exist in isolation but will complement existing tools already deployed along the enterprise value chain. A seamless integration will significantly reduce the complexity associated with scaling DPP adoption. Aspects such as **access right management** for transparency and Intellectual Property Rights will also be of paramount importance. A **blockchain-based approach** will enable brands to transparently share accurate information with end consumers and other stakeholders within the value chain. This transparency, ensured by the immutable nature of blockchain will allow the DPP to serve as a trustworthy Digital Twin of the product.

Beyond SOTA: RETAIN aims to innovate by aligning with the existing trend in Europe regarding the creation of valuable data ecosystems based on the adoption of data space technologies for long-term storage through the development of a digital product passport (DPP) for tarpaulins. Such a digital document goes with the product through its lifetime. The document contains detailed compositional info that can be used once the product has been collected for recycling. RETAIN approach will start from a similar one already validated by IRIS in the packaging value chain and which is a good starting point to be scaled and replicated

[4] Araujo-Andrade C. et al., 'Review on the photonic techniques suitable for automatic monitoring of the composition of multimaterials wastes in view of their posterior recycling', Waste Manag. Res., 2021, 39:631;

[5] Götz, et al. (2022). Digital Product Passport: The ticket to achieving a climate neutral and circular European economy?

for the circular tarp value chain (TRL8). By implementing the DPP we aim to gradually replace non-traceable tarpaulins with unknown composition by more sustainable certified and in the EU produced tarps containing a digital product passport providing the end user information about its fingerprint, e.g., composition, how it can be recycled and to assure the safety of these products. This to establish the foundations of a global DPP standard for the Plastics sector that can be adopted internationally, making the link with standards organisations (e.g. ISO, CEN-CENELEC, NBN) starting already in the project by relevant partners like CTB (officially recognised as plastic standardisation sectorial operator in Belgium).

Cascade recycling of the tarpaulins. TRL4/6→7/8

State of the art: ***Mechanical recycling*** methods based on grinding, milling and fibre separation using air treatment, vibration and sifting offer promising options, as explored in feasibilities studies by partners VWERF &KKF (TRL6) and also the option of shaving away the PVC by EKOTREND (TRL5).

Physical recycling using polar solvents, the so-called Polysep process has been tested in batch sizes by partner CTB (TRL4). The input material is first shredded, and then soaked with solvent which swells the PVC fraction. In the next step, the soaked material comes into contact with the catalyst, which is hot water or steam. This induces a flash evaporation of the solvent, leading to a popcorning effect. At this step, the PVC coating is detached from the PET substrate. Subsequent conventional separation steps (like zig zag air separation) allow to obtain different individual streams (in this case PVC and PET).

Dissolution. Some small-scale recycling initiatives exist based on dissolution processes such as the Creasolv[6] process and

Polyloop[7]. Especially The Polyloop process from partner SF (TRL6) is interesting because PVC is separated from the fibres by dissolving the PVC in solvents and it is later recovered by a precipitation step (see figure 2).[8]. The Vinyloop dissolution process, similar to the Polyloop process, turned out not to be economically feasible due to the presence of (hazardous) plasticizers[9]. This will be less of a problem in the future as legacy additives will phase out and promising extraction methods are being developed[14]. This already recently led to the start-up of two local PVC recycling lines in Belgium launched by INEOS Inovyn. The two pilots draw on Vinyloop dissolution technology developed between 2002 and 2018 and they will also, if present, extract the legacy additives.[10]

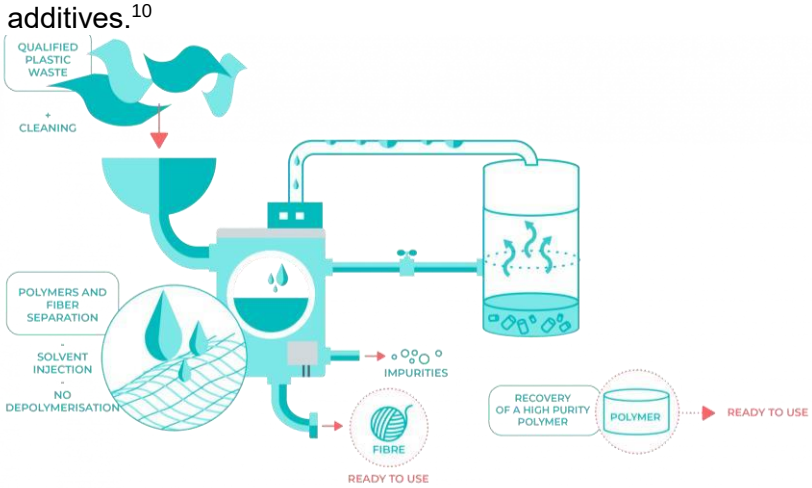


Figure 2: Polyloop process

Separation of PVC and PET fibres. A key element for both mechanical and physical recycling is the separation of PVC and PET fibres. This technology is industrially available for PVC-textile laminates but is

[6] https://www.ivv.fraunhofer.de/en/recycling-environment/recycling-plastics-creasolv.html

[7] https://polyloop.fr/

[8] https://polyloop.fr/strap-recycling/process/?lang=en

[9] https://www.plasteurope.com/news/Closure_of_operation_in_Italy_Phthalates_issue_under_REACH_brings_do_t240095/ [14] https://www.remadyl.eu/downloads-file/public-deliverables/

[10] https://www.recyclingtoday.com/news/ineos-pvc-plastic-recycling-europe-flooring-tarps-pilot-inovyn-plant/

not tested yet for tarps, hence a technology transfer is needed. At the same time, current separation efficiency is not sufficient as the PET fibres still contain too much PVC which, because of the Chloride leads to HCl formation in the extruder, leading in turn to corrosion. Further, at partner KKF it currently works only for short fibres, so an adaptation of the technology is needed for longer fibres (RETAIN goal is 50mm). Therefore, we estimate the current status TRL5.

Extraction of valuable additives such as plasticizers and flame retardants from plastic materials, with the aid of solvents and other extraction techniques has been studied in the past at lab scale, using different novel techniques, such as microwaveassisted Soxhlet extraction, ionic liquids, or supercritical CO_2[11,12,13,14]. However, these studies focus on the removal of lowmolecular weight phthalates and bromide flame retardants (legacy substances). To recover high molecular weight plasticizers and mineral flame retardants, as well as other valuable additives, a cascade extraction methodology will be performed[15]. These methodologies are used for the recovery of additives from other types of plastics, have to a certain level already been tried on PVC by CTB and AIMPLAS (TRL4-5) but will need to be further adapted to PVC tarps in terms of solvents selection and process conditions, aiming for high purity materials.

Cascade recycling material flow setup via AI-based decision support tool (DST). As is clear from the above, tarp recycling can be executed via various technologies. Depending on the quality of the EOL tarp material, and also on the equipment and infrastructure that is locally available, there will be an optimised route on how to recycle. For example, recycling locally with a technique that is locally available but that does not give the most valuable output stream versus shipping it to a location where a more optimal recycling techniques is available, but at an extra cost logistics cost. For such decision support, a strategy for optimal recycling will be worked out and a decision support tool will be developed. For this AI-based DST, techniques have already been investigated by RINA-C, but that involved other processes and/or parameters in a different environment, so tech transfer will be needed. Estimated current status TRL5-6.

Beyond SOTA: RETAIN aims to further develop the various complementary recycling technologies which are mechanical, physical and dissolution recycling and to show separation of PVC and PET fibres. It is also our ambition to further purify these recyclates to recover the plasticizers and other high value components such as the flame retardants, by using selective solvents and extraction methods, so that the purified additives and part of the PVC can be reused to obtain new plastisol formulations for tarpaulin production in a semi-closed-loop recycling process. RETAIN aims to bring those techniques at a level they mature enough to integrate them in the pilots and show their value (TRL8). An AI-based decision support tool will be investigated to optimize the processes into a cascade recycling process that optimizes for the various output streams (soft PVC/PET materials, pure PET fibres and a PET-free PVC fraction). A key element will be to optimize the required recycling resources in function of value of the output streams.

Conversion into PVC-based products with recycled content TRL4/6 →7/8.

State of the art: As can be seen from the recycled PVC tonnages table shown in the beginning of section 1.1.2, tarps are almost ***not*** recycled, meaning that soft PVC recyclates are coming from other waste sources such as EOL cable insulation and especially from post-industrial waste, e.g. for roofing and flooring for which partners RENO and BIG, respectively, have experience in their re-use (TRL5-6). The applications of EOL PVC recyclates are typically still limited, e.g. as filler material in composites or as additive to cement

[11] Pérez et al (2022). Improvement of PVC Compound Plasticizer Extraction Process. Ingeniería Industrial. 10.26439/ing.ind2022.n42.5865.

[12] Wang et al (2024). Environmental-friendly extraction of DEHP from PVC using liquefied dimethyl ether. Waste Management. 10.1016/j.wasman.2024.04.050.

[13] Lateef et al (2008). Extraction of components of composite materials: ionic liquids in the extraction of flame retardants from plastics. Journal of Chemical Technology and Biotechnology. 10.1002/jctb.1829.

[14] Shieh et al (2003). Supercritical CO2 Extraction of Phthalate Plasticizers from PVC. Journal of Applied Polymer Science. 10.1002/app.13183.

[15] Ügdüler et al (2020). Challenges and opportunities of solvent-based additive extraction methods for plastic recycling. Waste Management. 10.1016/j.wasman.2020.01.003.

and bituminous masses[16]. Regarding the polyester fibres, high quality recycled polyester fibres are mainly coming from recycled bottles, but there is also now equipment and process know how available to obtain high quality PET fibres from EoL textiles at CTB and SIOEN (TRL4-5) and this opens possibilities to develop high tenacity continuous PET multifilament yarns from EOL tarps.[17].

Beyond SOTA: RETAIN will demonstrate the conversion of recycled PVC from EoL tarpaulins into (new) high value products (i.e. recycling/upcycling). Recycled PVC, for instance, will be converted into dedicated PVC parts by means of injection moulding, will be used for production of PVC roofing and for tarps (both via calendaring) and for flooring (ideally via plastisol coating, if PVC quality too low also via calendaring). On the other hand, recycled PET *material* will be used for extrusion of high tenacity continuous PET multifilament yarns, recycled PET *fibres* (>50mm) will be used for high end technical non wovens.

FREEDOM TO OPERATE (FTO).

A preliminary freedom to operate analysis was carried out. It was clear that historically already many patents in the domain of PVC recycling of PVC coated textiles/tarpaulins were filed. However, most of them are now in the public domain and free to use. Especially most of the patents by Solvay related to the Vinyloop process are now free to use. These are for example EP0945481, EP1232204 and EP1268628 and their family members. In the search a limited number of living patent families were discovered.
These are EP1737904 by Fraunhofer, EP2276801 by Solvay/SF and EP3504037A1 by PVC Separation Holdings Pty Ltd. However, the Fraunhofer method will not be used in this project. The Solvay/SF patent is transferred to Polyloop, but SF is allowed to further optimize this recycling process for their products. The patent EP3504037A1 refers to the Polysep process which has been developed in Australia. A container-based demo line has been installed at the premises of CTB, and CTB is allowed to use it for trials at pilot scale within RETAIN. Thus, the active patents are not impeding the project's progress. #§PRJ-OBJ-PO§#

1.2 Methodology

1.2.1 Basic concept and assumptions

The basic concept to create a circular PVC tarpaulin value chain is based on the collection, screening and sorting of end-of-life (EOL) and postindustrial waste which will be 'reconfectioned' (term used for indicating re-using, repairing and remanufacturing) or recycled by scaling existing technologies. Within RETAIN we will develop and implement a very strict gate control procedure. As shown in the gate control scheme (figure 3), this procedure will distinguish between the various tarp qualities, thus guaranteeing minimal value reduction. Also, at manufacturer level, actions will be taken to enhance the European industrial sustainability. RETAIN will develop and integrate 1) a digital product passport for tarpaulins, and 2) integrate biobased and recycled products. If each EU tarp manufacturer integrates a DPP in their products (representing 65% of the EU market), it will ease and cheapen the circular tarp value chain because no screening for toxic components would be necessary anymore for these tarpaulins.

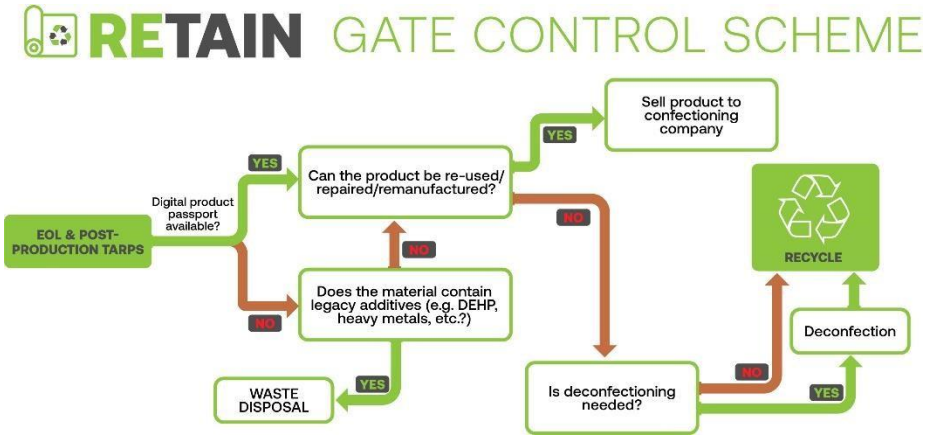


Figure 3: gate control scheme

[16] Lewandowski, K.; Skórczewska, K. A Brief Review of Poly(Vinyl Chloride) (PVC) Recycling. Polymers 2022, 14, 3035. https://doi.org/10.3390/polym14153035
[17] https://www.erema.com/us/pet-fibre-inline-recycling/

We will *not* focus on the recycling of 'old' tarpaulins containing legacy additives, there were and are several projects to (completely) remove these legacy additives, but it is still challenging to completely remove these additives in an economically feasible way. Even a small amount of legacy additive present in the recycled materials will pollute the end product making it impossible to sell these products. Therefore, as RETAIN is focusing on a toxic free and economically mature circularity chain of PVC tarps on short term, these materials will be sent to specialized waste disposal facilities and will not enter the reconfectioning and recycling facilities of the RETAIN. Further the DISSOLV project (with RETAIN partners SIOEN, BIG & CTB) is working on solutions to recycle soft PVC materials containing legacy additives, so the consortium will have state-of-the-art information on best handling of such waste.[18]

1.2.2. Overall approach

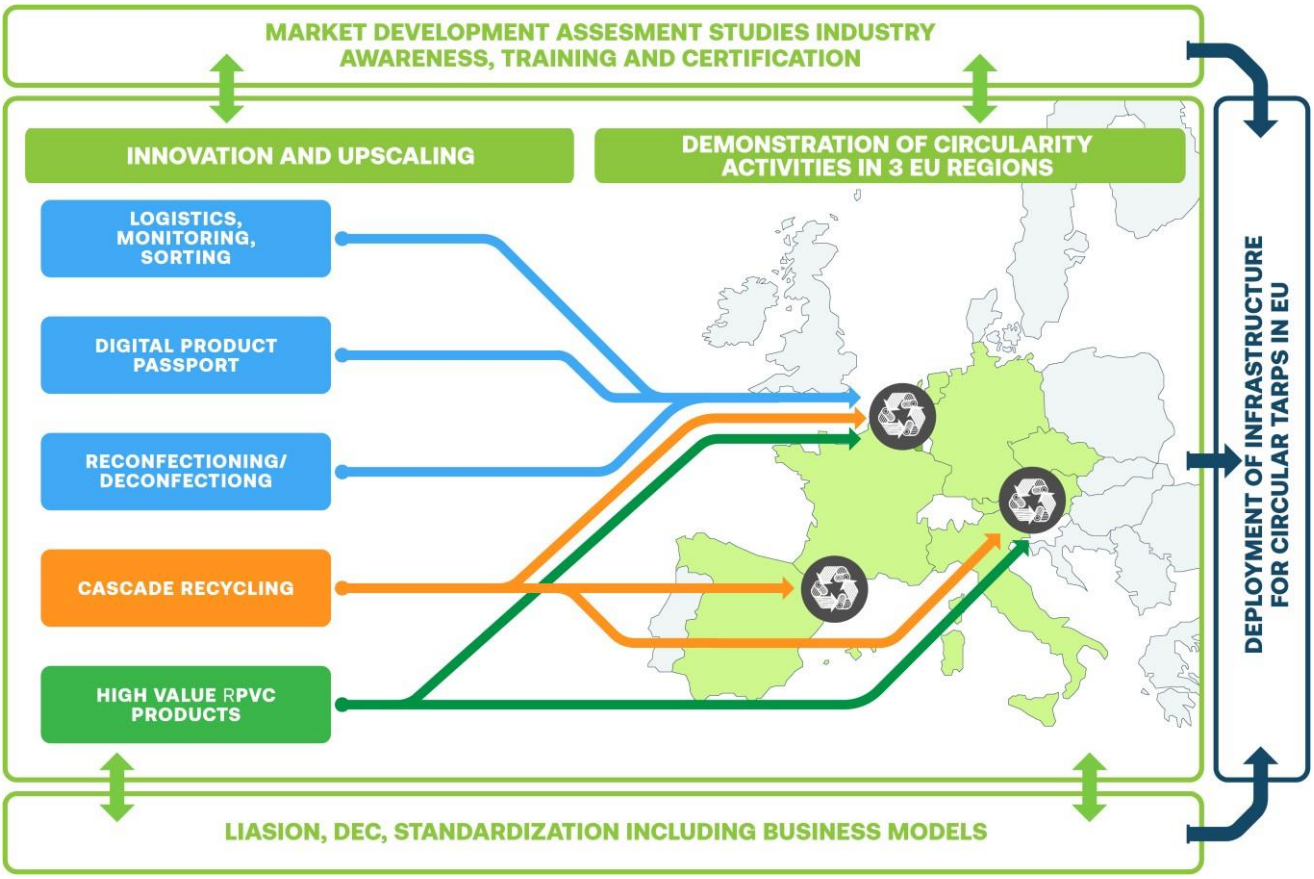


Figure 4: RETAIN approach

The RETAIN methodology for establishing tarpaulin circularity is divided into several activity blocks, as illustrated figure 4. In the **first phase we focus on the innovations and scale up of the EOL tarps needed to reach TRL 6**. Solutions for sorting and screening need to be further developed to develop processes capable of handling end-of-life (EOL) tarpaulins on a large scale. Additionally, a digital product passport (DPP) should be created. These elements will lead to the RETAIN gate control process. Further, we will develop reconfectioning and deconfectioning techniques for processing the EOL tarps. Next activity block will be the optimisation of a set of recycling technologies (mechanical, dissolution, extraction methods) leading to a strategy for cascade recycling. Next, existing products and manufacturing processes will be investigated and adapted for the uptake of secondary raw materials from EOL tarps, leading to high value products made from rPVC. In the **second phase which is the demonstration phase, the entire circularity chain of PVC tarps will be showcased**. This will involve the integration of each optimized component from the innovation and upscaling phase into one of the three demonstration pilots (TRL7). Furthermore, the ecological and economic feasibility of these technologies will be assessed to ensure they are both profitable and environmentally friendly. Finally, we will ensure that the new materials are of high

[18] https://www.catalisti.be/en/projects/dissolv

quality and comply with the proposed standards developed during the project, qualifying the recycled products for market entry (TRL8). Below we describe the different activity blocks and mention for each activity the most important challenge we intend to overcome. **Logistics, monitoring, and sorting**

Challenge: PVC tarps are often large and bulky, making them difficult to handle and transport. Monitoring their condition is essential, as they can degrade over time or become contaminated, complicating recycling efforts. Sorting PVC tarps by type, colour, and condition is labour-intensive and requires specialized knowledge to ensure that they are processed appropriately, whether for reuse, recycling, or disposal.
RETAIN Approach: A drop-off centre will be deployed during demonstration using the facilities of project partners. An AI based digital tool will be used to configure the collection stream optimization. The collected tarpaulins will be first checked in the presence of a digital product passport. If not available (this will be the case during the project phase as currently no DPP exists for tarpaulins), they will be screened for the presence of legacy additives and other tarp components using advanced monitoring techniques such as Raman spectroscopy & Laser-induced breakdown spectroscopy (LIBS). If legacy additives are present, the tarpaulins will be safely disposed of as waste. If not, it will be examined by a visual and mechanical check if they can be reused/repaired/remanufactured. If ok, they will be transferred to 'confectioners' (in RETAIN this will be SIOEN, beyond RETAIN this will also be other confectioning companies) which can bring these tarpaulins again on the market. If they are too much damaged, they will be sent to recycling. This concept will first be evaluated at small scale focusing on manual processing, and afterwards upscaled to an automated process.

Digital product passport
Challenge: The implementation of DPPs across supply chains is a requirement of the ESPR regulation, whereby all products sold in the EU (except food and medicines) will have to have a DPP by 2030. This is a massive undertaking involving the restructuring of data management across supply chains, whereby each product will require a unique identifier that will have a machine-readable mechanism that tracks the product through its lifespan and advises consumers on best practices for recycling, and foster trust between consumers and businesses, based on a DPP carrier with an embedded web link to a DPP database. IRIS (who will engage with standard's official bodies and operators like CTB or UNE in Spain, who they have an active working relation with) will ensure that all standards in relation to DPPs are understood and that we are in compliance. As part of our DPP architecture we will assess current data storing systems to ensure interoperability with what already exists, to make sure that the DPP aligns in as much as possible with what the industry is already doing.
RETAIN Approach: The Digital Product Passport that will be developed in the RETAIN project, will be underpinned by a cocreation approach with representative from the industry, from inside and beyond the consortium based on a series of participatory workshops and scoping sessions in an agile process to gather feedback in relation to current data management and traceability processes, relevant data and information sources (e.g. datasheets, databases, etc.), and connectable enterprise applications, etc. in order to map out the interoperability requirements. Throughout this process, an understanding of the standardization and regulatory landscape will inform a standards-based and compliance by-design approach, as well as ensuring compliance with NGSI, European Blockchain Service Infrastructure, and Gaia-X concepts. The core of the DPP concept is to digitize (or use already digitized) information from the product journey across the complex PVC tarpaulins value chain in a blockchain-based Digital Product Passport, which will be accessible via smartphones for full traceability.

A high-level architecture of the RETAIN DPP is presented in figure 5, and it provides a robust framework for secure, transparent, and traceable product information management, utilizing advanced technologies like blockchain distributed ledgers, and secure identity management systems. It is a comprehensive integration of various information technology components to ensure secure and efficient data handling throughout a product's lifecycle.

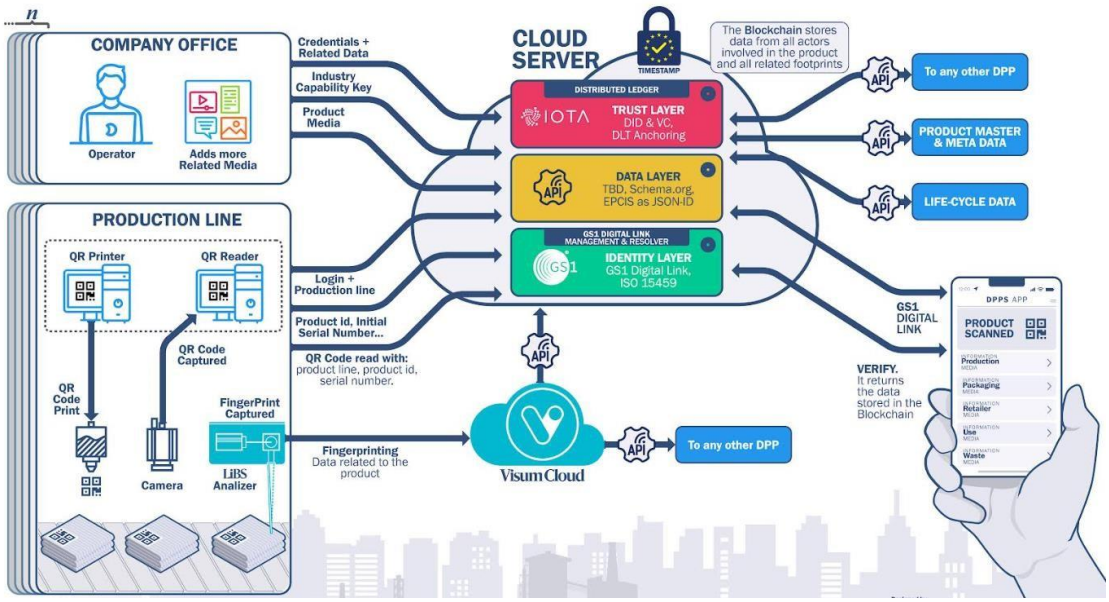


Figure 5: high level architecture of IDP and DPP

The production line level includes a data carrier management system (QR code, RFID or other tagging system), which will be based on the most appropriate tagging or carrier system. Login and production line data are transmitted to the DPP cloud server. Product fingerprinting can also be captured by appropriate inline analysers (such as, VisumHSI, VisumLIBS or other process analytical technologies trademarks from IRIS) installed in the line (or taken from handheld analysers along the value chain) for capturing spectral or image data in relation to chemical composition (unique fingerprints) about the material or product. The 'Visum Cloud' collects and processes fingerprinting data related to the products for obtaining DPP compatible information that can be sent to the RETAIN DPP or any other DPP in the future (as part of our strive for future interoperability across DPPs which will become widespread as regulatory compliance kicks in).

The Company Office component provides a feature for operators for managing credentials and related data, adding industry capability keys, and providing product media, which are securely transmitted to the DPP cloud server.

The DPP Cloud Server consists of a series of layers. The Identity Layer uses GS1 Digital Link and ISO 15459 standards for identity management and data resolution. The Data Layer manages schemas and product data, represented as JSON-ID for standardized data exchange, and an Application Programming Interface (API) enables data interaction between various components and the cloud server for product master and metadata management, lifecycle data management, interfacing with other DPP systems, as well as ensuring connectivity with existing enterprise and business information systems (as identified as part of the cocreation process). The Trust Layer employed Distributed Identity (DID) and Verifiable Credentials (VC) for decentralized ledger technology (DLT) anchoring, ensuring data integrity and trust. Blockchain Security ensures that all data transactions are securely timestamped and stored from all actors involved in the product lifecycle, ensuring transparency and immutability.

A Mobile Application (DPPS App) enables product scanning, whereby users can scan products using QR codes, and the app interacts with the DPP cloud server via the GS1 Digital Link to verify and retrieve product data stored on the blockchain, and displays information related to process of end-of-life complex PVC or tarpaulins, use, and waste management of such product. As described, the goal is to leverage a standardized, interoperable, secure and scalable DPP architecture as a replicable template for DPP developments, which could also could potentially contribute to a DPP standard in Europe going forward.

Reconfectioning and deconfectioning

Challenge: To increase the circularity of PVC tarps, one must avoid that tarps are disposed too early which is currently the case and it is important that actions are taken to extend their lifetime. Reconfectioning (i.e. repair/reuse/remanufacture) of EoL tarps is hardly done today as the logistics and sorting of these materials is highly underdeveloped. Manual de-confectioning is labourintensive and costly. Automated

processes for disassembly are not yet widely available or efficient for such complex materials. *RETAIN Approach*: **Reconfectioning** involves: i) repairing any holes, tears, or damage to ensure its effectiveness in providing protection from the elements. As already many repair kits (containing glues, vinyl cement, grommets and even vinyl-coated fabric patches) are on the market, focus will be on informing customers that such repair kits exist and can also be ordered from the tarp manufacturers. If customers cannot repair the tarps by themselves. They can send it to a company who is doing confectioning such as SIOEN. ii) Reusing/remanufacturing: EOL tarpaulins still in good condition will be reused/remanufactured meaning they will be sent to confectioners who can then confection and sell these second-hand tarp materials in 'new' products. Important here is to know the age of the tarpaulin to estimate the remaining lifespan. During the project, this info must be provided by the company that disposed the tarpaulin. Later this info will be provided by the DPP. **De-confectioning**: metallic parts such as buckles, zippers and eyelets will be removed, as they block/contaminate the recycling process. To be economically viable about 250m²/hour/person should be de-confectioned. Therefore, a semi-automated deconfectioning workstation will be developed.

Cascade recycling

Challenge: The recycling infrastructure for PVC tarps is limited, with many facilities not equipped to handle such materials because of their complex composition, leading to more waste ending up in landfills. This gap in infrastructure highlights the need for improved recycling systems tailored specifically to PVC tarpaulins. Also, especially during mechanical recycling microplastics could be unintentionally released in the environment.

RETAIN Approach: we will further work on different promising recycling technologies. i) *mechanical recycling*. Here the PVC tarpaulins will be shredded, washed and grinded. This material can then be used as such in injection moulding to produce PVC parts. Initial tests will be executed on kg scale, but later, it will be upscaled to ton scale. To separate the PVC from the PET we will optimize to dry mechanical separation processes. The first one is using air treatment, vibration and sifting (KKF), the second one a shaving process (EKOTREND). This leads to a PET free PVC fraction which can be used as secondary raw material and a PET/PVC fraction that will be further purified by dissolution processes to isolate the polyester and the PVC. Special exhaust systems will be used to avoid that potential harmful microplastics will be released in the environment.

ii) *optimizing soak-and-separate process* (see figure 6). The process begins by shredding the input material followed by a pre-separation of light fibres (not-bounded PET fraction) by a conventional cyclone system (dry process). Then, the multilayer material is soaked in a solvent and the PVC fraction swells. Next, steam (referred to as catalyst) is injected, triggering a rapid evaporation of the solvent (popcorn effect). As a result, the PVC coating separates from the PET substrate. To obtain different individual streams, the materials will be separated using conventional methods like zig zag air separation.

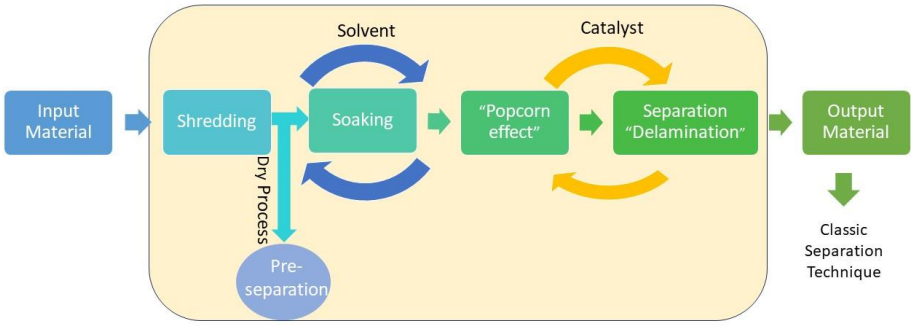


Figure 6: Soak-and-separate process

The solvent is recovered through a distillation process and reused in a closed loop system. This equipment is available at CTB, and it is targeted to scale it up from a few kgs/day to 50kg/day by optimizing the process parameters. iii) *Dissolution*. The PVC, including the plasticizer, will be dissolved and the PET fibre and other potential fillers such as flame retardants remain as solid. The PET fibres and fillers will be separated by a density separation process (see also the schedule shown in the state-of-the-art section of the Polyloop process of which SF is allowed to use it for tarp recycling). The PVC will be in a later stage precipitated and can be reused.

iv) *Selective dissolution and extraction methods* for plasticizers and other high value components. In this process, the tarpaulin will be shredded, and then dissolved in a suitable solvent, followed by the filtration of the insoluble fractions, that could be composed of flame retardants, fillers and PET fibres, that will be later separated by other techniques. The dissolved PVC will be then precipitated with an antisolvent and separated from the other solubilized components. Final dissolution will be distilled to recover the plasticizer. Each of the recovered fractions will be purified by different secondary extractions, washing and distillation processes, as well as with supercritical CO_2, to remove volatiles contaminants.
Initials test will be performed on lab scale and later upscaled to DEMO scale. The quality of all the recycled materials will be analysed by determining the exact composition, the purity, viscosity and thermomechanical properties of the materials.

High value rPVC products by uptake of secondary raw materials from EOL tarp
Challenge: Integrating recycled materials into new products requires careful management to ensure quality and performance, as recycled materials can vary in consistency and degrade over time. Designing with bio-content materials involves sourcing sustainable inputs and ensuring they meet durability and functionality standards while balancing environmental benefits. *RETAIN Approach*: The various recycled products (PVC, PET, plasticizers, etc.) coming from the various recycling technologies will subsequently be integrated into (new) products with at least 50% recycled content. Examples of such applications are PVC/PET fractions that will be used in injection moulding to produce unique fibre reinforced PVC pieces, recycled PET fibres will be processed into high tenacity multifilament yarns which can be again used as reinforcements in tarps, and recycled PVC that will be converted into roofing films by means of extrusion coating or calendaring. Furthermore, we will also integrate bio content (PVC and plasticizers) into these products to further increase the sustainability of these products.

Demonstration by piloting circularity activities in 3 EU regions
Challenge: show cost efficient, reproducible and effective circularity processes for EOL tarp at semi-industrial scale.
RETAIN Approach: we will demonstrate that post-production and EOL tarpaulins can be recycled at multiple tons scale at 3 locations in the EU in an efficient way. By applying technologies and processes which were optimized during the innovation and upscaling phase we will demonstrate every step of the RETAIN circularity chain from collection and sorting up to reconfectioning, recycling and conversion into new products. As shown in the sketch at the beginning of section 1.2.2, each DEMO site will focus on specific aspects related to gate control, reconfectioning, cascade recycling and eco-design of recycled products. These specific aspects of the circularity chain for each region were chosen based on the expertise of the companies present in these regions.
Also, we are familiar with local waste collection and transport peculiarities and we will align with local regulations. A summary table of the pilot and demonstrations activities is given on the next page:

Table 2: summary table of each stage of the pilot and demonstration activities

Task	Deliv. Nr.	Prototype/ Pilot/ Demo	Short description	Location (facility/ city/ region)	Lead partner(s)	Start & end TRL*	Type and volume of RETAIN input materials/ products	Type and volume of RETAIN output materials/ products
T5.2	D5.2	Demo	gate control process: logistics and process monitoring (incl. validation of identification and sorting technologies)	BENELUX	VEOLIA	6 → 8	10 tons of collected postproduction and EOL tarps per day	10 tons of sorted post-production and EOL tarps per day
T5.2	D5.2	Demo	gate control process: deconfectioning & sorting	NL	VWERVEN	6 → 8	1 ton of collected postproduction and EOL tarps per day	700 kg of recyclable tarps per day
T5.3	D5.3	Demo	repair/reuse/remanufacturing	BE	SIOEN	6 → 8	to be determined during project in T5.1	to be determined during project in T5.1
T5.3	D5.3	Demo	mechanical recycling	DE, CZ	KKF, EKOTREND	6 → 8	10 tons of recyclable tarps per day	6-7 tons of recycled PVC/plasticizer, and 2-3 tons of rPET per day
T5.3	D5.3	Demo	solvent-based recycling	FR	SF	6 → 7	300 kg of recyclable tarps per day	100-200 kg of recycled PVC/plasticizer, and 50-100 kg of r-PET per day
T5.3	D5.3	Prototype	Soak-and-separate process	BE	CTB	TRL 5 → 6	50kg of recycled tarp	40kg of r-PET and PVC
T5.3	D5.3	Prototype	Extraction process for specific tarp components (e.g. ATO)	ES	AIMPLAS	TRL 5 → 6	10-50kg of recycled tarp	1-5kg depending on the specific tarp components (e.g. ATO)
T5.4	D5.4	Demonstrator	Uptake into recycled products	BE, FR, DE	RENOLIT	TRL6 →8	5 tons of recycled PVC/plasticizer, and 2 tons of r-PET per day	Multiple tons of roofing and nonwoven materials per day

*The start and end TRL levels refer to the TRL progress targeted in WP5.

Supporting activities: Liaison, DEC, Standardization including business models and market development, Assessment studies, Industry awareness, Training and Certification

Challenges: establishing an EU-wide circular tarp value chain will require to educate and convince all stakeholders to make the transition. Indeed, raising industrial awareness requires overcoming scepticism about the benefits and feasibility of circular practices, which can be difficult given the entrenched norms and practices in many industries.

RETAIN Approach: **implementation of several supporting activities to gain evidence for showing the environmental and economic benefits of the transition as well as to bring this message to all relevant stakeholders. A brief outline of the key elements in the next paragraphs.**

For **liaison**, we will align with the European PVC industry's commitment to sustainable development and explore if we can find synergies with other projects dealing with circularity of (complex) plastics, thus maximizing the exploitation and replication of the RETAIN results. Developing and implementing a consistent **communication strategy** to promote the advantages of circular PVC tarps and educate stakeholders on their benefits and usage requires a coordinated approach and effective channels. This will be elaborated in the **DEC plan**, clearly defining specific target groups and related key messages. This way, we will create general industrial awareness among stakeholders in the broadest possible way. We will select and further define **circular business models**, commercialization, and partnerships strategies by developing a complete overview of the reuse and recycling markets, associated legal and regulatory aspects, and financial incentives. Turning innovative circular business models into viable commercial opportunities involves navigating market challenges, such as securing investment, scaling operations, and proving the economic benefits of circular practices. We will share information about circular business models and the benefits of recycled PVC tarps creating clear, compelling messages that resonate with a variety of audiences and overcoming resistance to change. **Environmental and socio-economic assessment (LCA & LCC & SLCA)***:* accurately measuring and quantifying the environmental impacts of products throughout their entire lifecycle, from production to disposal. We will collect comprehensive data on resource use, emissions, and waste, which can be difficult to obtain and analyse. During the innovation phase of RETAIN, the most appropriate technologies and processes will be identified based on efficiency, ecological impact, costs, etc.

Establishing and aligning industry-wide standards for circularity in PVC tarps involves overcoming the complexity of different regulations and practices across regions, which can hinder the creation of uniform standards and practices. We will support and adhere to certifications and standards that promote sustainable and circular practices in the PVC industry. Standards on circular plastic products are developed in CEN TC 249 (Plastics). We will build on this existing knowledge and existing standards for PVC to develop the circular tarp value-chain and develop a blueprint for certification.

Hands-on training programs will be developed to educate employees on how to increase the circularity of PVC tarps. Employees of waste management companies will be trained on how they can efficiently sort and screen tarpaulins and see which are reusable and which must be recycled. Further, we will prepare education material for employees at relevant stakeholders. Indeed, during the demonstration phase findings will be integrated at the demo sites. tarp confectioning companies and end-users of tarp products (e.g. trailer transport companies) will be educated about collection, sorting, reuse, and recycling options for postproduction and EOL tarps.

1.2.3. Additional methodology aspects

Compliance with 'do no significant harm' principle. The "do no significant harm" principle refers to the six environmental objectives of the EU Taxonomy Regulation. RETAIN aligns with the Do No Significant Harm (DNSH) principles, ensuring that the methodology does not cause significant environmental or social damage. Here are key motivations for why our project is beneficial and does not cause significant harm:

- **Waste Reduction**: Recovering and Recycling PVC tarps reduces the amount of waste sent to landfills, mitigating the environmental impact associated with landfill usage, such as leachate production and methane emissions.
- **Resource Conservation**: By recycling PVC tarps, valuable raw materials are reclaimed and reused, reducing the need for virgin material extraction. This conserves natural resources and reduces the environmental degradation associated with mining and production processes.

- **Energy Savings**: Recycling PVC tarps typically requires less energy compared to producing new PVC materials from scratch. This leads to lower greenhouse gas emissions, supporting climate change mitigation efforts.
- **Pollution Prevention**: Proper recycling processes prevent PVC tarps from breaking down in landfills, where they can release harmful chemicals into the soil and water. Recycling ensures that the materials are handled safely and responsibly. Also, during the recycling process, it will be monitored that no microplastics are released in the environment.
- **Economic Benefits**: Recycling creates jobs and stimulates economic activity in the recycling industry. It also provides cost savings for businesses by reducing waste disposal fees.
- **Health and Safety**: Recycling PVC tarps involves controlled processes that minimize health risks to workers and communities. It ensures that harmful chemicals and additives in PVC are managed appropriately. Within this aspect, solvents used in the recycling processes will be recovered and reused and will not be discharged into the environment. Also, no toxic solvents will be used in the recycling process.
- **Promoting Sustainability**: By recycling PVC tarps, companies and consumers contribute to a circular economy, promoting sustainable practices and reducing the overall environmental footprint.

Overall, transitioning to sustainable and circular PVC tarps is a responsible and beneficial practice that supports environmental protection, resource conservation, and sustainable development, aligning with the DNSH principles.

Artificial intelligence (AI) based systems and/or techniques. The proposed AI-based solution considered for the implementation of specific calculation methodology, follows the EU framework for Trustworthy AI, which defines 3 complementary concepts, Lawful AI, Ethical AI, and Robust AI. The Assessment List for Trustworthy AI (ALTAI) will be used to assess the level of compliance to the key principles of Transparency, Traceability, Explainability, Robustness, Safety, Resilience, Accuracy, Reliability, Reproducibility, Privacy and data governance, Diversity, Non-discrimination and Fairness, Accountability and Redress in case of adverse impact. For AI-based algorithms, particularly optimization routines (e.g. evolutionary algorithms) and datadriven modelling approach such as Artificial Neural Networks (ANNs) or regression generation used for process optimization with no access to personal data or involvement in direct control of recycling processes machineries, adherence to the Trustworthy AI principles can be described in terms of: Technical Robustness and safety ensuring the proposed AI solution will be reliable, traceable and reproducible, able to report on its behaviour even if not involved in direct control of recycling processes machineries. The system will have to be technically robust to handle uncertainties in the operative environment; Privacy and Data Governance ensuring that the AI system does not have access to personal data: Transparency granting the AI-based algorithms functioning will be transparent and understandable even if not involved in direct control of recycling processes machineries; Diversity and Fairness: Ensuring the training data used for the data-driven models are diverse and free from biases, even if it's not dealing with personal data directly. The validation of the proposed AI solution will be performed using specific accuracy metrics relevant to the signal processing task (including metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), or any other relevant metric), to ensure that the predictions remain reliable and consistent as data evolves and conducing a test campaign on relevant scenarios to verify the safety of the provided suggestions. More details in T3.4.

Linked (inter)national research and innovation actions. The table below gives an overview of research projects which have a link to the RETAIN project and also show how RETAIN will benefit from these projects.

Table 3: overview of research projects which have a link to the RETAIN project

Project Acronym (type of project[19]) + short description	Partner involved	Link with retain (related KO)
CISUFLO (HEUR): Route to circular & sustainable floor coverings /	BIG, CTB, RINA-C, EuPC	RETAIN will work out a circular business model for tarps (KO7)

[19] HE = Horizon Europe; H2020 = Horizon 2020; I3 = Interregional Innovation Investments

Cisutac (HEUR): Increasing Circularity and Sustainability in Textiles and Clothing in EU	CTB, AIMPLAS	RETAIN will rely on knowledge generated for high quality rPET multifilament fibre extrusion (KO5)
Digintrace (HEUR) Solutions to promote plastics circularity /	SIOEN, IRIS, CTB, EuPC	Input for digital solution development in RETAIN (KO1)
Dissolv (BEL-COO) – recycling PVC waste from flooring, carpets and tarpaulin applications	SIOEN, CTB, BIG	Recycling solutions for complex PVC waste (KO3)
ECOFACT (H2020) support manufacturing in optimizing energy performance https://ecofact-project.eu/	RINA-C	Will be used in RETAIN to optimize recycling processes (KO3)
FLEX-ID (VinylPlus project): setting up data structure for a DPP for soft PVC value chains	VINYLPLUS, SIOEN, RENO, BIG	RETAIN will build on the blueprint for a DPP for plasticized PVC products as developed within FLEX-ID. As such, DPP development will be able to start from the concepts for which data is needed for soft PVC value chains (KO1).
PULPACKTION (H2020) Optimised moulded pulp for renewable packaging solutions	IRIS	Build further on the developed integrated traceability system (DPP), involving software, QR and a vision system and imaging analysis software (KO1).
REMADYL (H2020) &: Removal of legacy substances from polyvinyl chloride (PVC)	CTB, IRIS, AIMPLAS, VINYLPLUS	RETAIN will build further on the screening methods of legacy additives (related to efficient monitoring and sorting KO1) and plasticizer removal (KO3)
TARPAULIFE (LIFE) demonstrate possibility of POLYOLEFIN tarps	RINA-C	Input for ecodesign of tarpaulins (KO4)
CIRPASS (Digital Europe Project): Digital Product Passports Enabling At-Scale and Real-Life Circular Economy in several value chains	No direct partner involvement	RETAIN will follow such baseline to comply with the existing pilot and standards (KO5)
Circular Flooring (H2020) – recycling of PVC floor coverings		Potential channel for actual recycling of RETAIN preprocessed material (KO4)

Interdisciplinary aspects. A close interaction between scientists and innovators belonging to diverse disciplines is essential to reach RETAIN objectives and high impact. To achieve a successful exploitation of project results, it is essential to translate project outcomes into marketable solutions. For this, the RETAIN Consortium is composed of partners with different background and competencies, covering the whole value chain from design to the manufacturing complemented by other partners providing the expertise which is needed to develop and deploy the project. In the consortium we have partners dealing with 1) Material Science and Engineering to optimize the monitoring, sorting, recycling and manufacturing processes: Veolia, IRIS, VWERVEN, KKF, EKOTREND, SF, CTB, AIMPLAS, RENO, BIG, SATTLER, MEHLER 2) Environmental Science: RINA-C, Economics: RINA-C, industry partners, Policy and Regulatory Studies and Sociology and Public Awareness: EUPC, VINYLPLUS 4) Expertise in digitalization, data management, cloud deployments, blockchain technology and application development will be leveraged for building the digital product passport: IRIS, RINA-C and 5) Certification: CTB. By integrating expertise and methods from these various disciplines, the PVC tarp recycling initiative can achieve its

objectives in a holistic, efficient, and sustainable manner, ensuring that the recycling processes are environmentally friendly, economically viable, and socially responsible.

Social Sciences and Humanities (SSH). An effective contribution will be given at two different stages before and after the solution development. In the first stage, an early involvement and engagement of end users to obtain their feedback for the novel products design. Different methodologies (questionnaires, interviews, participatory workshops, etc.) will be evaluated and considered for stakeholders' engagement. Furthermore, open challenges such as cost-effectiveness, environmental sustainability, privacy, security, trust, and ethical issues will be addressed to find shared solutions and enhance the societal impact of RETAIN. In the second stage, a survey to analyse the stakeholders' feedback on the developed solutions will be carried out. This analysis will allow to evaluate the social impact of the RETAIN developed solutions and their replicability potential.

SSH will be included in the RETAIN project covering a wide portfolio of actions: (a) Sociology and Human Behaviour for analysing acceptability of RETAIN solutions through life cycle analyses, life cycle costing and social life cycle assessment; (b) Economic and Business: business strategy for the exploitation and future market uptake of the project outcome, including market and competitor research and business modelling extending the research frontier. Detailed roadmap for replication in sectors beyond those targeted by RETAIN use case will be also performed; (c) Media and Communication: a carefully designed societal outreach campaign, where effective communication strategies will be implemented to target the different stakeholders.

Gender analysis. RETAIN gender policy supports Article 141(3) of the EC Treaty to protect male/female members exercising the rights inherent in fatherhood/motherhood or the combination of professional and family lives and ensure that women participate equally and actively alongside men, while links are established with the EU platform of women scientists to promote gender equality in research. The work also considers the gendered innovations proposed by EC. Gender aspects are considered and include targeted objectives for gender equality in the consortium. The objectives of the project equally address the needs of all genders in relation to plastics recycling and new products development via upcycled materials. Special attention will be paid to data collection from end-users ensuring a balanced representation of gender, age and cultural differences, and their needs, since these factors influence acceptance, expectations and experience. People react in fact differently based on gender, age, social and economic group in regard to degrees of risk aversion that will affect their perception of threats, or in regard to the significance of impacts of the various threats. The above will be taken into account in the development of interviews, questionnaires, focus groups, and also training, that will include both men and women, and dissemination, where project's results are addressed in all genders equally and are adjusted according to their profile and needs.

Open science practices. Open science principles are established as an integral part of RETAIN methodology and Management envisaging a Data Management Plan (DMP) produced to guarantee that data is 'as open as possible' and in line with the FAIR principles. The data/research outputs will be handled in fact according to FAIR principles (Findability, Accessibility, Interoperability and Reusability), unless for the information classified as sensitive. All data and results are shared, as early as possible, in a systematic way between the RETAIN target groups, pilots, third parties, and internationally. From the very beginning all stakeholders are involved order to develop better, more innovative and relevant outcomes and to increase societal trust in the RETAIN process and outcomes delivering re-used, recycled and upcycled novel products. The consortium will participate to open peer-review with direct exchanges with scientific community by means of Open Research Europe. Open access is offered to all results (algorithms, models, data, etc.) that are released as open source, public and free. Moreover, RETAIN participates in open peer-review, the publications of the project will be published in Open Access Journals and Open Access Repositories. In addition, all project reports, dissemination material and research datasets classified as Public will be available on the project website. The project has dedicated part of its budget (under the dissemination manager) to cover the fees associated with open-access publications to facilitate dissemination and reuse of the project results.

Management of Research Data and other research output. The preliminary Data Management Plan (DMP) addressing FAIR principles has been outlined.

Table 4: FAIR principles outlined

Type of data	Findability	Accessibility	Interoperability	Reusability
LCA data (materials, scope, energy flows, processes)	Project Teams setup, interaction among partners (e.g. email, questionnaire)	Data owned by each Company, not public upon consensus	Standard data files: excel, doc	Results are made public, but source data are confidential
Technical data, product data	Teams environment, project	Specific naming & labelling	Standard data files: txt, doc, xls, csv…	Offered in appropriate format & data sharing licenses
Liaison data (e.g. from events)	Only in restricted MS Teams environment	Restricted to organiser	Audio/video recordings, registration	Anonymised form only
Data from questionnaires	Only in restricted MS Teams environment	Restricted to organiser	Online survey tool	Anonymised form only

2. Impact

2.1 Project's pathways towards impact

2.1.1 Unique contributions to topic outcomes and longer-term wider impact

The impact of tarpaulin recycling involves several environmental and economic considerations. Tarps are widely used for various applications, including covering goods during transportation, providing shelter, and protecting materials from the elements. Here are some key aspects of the expected impact of RETAIN tarp recycling:

Scientific impact: RETAIN will create a strong scientific impact, including via:

- Successful development of selective solvolysis and extraction techniques to isolate specific components from tarps will allow RTD partners to expand and exploit their knowledge to other recycling areas.
- Non-confidential project results will be published such that these results are available to other researchers.

Target group: R&D&I personnel in recycling industry, RTD centres, (non) governmental organizations and others involved with sustainability, circularity, recycling, etc.

Economic/technical impact:

- Identification and demonstration of successful recycling technologies will enable tarps recycling at economic scale, thus contributing to EU's circularity and sustainability targets, and creating new companies and jobs along/in the recycling chain.
- Development of (new) products based on recycled materials, and corresponding sales. Using recycled components from tarps in the manufacturing of new products can result in cost savings for businesses. Recycled components are typically more cost-competitive than virgin chemicals, providing economic incentives for manufacturers.
- Recycling of tarps contributes to the development of markets for recycled plastic materials. As demand for recycled plastics increases, it stimulates economic activity in the recycling sector and provides opportunities for businesses to enter the recycled materials market.
- Implementing tarp recycling programs aligns with Extended Producer Responsibility initiatives, where producers take responsibility for the entire life cycle of their products. This can enhance corporate responsibility and compliance with regulatory requirements.

Target group: companies and organizations along the recycling and circular value chains, including recycling companies, process industry, logistical sector, converters of recycled materials, etc.

Societal and environmental impact:

- *General:* The objectives as stated in this RETAIN proposal will strongly support the need for sustainability and circularity expressed in the EU Green Deal, the new Circular Economy Action Plan (CEAP), and the Eco-design for Sustainable Products Regulation (ESPR).
- *Waste Reduction:* Today <1% of post-production and end-of-life tarp materials are being recycled, 36% end up in landfills and 63% of these materials are being incinerated. RETAIN targets to bring the recycling of tarp materials to over 90% by 2036. As such our project will have a strong impact on the reduction of the unsustainable 'solutions' of landfill and incineration. Tarps contain durable materials, and recycling of these materials allows for the extension of the functional lifetimes, thus contributing to circularity of plastics in the EU, and preventing them from becoming a long-term environmental burden.
- *Sustainable Development Goals (SDG's):* The impact of a successful RETAIN project will strongly contribute to the UN sustainable development goals. Although this impact will be on multiple SDG's (4, 8, 9, 12, 13, etc.) we explicitly mention the following: (i) **SDG12** - Responsible Consumption and Production: RETAIN will pursue a business model and product development strategy compliant with the principles of the circular economy. Waste is avoided through upcycling, recycling and optimized production processes, and the life cycles of products are extended through circular services such as repair and reuse. Ensuring tarp quality and developing circular materials addresses other aspects of this goal; (ii) **SDG8** - Decent work and economic growth: the proposed RETAIN circular tarp value chain (i.e. from collection, monitoring, sorting and deconfectioning to remanufacturing and manufacturing of (new) products from recycled tarp components) will bring (new types of) jobs and contribute to the

development of the circular economy with a targeted yearly recycling volume of over 100kton of post-production and end-of-life tarps.

Target group: environmental organisations, policy makers, all EU citizens

Below the unique contribution of our project results related to the specified outcomes of this topic and the wider impacts, are described.

1) Enhance European industrial sustainability

This is related to KO5 which is the demonstration of circularity for complex PVC value chains at TRL 8 via 3 dedicated pilots in the EU. This result will demonstrate that solutions at scale are available to increase the sustainability of PVC tarps. The related knowledge gain and expertise built-up will give major RETAIN companies, e.g. VEOLIA, BIG, RENO, SIOEN, a unique competitive advantage **Wider impact**: on the longer term, the technologies as developed and demonstrated within RETAIN have potential to be applicable to other complex PVC products such as roofing materials and flooring materials. Successful first implementation of circular tarps will strengthen the EU leading role in producing these technical products, giving also export leverage to the companies.

2) Improve on consumer and citizen benefits

Although RETAIN is focusing on industrial partnerships, consumers and citizens will also benefit from the project. The aim is to avoid potentially hazardous materials re-entering the market (KO1 gate control) and via certification consumers and citizens will be informed which products are safe to use (KO4). **Wider impact:** On longer-term making also other applications circular such as consumers good (bags) or small tents should be feasible.

3) Increased deployment and demonstrated benefits of advanced digital solutions (e.g., through AI, robotics, IoT and blockchain) in circular businesses including waste management and recycling;

Optimization of the use of recycled tarpaulins in RETAIN is based on deploying digital technologies to enhance their chemical upgrade. At product and process level, optimization to efficiently valorise the materials and consume less energy, by applying novel digital technologies (e.g. data driven optimization, sensing, and sorting mechanisms) and chemical innovations for recycling, repair or reuse is to be employed. A set of demo-specific sensing and sorting technologies will be applied to ensure material suitability to valorisation routes and to contribute directly to the optimization of waste materials, while also unlocking economic opportunities (KO1 & KO5). The use of DPPs offers a possibility to examine how transparency through digital traceability may assist, not only the improvement of use of the specific waste materials, but also the reinforcement and optimization of methodologies for tracking materials and products across and within different value chains (KO1). **Wider impact:** these developed and demonstrated technologies will also be applicable to other plastic circularity chains.

4) Emergence of new value chains using upcycled and/or recycled resources, e.g. through industrial symbiosis;

The obtained recyclates and extracted components (KO3) are available for other sectors such as flooring (e.g. cushion vinyl), roofing, injection moulding parts as will be demonstrated via our ecodesign approach (KO4). To **facilitate such symbiosis**, we have included partners from these industries (BIG and RENO) in the RETAIN consortium and have the associations ESWA (roofing) and IVK (foils and films) in the Transition Support Group. **Wider impact:** increased use of secondary materials in related sectors, but also the tarpaulin manufacturers themselves will reuse the extracted tarp components (PVC, PET, additives and plasticizers), leading to a truly circular value chain.

5) Increased uptake of recycled material and upcycling to new high value products;

RETAIN will demonstrate at least 3 recycling technologies to TRL8 level, thereby converting EOL and post-production tarps at tons scale into recycled products such as PVC, PET, and plasticizers (KO5). These resulting products will subsequently be converted into existing and/or new high value products like roofing, flooring and in tarps (KO4). **Wider impact:** By demonstrating the feasibility to obtain large quantities of fair quality recycled materials, the conversion of tarp based secondary raw materials into recycled and upcycled products will be accelerated, also reducing the need for non-renewable resources.

6) Increased resource efficiency along and across value chains, causing a measurable reduction in GHG emissions and other environmental pollution and an increase of carbon removals;

Resource Conservation: Recycling and reusing of tarps mean that raw materials resources are being conservated (KO3 & 4). Production of virgin tarp components such as PVC, PET and plasticizers requires massive amounts of raw materials (e.g. chlorine, ethylene, etc.) and energy (see below). RETAIN will strongly contribute to resource conservation.
Energy Savings: The recycling process for tarps can potentially save energy compared to the production of new tarps. The exact energy reduction will depend on the type of recycling technology that will be applied. Manufacturing of e.g. virgin PVC involves energy-intensive processes, and recycling of PVC can, as such, be a more energy-efficient alternative. **Wider impact:** *Emission Reduction*: The global warming potential of producing a 900 g/m^2 tarpaulin is 4.65 kg CO_2eq/m². When these materials are incinerated at their end-of-life an additional 2.13 kg CO_2eq/m² are released. Hence, the disposal of the tarpaulin encompasses approximately 30% of its total impact. Considering in EU an annual estimated disposal of 200 million m² tarpaulins and assuming landfill would be no longer allowed, this **tarp incineration would encompass an EU-wide impact of over 400kton of CO_2eq**. **By recycling PVC tarps this will be avoided and we will avoid as well greenhouse gas emissions associated with the production of virgin PVC** as recycling requires less energy and produces fewer emissions than the manufacturing of virgin PVC; *Avoidance of Hazardous Waste*: Proper recycling and disposal of PVC tarps containing legacy additives can help avoid the generation of hazardous waste associated with the current incineration or landfilling of such materials.

7) Increased diffusion of new circular business practices, in particular in the uptake of repair, reuse, and remanufacturing, but also practices that form part of the sharing economy.

RETAIN aims at providing validation of circular business practices via the pilot demonstrations which will cover all the stages of the novel circular value chains: collection, screening & sorting, repair, reuse, remanufacturing and recycling will be piloted (KO5). These flagship pilot activities are expected to boost the adoption of circular business practises for tarps and related products. Moreover, activities as replicability assessment, training, stakeholder engagement, are supposed to further boost the deployment in real cases of RETAIN approach (KO7). **Wider impact:** RETAIN will focus on extending the lifecycle via repair, reuse, and remanufacturing. That will lead to new circular business practices and a related job creation of ca. 800 FTEs (Full Time Equivalents), see also section 2.1.2.

2.1.2 Scale and significance

RETAIN markets and scope scale and significance.

Tarp market. As mentioned above, **the total EU production of tarps is appr. 200mio m², representing a value of €600-800 mio for a** common 600g/m² tarp with a typically *materials* value of ca €3-4/m². **The RETAIN tarp manufacturers SIOEN, SATTLER, MEHLER and SF represent together over 90% of the EU production capacity**. These tarp materials are confectioned into applicable products such as trailer curtains (60mio m²), tents (appr. 30mio m²), pool covers (20mio m²) as main applications but also in other applications like textile architectures, biogas installations or packaging (wrapping). Confectioning is an activity with relevant added value: after confectioning, the tarps *application* value is typically around €10/m², and up to €20/m², depending on the degree of confectioning. The **confectioned tarp market thus represents ca. €2-4billion in EU**. The global tarpaulin market is valued at approximately €8.0billion (2022). The tarp market is projected to **expand at ca. 4.5%** (CAGR) from 2023 to 2028, reaching worldwide ca. €10billion by 2028. The growing adoption of tarpaulin sheets for protecting a wide range of products (e.g. industrial equipment) from extreme weather conditions is anticipated to play a crucial role in this market expansion.

PVC Recycling: The **global PVC recycling market is valued at ca. €3.0billion** (2023) and is expected to grow to **ca. €4.5 billion** by 2028, **at a compound annual growth rate (CAGR) of 8.0%**. This growth is driven by increasing regulatory pressures and demand for sustainable materials across industries like construction and packaging (MarketsandMarkets, GlobeNewswire). Within this global market, **Europe** represents a significant portion of PVC recycling efforts due to the region's strict environmental regulations and focus on sustainability, especially in **Germany, France, and the UK**, where companies like Veolia, who is partner in the project, are leading players (MarketsandMarkets).

Digital solutions (including DPP) for circular economy: **EU circular economy market**, driven by increasing regulatory compliance needs, like DPP, and taken across industries such as textiles, electronics and construction, is estimated over **€600 billion** by 2030 (Textile World, euronews, Worldfavor Sustainability

Blog). **Global market** for digital sustainability solutions expected to surpass **USD 1 trillion** by 2030 (Worldfavor Sustainability Blog).

RETAIN expected outcome scale and significance.

As mentioned, today less than 1% of tarps is recycled. RETAIN goal is to have circular use of 90% of the EOL tarps that are produced in EU. This means ca. 180mio m2 or ca. 110kton (at 600g/m2). Realising this expected outcome will lead to a very meaningful reduction of impact on the environmental, create a novel market of circularising tarps which in turn will generate jobs. *Environmental benefit*. Production of a typical tarp requires ca 4.65kg CO_2 eq/m², of which ca 4.15kg CO_2 eq/m² stems from the raw material production. The remaining 0.5kg CO_2 eq/m² is mainly due to the materials processing (energy) and logistics. Incineration at EoL of the tarps add another ca. 2.1kg CO_2 eq/m². As motivated before (section 2.1.1 item 6) **prevention of tarp incineration corresponds to a reduction of ca 400kton CO_2 eq annually**. Further, if we assume remanufacturing can be applied to 10% of the EOL tarps, **remanufacturing will lead to over 55kton CO_2 eq reduction**, assuming 18mio m² remanufactured tarp and the reduction being 3.15kg CO_2 eq/m² (i.e. 4.15kg CO_2 eq from tarp production minus 1.0kg CO_2 eq/m² as estimate for the remanufacturing process. Recycling will also reduce the CO_2 emission but as it is difficult to assess (the technology still has to be scaled up), we do *not* take it into account here although **RETAIN cascade recycling will further boost the estimated reduction of CO_2 emission**.

Market creation. RETAIN processes will lead to various secondary raw material streams, mainly of rPVC and rPET but also e.g. plasticizers and (FR) additives. Because of the mix, it is difficult to estimate the exact value, especially because the market value of the secondary raw materials will largely depend on the level of purity we can reach. Indeed, current industrial scale recycling of PVC products is typically mechanical recycling, leading to relatively low quality secondary materials that are used for low value (bulk) products, e.g. brick-type structures. To give nevertheless an estimate, we assume an average revenue of €1000/ton of secondary raw material. As the goal is to process 90% of the EoL tarps by 2036, which is estimated to correspond to 110kton (see above), this translates into **a market for secondary raw materials from EoL tarp of ca. €100mio (80% share of EOL tarps) and a remanufacturing market value of ca. €300mio** (**10% share of EOL tarps**) (see '**BC #3 – Remanufacturing and repair**' from SIOEN below).

Job creation: for estimating the number of Full Time Equivalents (FTE), we consider the most labour-intensive aspect of circularisation of tarp use: the deconfectioning step. We aim to develop semi-automated workstations that enable to process 250m2 per hour per person (see KO2) which corresponds to ca 200ton/year. Consequently, **ca. 550 FTE will be needed to deconfection 90% of EoL tarps** (estimated at 110kton). All the other steps in the circular value chain (collection, transport, screening & sorting, running the recycling processes) will of course also require adequate staffing so that **we estimate the total job creation ca. 50% higher, or ca. 800FTE**.

Socio-economic: The PVC industry and tarp manufacturing industry is strong in EU, with still a high share of manufacturing within EU, e.g. **for tarp 2/3 of what is yearly put on the market is EU produced**, a ratio of EU production that many sectors no longer have. But, on the other hand, PVC and PVC-based products are under scrutiny, e.g. from ECHA. Often actually the additives pose the main issue from health and safety side, not the PVC itself. Therefore, RETAIN is crucial: its aspects of ecodesign and circularisation of tarp use will show that PVC is intrinsically well-fit for recycling and for sustainable products. Preventing tarpaulins from ending up in landfills or being incinerated conserves natural resources by recovering useful materials, thereby reducing the need for new raw materials. As shown above, the RETAIN processes will lower CO_2 emissions associated with the production of new plastic products, and will promote sustainability and social responsibility, **enhancing the reputation of companies that adopt these circular tarp practices**. Additionally, it can lower disposal costs ('gate fees') for companies and municipalities by reducing the volume of tarpaulins that need to be managed. *2.1.3 Requirements and potential barriers*

Requirements and Potential barriers	Mitigation measures

Not enough end-of-life tarpaulin available to feed the tarp circularisation hubs	Initially use of post-production waste to optimise circularisation technologies. Once RETAIN gate control system in place, extend to actual tarp waste.
There are many different qualities (i.e. compositions) of tarps on the market	Apply fast and affordable cost analysis method(s) to differentiate among different tarp qualities. Further, introduction of digital product passport to monitor and sort post-production and end-of-life tarps in a cost-efficient way on long(er) term.
Presence of legacy additives in recycled products will limit their use	Within RETAIN we will develop a gate control system that will allow us to exclude legacy additives from recycled products
There are many components in tarps, making these materials difficult to separate	We will apply cascade recycling of components from tarps. We will first focus on the 3 main constituents of tarps (PVC, PET, plasticizer) and then investigate potential separation of stabilizers, colorants and/or other components
The final recycling process(es) must be economically feasible	RETAIN will prioritise cost effective and scalable technologies (selection after RETAIN development phase) and will sort out qualitative EOL products.

2.2 Measures to maximise impact - Dissemination, exploitation and communication

This section outlines a credible and realistic path to deliver RETAIN innovations to all multi-actors and maximise impact during and after the project. The **Plan for Exploitation and Dissemination of Results including communications tools (DEC)** is drafted at the proposal stage by the consortium and will be fully developed in WP7 during the first months of the project and regularly updated and evaluated during the full project. In the next subsections, we first outline the whole plan, covering the aspects of dissemination, exploitation and communication (section 2.2.1). Then, we detail more the exploitation part (section 2.2.2) given its importance followed by our IPR strategy (section 2.2.3).

2.2.1 Plan for the dissemination, exploitation and communication (DEC) activities

Dissemination. A primary objective of the dissemination strategy will be stakeholder engagement tailored to a wide and diverse array of relevant actors, including, but not limited to researchers, business operators, industry representatives, and European and national authorities. The activities will be monitored and reviewed by using an online tracking tool where all performed and planned dissemination activities will be incorporated by the partners. With each DEC update, the various activities and approaches will be modified if necessary. A structured approach to identifying and engaging relevant stakeholder groups will be pivotal for the purpose of maximising impact and engagement. To this end, the **consortium partners will identify and agree on the relevant stakeholder groups and their intended level of engagement:** identification of the parameters aimed at **categorising the relevance of the stakeholders, e.g. industry, position along the value chain, foreseeable added value for the maximisation of the RETAIN project's results**. This cooperative effort will be initiated through the organisation of workshops involving the entire consortium during the first months of the projects to take full advantage of each partner's network.

The **agreed list of stakeholders will be documented**, concurrently detailing how each of them will be engaged throughout the project lifecycle. RETAIN will encourage the **dissemination** of research results by participating in a series of events and publications, creating synergies with other relevant EU projects, and exchanging best practices with similar projects funded by the industry. In addition, **training materials** will be created in the form of manuals and will be available via an e-learning platform that will be incorporated into the project's website.

Exploitation. The **Exploitation Plan (EP),** as **part of the DEC**, will be designed to multiply the impact of the proposed solutions. The EP will describe the activities to be undertaken (how and by whom) in order to ensure the exploitation beyond the project itself. The target users will be precisely identified and analysed in terms of specific needs and objectives. The exploitation activities will be coordinated by EUPC. **A preliminary exploitable results plan of the project is described in the table below.** The **Key Exploitable Results (KERs) will be regularly revised and updated to determine the RETAIN exploitation priorities** based on: (i) ***degree of innovation***, contribution to advancing RETAIN solutions

beyond the state-of-the-art; (ii) ***exploitability***, feasibility for market deployment, including assessment of IPR & commercialisation readiness; (iii) **(industrial) impact**, including assessment of potential to address current challenges, meet current needs, and contribute to sustainable development. When applicable, key exploitable results will be promoted through the Horizon Results Platform.

List of preliminary Key Exploitable Results (KERs) foreseen in RETAIN

Legend type of exploitation: **KG - Knowledge Generation: partner intends to apply the KG to establish consulting services and/or to deepen understanding; **U - End user**: partner intends to internally use the project output as competitive advantage and/or to solve internal technological issues ; **COMM - Commercialization**: partner intends the direct industrialization and commercialization of the product/process/solution, e.g. via licensing ; **IPR -** the partner is interested in patenting the result.*

Key Exploitable Result / (Responsible partner)	Type*	Target end users	Exploitation plan and time to market / y (years after project end)
KER1 – **Interoperable data platform (IDP) and standards-based digital product passport (DPP)** (IRIS), linked to KO1	COMM	Textile (coating) industry; flooring & roofing industry	Offering RETAIN software platform, own & offering it as module to exiting software platforms/ **1y**
KER2 – **repair/reuse/remanufacturing** concepts, technologies and related know-how (SIOEN, SF), linked with KO2	KG, U	Companies involved in confectioning and deconfectioning of tarps	Optimise RETAIN tarp processing technologies and concepts / **2y**
KER3 – Deconfectioning process for postproduction and EOL tarpaulins (VWERVEN), linked with KO2	COMM	Plastics recycling industry; textiles recycling industry	Scaling up of process; communication about process / **1y**
KER4 – **Mechanical recycling** technologies and related pretreatment (EKOTREND, KKF), linked with KO3	IPR, U, COMM	(SME) companies (starting to be) active in PVC recycling, PVC convertors	Sales of recycled streams after scale up /**2**y; licensing to further companies / **1y**
KER5 – **dissolution recycling** technologies (SF, CTB), linked with KO3	IPR, U, COMM	(Large) companies interested in PVC recycling	Sales of recycled streams after scale up – SF / 2y; Licensing technology – AIMPLAS&CTB / **0y**
KER6 – **valuable additives extraction** (AIMPLAS & CTB); linked with KO3	IPR, U, COMM	(Large) companies interested in PVC recycling	Sales of recycled streams after scale up – SF / 2y; Licensing technology – AIMPLAS&CTB/0y
KER7 – **ecodesign solutions** for complex PVCbased products (tarp: SIOEN, MEHLER, SATTLER, SF; other products: RENO, BIG), linked with KO4	U	Producers of PVCproducts, e.g. tarp, flooring, roofing	Applying hen bringing novel circular products to the market / 0y
KER8 – Tarp circularization hubs, including gate control process and first processing and separation (VEOLIA), linked with KO5	U, COMM	PVC Waste collectors and first processing	Own use – setup in several EU regions, consider partnerships with other waste collectors

KER9 – Environmental and techno-economical assessment of the RETAIN circular tarp technologies, processes and products (RINA-C, EUPC, VINYLPLUS & RETAIN companies), linked with KO6	KG, U	Producers of PVCbased complex products, e.g. tarp, flooring, roofing	Communication of public results (RINA-C & EUPC, VINYLPLUS to their members)/0y; internal use for decision making (RETAIN companies)/0y
KER10 – Training and learning materials on 'circular complex PVC-based products' (EUPC, VINYLPLUS), linked with KO7	KG	Plastics industry & users; higher education & VET providers	Offering own courses + interaction with education & training providers to integrate in curricula **/ 0y**
KER11 – **Standardization input**, e.g. workshop agreement (CTB), linked with KO4	KG	EU, (inter)national standardisation bodies	Present to standardization bodies / 0y, already during project

Communication. The communication activities will include practice-oriented elements covering items from clean messages to visual aids, success stories, and feedback mechanisms. These activities will cover the following: brand identity, communication kit, project website, social media channels, publications, networking activities & synergies as explained below.

Brand identity. Core elements of RETAIN branding will be a logo, colour palette, typography, project description, document and presentation templates. Consortium partners will be provided with style guidelines for consistency and coherence. **Communication kit.** Printable and digital versions of dissemination material (leaflet, roll-up banner, poster, general project presentation, etc.) will be designed in line with brand identity to present the project objectives, consortium and expected impact. If needed for regional actions, materials would be translated into local languages. **Project website.** The RETAIN website will serve as main information source about the project and the latest project updates and news and offer the public deliverables, and any other relevant information about the project. **Social media channels.** A LinkedIn account will promote the latest news and updates, with a goal to reach a wider audience and drive online trafficking towards the project website. Specific social media campaigns will be regularly conducted, e.g. highlighting team members, important realisations & results or key events. **Publications.** RETAIN outputs and results will be considered for publication in technical journals and conference proceedings.

Partners will also be encouraged to present in conferences and innovation fairs. **Networking activities and synergies.** The consortium will pursue networking opportunities in order to build connections and create synergies that can strengthen the impact of the project. To this purpose, the project partners will collaborate with other Horizon Europe and EU-funded projects as well as national, local, and industry initiatives.

2.2.2 Business case and exploitation strategy

Innovations resulting from RETAIN will contribute to market deployment of circular tarps and establishing related circular value chains for complex PVC products. For the exploitation strategy, we focus on all elements required to establish these circular value chains, i.e. from (logistics for) collection over pretreatment and recycling to uptake of secondary raw materials into products. A core element for the business case are the so-called tarp circularization hubs, which are introduced below, followed by highlighting some of the related business cases.

RETAIN unique value proposition: tarp and complex PVC products circularisation hubs

Market potential for circular PVC products. For products made from recycled PVC, RETAIN targets the same properties as for virgin material, so the current markets are targeted. Different situation for repaired and remanufactured tarps where the properties/possibilities will be different as for virgin materials. Nevertheless, various markets are expected to be interested: in agriculture and livestock, farmers use tarpaulins to safeguard crops and equipment; the logistics and transportation sector relies on tarpaulins for load protection during transit, and a recycling system would align with their sustainability goals; the event and exhibition industry, which uses tarpaulins for temporary structures, could enhance its image and

meet sustainability expectations through recycling initiatives; the construction sector, telecommunications and renewable energy sectors use tarpaulins to protect outdoor equipment, and recycling these materials could support overall sustainability efforts. Governments and the public sector could promote and incentivize the use of such tarpaulins. As for **sales channels**: same as current existing, but also focusing more on public procurement.

Customer Relationships. To build and maintain customer relationships, **education and awareness initiatives** will be carried out, providing information on the benefits of recycled tarpaulins and sustainability practices to build trust. Robust customer support services, including product information, recycling program details, and after-sales assistance, are essential. We also implement loyalty programs to incentivize repeat customers and bulk buyers. Engaging with local communities and businesses through workshops, events, and sustainability initiatives further strengthens our customer relationships.

SWOT Analysis. *Strengths*: unique combination of partners representing large share of current tarp manufacturing capacity in EU (i.e. 100% of EU and 66% of global tarp production) with logistics and waste management infrastructure and expertise. Possibility to develop fitting digital platform according to the specific sector need. *Weaknesses*: high initial costs are required for infrastructure and technology investments, and creating stable demand for recycled materials can be challenging. Ensuring consistent quality of recycled products and scaling the recycling process to meet large demand are other hurdles. *Opportunities*: development of new markets and products, using recycled materials to create items like bags, sports equipment, and building materials. Investing in recycling technologies can lead to new discoveries and advancements, fostering innovative solutions for other difficult-to-recycle materials. Companies adopting advanced recycling technologies can gain a competitive edge by positioning themselves as sustainability leaders. Furthermore, partnerships and collaborations among companies, research institutions, and governments can strengthen the sustainability network, while regulatory support and financial incentives could facilitate the development and use of sustainable recycling technologies. *Threats*: the presence of cheaper or easier-to-recycle alternative materials, variable legal and regulatory frameworks, and fluctuations in raw material prices. Economic uncertainties and risks associated with adopting new technologies also need to be addressed. *Conclusion*: the tarpaulin market is growing significantly, driven by increasing demand across various sectors. While there are challenges and risks associated with introducing circular use and recycling, the potential to reduce environmental impact, conserve natural resources, and gain competitive advantages makes **investing in setting up circular tarpaulins strategically attractive for multiple industries**.

Business case for tarpaulin circularisation hubs.

Value Proposition. The sustainable products derived from the recycling process, which include recycled tarpaulins and related items such as bags and outdoor gear, cater to the growing consumer demand for eco-friendly solutions. By offering effective recycling solutions, the aim is reducing the volume of tarpaulins ending up in landfills or being incinerated, thus contributing to environmental protection. This approach also translates to cost savings for businesses and municipalities by lowering disposal costs. Furthermore, adopting recycling solutions enables businesses to enhance their brand image by showcasing their commitment to sustainability, thereby attracting environmentally conscious customers.

CAPEX for 10kton TCH		
gate control line	1	mio€
deconfectioning stations	2	mio€
logistics	0,5	mio€
		mio€
Total core investment	**3,5**	**mio€**
OPEX per ton		

***Business case financial outline**:*

Crucial for realising the RETAIN impact is to make tarp EOL waste streams available in suitable form, in sufficient amounts and in a safe and cost-efficient way. Indeed, PVC recycling industry is already well-established for several products but not for tarps. Main bottleneck is the difficulty in handling the initial material: large sheets. Hence, the **focus on the collection, screening and first processing, including the deconfectioning**. Once this step is taken, the pre-processed EOL tarp material can be further channelled, be it large sheets for remanufacturing or smaller pieces for various recycling methods. **This preprocessing will be done in the so-called Tarp Circularisation Hubs (TCH). In the next paragraphs we sketch their business case.**

An operational TCH is envisioned as covering the collection, screening & sorting (RETAIN gate control process), first tailored processing (e.g. deconfectioning) and dispatching tarps. Typical scale of the operations is estimated at ca. 10kton. An exercise has been done to outline the simplified investment case for these activities assuming a plant with a capacity of 10kton /year as shown in the table on the side. **Assumptions made for this calculation:** *CAPEX*: simplified to the gate control line (€1mio), the deconfectioning workstations (€2mio) and extra logistics (€0.5mio); *OPEX:* main part is the actual deconfectioning (€100/ton, see also 'BC #2 – Deconfectioning' from VWERVEN below) and further the logistics (€50/ton) and the handling for sorting (€50/ton); *REVENUES:* stem from gate fees (€100/ton) and from selling the pre-processed material (€150/ton). The table shows that for the given scenario, a **payback time within ca. 7 years** can be expected.

logistics	50				50 50	€/ton
sorting	50				100	€/Ton
deconfecti	100					€/Ton
						€/ton
Cost for 1 ton of preprocessed tarp					**200**	**€/ton**
REVENUE per ton						
fee for input material (gate fee)					100 150	€/ton
output matrial streams						€/ton
						€/ton
Cost for 1 ton of preprocessed tarp					**250**	**€/ton**
Tarp collector & pre-processor - extra turnover from RETAIN technology						
Cost for 1 ton preprocessed tarp					200	€/ton
Revenu for 1 ton preprocessed tarp					250	€/ton
Benefit RETAIN per ton preprocessed tarp					**50**	**€/ton**
Processed amount of tarp waste					10000	ton/yr
RETAIN created turnover					2,5	mio€/yr
RETAIN created revenu					**0,5**	**mio€/yr**
Investment needed for the plant					3,5	mio €
Earn back time of investment					**7,0**	**year**

Roadmap towards >90% collection and processing of tarps. Partner VEOLIA is the main partner for the deployment of the TCH. At the end of RETAIN, the goal is to demonstrate a pilot at semi-industrial scale (WP5). In 'BC4 – Tarp circularisation hubs' of VEOLIA (see below), an outlook is given how VEOLIA can invest in setting up several TCHs, leading to a capacity of 75kton by 2036. In capacity terms (financially and operationally) this is feasible for VEOLIA, especially given its current activities (over 600kton plastics waste managed) and investments (ca €200mio in 2023). So, to cover 90% of the EOL tarp market (ca 110kton), the key goal of RETAIN, a further tarp processing capacity of ca 35kton is needed. This is expected to be taken by competitors of VEOLIA as the successful demonstration of the RETAIN pilot will trigger competitors to move into this field.

Even though these are simplified calculations, **it shows that the pathway to the industrial deployment of the RETAIN tarp circularisation hubs to reach over 90% collection and processing by 2036 is realistic, can be achieved with investments within reach of the consortium partners and that the outlook for the operation is economically feasible**, even though it will not be a high profit business.

Summary of related RETAIN preliminary business cases and exploitation strategy

Below, we present a preliminary analysis of RETAIN business cases (BCs) to assess and define the business potential that the newly determined technology will have on industries in terms of enhanced business opportunities. For this we focus on the preliminary business plans for some core developments, namely IRIS, VWERVEN, SIOEN and VEOLIA.

BC #1 – Interoperable data platform (IDP) and standards-based digital product passport (DPP) (IRIS – KER1)	
Target markets	The **Digital Product Passport (DPP) market** in EU due to the **Ecodesign for Sustainable Products Regulation (ESPR) for tarpaulin circular value chains** and related need for the **digital tracking**.
Business opportunity	**EU circular economy market**, driven by increasing regulatory compliance (like DPP) needs across industries such as textiles, electronics, and construction, is estimated over **€600 billion** by 2030 (Textile World, euronews, Worldfavor Sustainability Blog). **Global market** for digital sustainability solutions expected to surpass **USD 1 trillion** by 2030 (Worldfavor Sustainability Blog).
Competing solutions	(Competing) solutions for **Digital Product Passport (DPP) implementation are emerging**, e.g. **Worldfavor** provides a comprehensive sustainability platform to track supply chain data and product lifecycle information, ensuring adherence to EU regulations, particularly for textiles and electronics (Worldfavor Sustainability Blog). Similarly, **Ramboll** offers tools for implementing DPPs for high environmental impact products, that enable businesses to meet transparency and sustainability requirements across their supply chains.
Value proposition	Offer an **interoperable data platform (IDP)** tailored specifically for the **tarpaulins value chain**. The unique competitive advantage of IRIS's secure IDP and DPP is its ability to provide **real-time, end-to-end traceability and secure data exchange across the tarpaulins value chain**, ensuring regulatory compliance and data transparency while facilitating sustainability goals. Unlike its competitors, IRIS's platform is **highly adaptable to complex material flows** and integrates seamlessly into the circular economy, providing unparalleled transparency and sustainability benefits to users.

Preliminary exploitation strategy and business plan	For industrial take-up, IRIS will target **early adopters** in industries, starting within the RETAIN pilot companies followed by other companies in the tarpaulin value chain. Indeed, further partnering with established **EU-based recyclers** and/or manufacturers to implement **pilot projects** beyond RETAIN will help demonstrate the platform's benefits in real-world settings. Next, IRIS will target related sectors where organizations experience high regulatory pressure, such as the **textile and construction sector.** Further also the **electronics sector. Scaling partnerships** with larger industry players will drive broader adoption, also as regulations will tighten by **2026-2030.** IRIS will require an estimated **€2-3 million** for **R&D** to further develop and enhance its **DPP and IDP technologies**, focusing on security, scalability, and compliance: **€0.5-1 million** for **pilot projects** with industry partners to validate the platform in real-world scenarios; **€200k** for **regulatory certifications** and compliance audits; **€300k** for **marketing and business development**. For these investments, IRIS has several options: seeking financing through the **EIC Accelerator** (up to €17.5 million), **EIT Climate-KIC** offers tailored funding, business acceleration, and partnerships focused on circular economy solutions, **European Investment Bank (EIB)** loans and national programs like **CDTI** (Spain) can provide further support. IRIS can also explore **private equity** or **venture capital** for sustainability innovations. For an SME like IRIS, capturing **2-5%** of the **EU Digital Product Passport (DPP) market** by 2030 is an ambitious yet realistic goal. Given the projected growth of the circular economy and sustainability markets, which could reach **€600 billion** by 2030, this could translate to potential revenues of **€12-30 million** for IRIS in DPP-related services, corresponding to ca. 20 Full Time Equivalents (FTE). For IRIS, **trademarking** should focus on protecting the **name**, **logo**, and **specific product features** of its **interoperable data platform (IDP)** and **Digital Product Passport (DPP)** solutions. By securing trademarks on **distinct product names** and **visual branding elements**, IRIS can build strong brand recognition in the **sustainability and recycling sectors**. IRIS will check to **offer certain elements as open source, e.g. the data structure and the user interface** to further increase uptake. Additionally, IRIS could trademark unique **data visualization features** within the platform that set it apart from competitors. This will help IRIS safeguard its market identity and ensure brand loyalty as its platform expands in the EU market. This will give IRIS a strong position while leaving open the option to potentially integrate the software into a large existing platform.

BC #2 – Deconfectioning (VWERVEN, – KER3)	
Target markets	The **recycling industry requires more and more specific pretreatment before actual recycling**. VWERVEN wants to address this for tarps, but results will also be useful for other soft PVC markets.
Business opportunity	**Need for post-consumer recycling of tarps**: currently virtually no tarp recycling, so, a huge need to find ways to process discarded tarps. **Need for secondary (mechanical recycled) PVC material**: VWERVEN currently already pretreats and recycles plastic mixtures, mainly PE, PP and rigid PVC. By pre-processing soft PVC materials in the RETAIN way, it aims to open up an extra market segment currently not accessible.
Competing solutions	Currently, post-consumer soft PVC is only limitedly mechanically recycled, and if so in low value bulk components, e.g. GRC-Blocks. Alternative recycling methods (e.g. dissolution, chemical recycling) could become competitors but in practice focus on other types of PVC waste (e.g. non-REACH compliant).
Value proposition	Soft PVC products include materials that prevent their recycling, like metal buckles. RETAIN **semi-automated technologies for tarp deconfectioning** will remove this hurdle and enable preprocessing.

Preliminary exploitation strategy and business	KER3 final TRL at RETAIN end will be TRL8. VWERVEN will lead the exploitation by further development and industrial scale-up, which will be supported by **investments in deconfectioning equipment** (exact amount will have to be determined but we assume ca €50-100k for a semi-automated workstation enabling a throughput of ca 1ton per person per day), in collaboration with industrial automation partners. The target is to
plan	reach a deconfectioning cost of ca. €100/ton material. The development will take approx. 1 year after project end to reach exploitation phase at a first plant. VWERVEN operates in several European countries (NL, BE, PL, SE and UK). It will **first apply the RETAIN results in 1 plant in The Netherlands, and then spread its use to the other plants**. That way, VWERVEN aims at **reaching a market share within 3 to 5 years after RETAIN of 7-8% of the European market** of discarded tarps, i.e. ca. 15kTons. This is estimated to lead ca. 70FTE (Full Time Equivalents). Given the manual labour needed for the semi-automated workstations, VWERVEN expects that the majority of these FTE will go to people employed at social enterprises, with whom VWERVEN plans to collaborate. VWERVEN plans to use the novel deconfectioning principles and preprocessing solutions **also for the recycling of other types of soft PVC products, like resilient flooring**. This way, the expected turnover and number of jobs will be further increased by ca 20 to 50%. VWERVEN will protect its know-how via secrecy but, in case of successful RETAIN deconfectioning**, it will consider licensing** its technology to other mechanical recyclers.

BC #3 – Remanufacturing and repair (SIOEN – KER2)	
Target markets	**Markets for confectioned tarp** are trailer curtains (60 Mio m²), tents (appr. 30 Mio m²), pool covers (20 Mio m²), etc., which requires confectioning of the tarp materials. That is an activity with relevant high added value: the average value of a typical 600g/m² tarp materials value is €3-4/m². After confectioning, the tarps application value is typically around €10/m², up to €20/m² depending on the degree of confectioning.
Business opportunity	RETAIN will enable to develop **1) a remanufacture market** for good quality used tarps that can be applied in new applications (e.g. heavy packaging for wind turbine parts, big bags, etc.). Assuming 10% of the 200mio m² tarp market replacement by deconfectioned/used tarps, the potential remanufacturing business equals a sales potential of €200-400 Mio for EU tarp confectioning (confectioned tarp value estimated at ca. €10-20/m², as above). **2) a repair market for tarps** (e.g. tents, trailer curtains). difficult to estimate as it is nearly nonexisting today. Roughly: if 10% of the tarp products can be repaired and that way extend their lifetime with 25%, that is estimated to represent a market value of ca. €50-100mio.
Competing solutions	Currently there is **no commercial scale repair in the sector**, **remanufacturing is only minimal**. The real competition is from new tarps, because currently remanufacturing and repair is hampered by the following limitations of deconfectioned tarps: limited availability of good quality deconfectioned tarps and their sizes; their cost; presence of (legacy) additives in deconfectioned tarps.
Value proposition	Repair kits for ecodesigned sustainable tarps. Offering in a cost-efficient way remanufactured EoL tarps, thereby relying on the capability to sort out high quality EoL tarps (KER1) and their efficient semi-automatized deconfectioning (KER3).

Preliminary exploitation strategy and business plan	KER2 final TRL at the end of the project will be TRL7-8. SIOEN will lead the further commercial development and industrial scale-up, which will be supported by internal investments. The needs for further research and investments is limited but installation of the necessary infrastructure and capacity will take some time, commercial exploitation is expected within 2 years after RETAIN. For **repair**, focus will be on tents and other industrial uses (e.g. in biogas installations, roof covers,…). Here, enabling repair will not only lead to extending the lifetime of the product but **because of the sometimes rather complex installation, repairing the tarp can also be cost effective**. The anticipated **tailored repair kits will create a novel market**. For **remanufacturing**, focus will be on technical applications where already quite some confectioning and/or customization is needed, meaning the extra step of deconfectioning can be more easily integrated. Examples are **big bags and custom packaging of large industrial equipment** (e.g. wind turbine parts, heavy machinery, large machine parts) SIOEN **currently confections ca. 1mio m² of tarps** per year for various applications, corresponding to a turnover of ca €15mio (average of €15/m²) and employing ca. 25 people. By adopting RETAIN remanufacturing and repair solutions, SIOEN estimates to **triple its confectioning activities,** especially as expected EU legislation will boost circular thinking within this industry and within society.

BC #4 – Tarp circularisation hubs (VEOLIA, – KER8)	
Target markets	**Market of waste logistics management**, including collection, sorting and dispatching.
Business opportunity	Pioneering role into systematic handling of EOL tarps, benefit of **first mover advantage in the market of EOL tarp handling**.
Competing	No comprehensive solutions exist for handling EOL tarps at the moment.
solutions	
Value proposition	**Blueprint for setting up strategic places of tarp circularization hubs.** Such a hub will cover collection, screening & sorting (RETAIN gate control process), first tailored processing (e.g. deconfectioning) and dispatching tarps.
Preliminary exploitation strategy and business plan	**VEOLIA will lead the exploitation of KER8 beyond RETAIN as follows**: VEOLIA plans to test run the RETAIN pilot during ca. 1 year, followed by the investment in a first pilot of 10kton, which should be built by 2030, i.e. within 2 years after project end. For this pilot plant, a test period of 2 years is foreseen. When successful, the goal is then to invest further to have by 2036 5 more pilots (estimate: 2 further of 10kton and 3x 15kton) operational, which means that in this scenario the total capacity of VEOLIA by 2036 is 75kton (3x10kton + 3x 15kton). From **resources point of view**, the roadmap for VEOLIA is deemed realistic and feasible: for comparison, in 2023 invested ca €200mio. VEOLIA processed ca 610kton of plastics. So, the addition of 75kton tarp requiring an investment of ca. €20-25mio over 5 years is absorbable. **The RETAIN Circularization hub concept and the experience from their piloting will strengthen the market position of VEOLIA by accelerating VEOLIA knowledge and expertise in the pioneering field of EOL tarp handling**. VEOLIA will get a key market advantage as first mover and it will in general reinforce its reputation of leading waste management company by setting up these tarp circularization hubs, among others because it will do so in partnership with the processing companies. Indeed, the projected EOL tarp management capacity would mean that VEOLIA can cover ca. 60-65% of the

	expected EOL tarps. This will generate a turnover of ca €3-4mio and correspond to several hundreds of jobs by 2036. Profitability could increase significantly as it is expected that EU environmental targets will further push handling of EOL PVC products to reduce their disposal via landfilling and/or incineration.

2.2.3 Strategy for the management of intellectual property

General Principles. The IP management will be described in the RETAIN Consortium Agreement (CA) in line with the Grant Agreement. The General Assembly (GA) will be responsible for ensuring implementation of the guidelines established within the CA. Mechanisms will be implemented to ensure adequate communication concerning IP among partners and confidentiality.

Protection, use and dissemination. The partners will agree on IP ownership and management arrangements as part of the CA. The overarching principle will be to manage IP in the most effective way towards RETAIN results exploitation. Knowledge will be captured in tangible formats such as technical reports and patents, when applicable. Where appropriate, IP which is *not* commercially critical will be disseminated to promote continued collaboration beyond RETAIN. All publications will be preliminary peer-reviewed by relevant partners to ensure that no commercial IP is disseminated without appropriate protection.

IP management strategy. All identified IP will be recorded centrally by EUPC (see also Task 7.2). The identified IP shall be maintained as long as it retains value, also beyond RETAIN. Following key elements of the strategy: (i) *Background IP* introduced by RETAIN partners will remain property of the IP owner; (ii) *Access to another participant's results or background IP* shall be granted based on conditions stipulated in the CA. Access rights for academic research purposes shall be granted royalty-free, also beyond RETAIN; (iii) *Improvements or enhancements to background IP* that occur as a result of work completed in RETAIN will be treated similarly; (iv) *Knowledge acquired during the project* shall be considered as a property of the partner(s) generating it, and in this sense, the originator is entitled to use and to license such right without financial compensation to other partners. If the features of a joint invention are such that it is not possible to separate them, the contributors shall agree on joint exploitation.

Table: Overview of RETAIN dissemination and communication plan

Action - Goal	Audience	Channel – Content	Organization
Continuous - Ensure that 'external stakeholders' remain committed to following RETAIN progress and gain increasing exposure	Depending on topic: focus on 'core' stakeholders (e.g. collectors, recyclers) but also on the 'wider' stakeholders (e.g. policy makers) and potentially the general public.	Newsletter, social media, presentations, popular publications – progress updates (small news items) and public results.	EUPC will coordinate. Material collected from ALL; set-up a schedule for contributions and activities, avoiding overlaps
Create awareness for RETAIN and **ensure the external stakeholders understand our goals and vision** by sharing objectives and consortium composition.	RETAIN TSG members and various stakeholder groups, with focus on reaching EC policy makers, tarp users, technical textile media & influencers & crosssectoral multiplicators.	Press release linked to kick off meeting; RETAIN website; first (digital) dissemination material (e.g. common presentation), all following a distinct project visual identity.	By M4 - Responsible: SIOEN (as coordinator) & EUPC (WP7 leader). *KPI*: >100 subscriptions for newsletter, >100 followers on social media
Call for collaboration for pick up of RETAIN results that need external partners and/or follow up projects, focus on use of WP1,2,3&4 results *outside* domain of RETAIN pilots.	RETAIN TSG members and especially innovators & entrepreneurs, large enterprises with own R&D, innovative SMEs, product designers & developers, investors. Also focus on other (EU) R&D projects.	Setup of own (online) events and participation in existing fairs and conferences, dedicated to a specific result/target group. Examples: TechTextil, Plastics Recycling Show Europe, KFair,… Use of compilation of RETAIN results and findings, based on public deliverable content.	Within 3 months after reaching MS5 (M30). Responsible: EUPC (WP7 leader), strong support by IRIS, VWERVEN, CTB & SIOEN (as WP1,2,3&4 leaders). *KPI*: > 8 project offspring activities (e.g. meetings with stakeholders, investment case, collaboration agreement)
Final event: promotion of RETAIN results for maximizing visibility, uptake and impact	Tarp manufacturers & end-users (in construction, mobility & technical applications), policy makers, consumers (organizations), citizens & influencers.	Dedicated final event with content focus on pilots. Accompanied by press release & final media campaign and peer-reviewed and techno-commercial articles.	In period M38-42. Responsible: SIOEN, in strong interaction with VEOLIA (WP5 leader) and EUPC (WP7 leader). *KPI*: > 150 participants in final event, uptake by at least 3 media

Tools and Channels	Description	Timing	KPIs	Responsible Partners
Logo & graphic identity, promotional materials	A consistent, impacting graphic identity to make RETAIN easily recognizable	M4	Imagery included **(a must)** in all communication materials	EUPC
Website & Social Media (LinkedIn, YouTube)	Interactive, dynamic online presence, showing RETAIN objectives & public result	Online at M4	By project end: > 20.000 page views; >400 followers by M24	EUPC, contributions to items from ALL
Newsletter	8 e-newsletters with RETAIN progress	6-monthly from M6	>150 subscribers by M24	EUPC & SIOEN, input by ALL

Press releases	Promote specific activities, events, results	At 'highlights'	>4, pick up by media	Partners involved in 'highlight'
Webinars	Knowledge transfer & training, based on public results & learning material from WP1 to WP6. Go for Circular Economy Webinar Series (EU Commission)	From M24 on	>6, i.e. one per WP >100 views per webinar	WP leaders of WP1 to WP6, EUPC support of common format
Visit to pilots	Wider stakeholders & general public	From M36 on	> 40 participants for demo visit	VEOLIA
Fairs and conferences	Plastics Recycling Show Europe, K-Fair, European Coating Show and Conference, Go Circular Summit, TechTextil, Waste management Europe, European Circular Economy Stakeholder Conference, Ecomondo	From M12 on, most intense in M24-42	>6 participations with presentation (oral/poster), 2 common RETAIN booths in a fair	EUPC & SIOEN, lead partners of key exploitable results
Publications	Plastics News Europe, RecyclePRO, Recycling Magazine. Go for entry in Sustainability Matters Newsletter (EU Commission), Circular Economy Newsletter (Ellen MacArthur Foundation),	From M12 on, most intense in M36-42	>5 publications	EUPC, lead partners of key exploitable results

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2.3 Summary

	Specific needs	Expected results	D&E&C measures
Collection & sorting	• **Efficient collection** of post-production and EOL tarps needs to be developed and set up to initiate circularity of discarded tarps (mainly PVC/PET materials, ca 200kton) • **Fast inline monitoring** techniques need to be developed to identify legacy additives • **Efficient sorting** needs to be developed yielding dedicated material streams for further processing	• **Unique gate control process for tarps** incorporating inline monitoring and efficient sorting into dedicated material streams (D1.1) • **AI based inline spectroscopic monitoring** to identify and screen tarps based on quality and presence of legacy additives (D1.3) • **Digital product passport (DPP)** for tarps to efficiently sort tarp based waste streams (D1.2)	• **D**: Publication in technical journals, conferences and trade fairs (e.g. PRE Show). Seminars and partner networks. Dialogues with adjacent projects. • **E**: Own use of DPP and IPD as monitoring tool, apply also in other sectors (eg construction, electronics) • **C**: Visibility in public media, websites, SoMe and during final event RETAIN Day
Re- / de-confectioning	• **Reconfectioning (=repair/reuse/remanufacturing)** of qualitative discarded tarps needs to be developed and demonstrated to the EU tarp confectioning industry • **High-throughput effective deconfectioning as pretreatment for recycling** needs to be made available • **Make market aware of the potential of high R-strategies** like repair and remanufacturing	• **Circular design approach for tarps** that will result in various repair/reuse/remanufacturing cases (D2.1, D2.2) • **(Semi-)automized deconfectioning workstation** to remove all kinds of parts (eyelets, buckles, zippers, seams,...) from EOL tarps at a capacity of 250m2/hour/person (D2.3)	• **D**: Publication in technical journals, conferences and trade fairs (e.g. Techtextil). Seminars and partner networks. Dialogues with adjacent projects. Workshop on reconfectioning • **E**: Own use of (semi)automized deconfectioning system; use of circular design protocol by EU tarp confectioning industry • **C**: Visibility in public media, websites, SoMe and during final event RETAIN Day
Cascade Recycling	• **Scaled up demo of promising recycling processes** (eg mechanical recycling, dissolution) is needed **to derisk large scale investment** by companies • **Extraction technologies to isolate valuable additives** (eg plasticisers, FR, stabilisers) **to reduce EU's dependency on non-EU import** of such components • **Method to maximise recycling value by selecting the most appropriate cascade recycling option** depending on the properties of the EOL tarp (quality, age, cleanliness,...)	• **Various mechanical recycling technologies** will be tested and compared. Best results will be demonstrated at scale up to 50 tons per day. D3.1, D5.3 & **Dissolution recycling** of deconfectioned tarp materials will be developed and demonstrated at TRL8 level at 3 tons per day. D3.2 and D5.3 • **Selective extraction of valuable additives** will be developed and demonstrated at kg scale. D3.2 and D5.3 • **Cascade recycling decision support tool** (D3.3 & D3.4)	• **D**: Publication in recycling journals, at recycling conferences and trade fairs. Dialogues with adjacent projects. Visit to mechanical PVC recycling plant with stakeholders • **E**: Own use of mechanical and dissolution recycling technologies, and selective extraction. License know-how. Potential to patent equipment design. Sales of secondary raw materials from EOL tarps • **C**: Visibility in public media, websites, SoMe and during final event RETAIN Day. Visit to pilot with stakeholders
Uptake of recyclates	• **Existing products need to be redesigned**, for making them more easy-to-recycle & for better fit for uptake of secondary raw materials • **Existing manufacturing methods and processes need to be reconsidered to enable and ease the uptake of secondary raw materials** • **Novel standards to characterise secondary raw materials** need to be developed to assess their specific properties	• **Circular design** for tarps and related products (D4.1) • **Valuable products from recycled tarp components** will be developed and demonstrated at 1-10 tons per day. D4.2 and D5.3 • **Sustainable and circular tarps** will be developed containing at least 50% recycled/biobased materials, and production demonstrated at 10 tons / day (D4.2 & D5.3) • **Contribution to (CEN) standardization** work groups (D4.3)	• **D**: Publication in technical journals, conferences & trade fairs (e.g. K-Messe – plastics focus and TechTextil – textile focus). Dialogues with adjacent projects and with relevant standardisation groups. • **E**: Own use of secondary raw materials from EOL tarp for roofing, flooring and tarp production. Explore further application areas. • **C**: Visibility in public media, websites, SoMe and during final event RETAIN Day.

	Target groups	Outcomes

Collection & sorting	• **Plastic waste collection and waste management actors** (both NGOs and companies)
	• **Plastic waste sorting** companies
	• **Plastics recycling** companies
	• **Tarp manufacturers and confectioning** companies, incl. for post-production tarp waste
	• **Municipalities & local authorities** with waste collection points
	• **Social enterprises** offering circular services
		• **EU tarp sector adopts DPP** within 5 years after project end, starting from the adaptation by RETAIN manufacturers (who represent ca 90% of EU manufacturing capacity) who will boost its use also to their clients and suppliers.
		• **EU deployed capacity for collection, screening, sorting and deconfectioning of 110kton in EU by 2036**. Realised via ca. 10 plants of 10-15kton, requiring a total CAPEX investment below €100mio.
		• **First mover companies (incl. RETAIN partners) offer high R-strategy services**:
		  • At least 10 tarp conversion companies in 5 EU countries will extend their business with repair/reuse/remanufacturing services
		  • At least 2 permanent tarp deconfectioning points in EU at the end of the RETAIN project

Re- / de-confectioning	• **Tarp converting companies** wanting to extend their business with repair/reuse/remanufacturing services • **Plastics and textile recycling companies** that want to offer additional deconfectioning services • **Social enterprises** wanting to offer deconfectioning services • **Unions/labour organisations**: opportunity for jobs & training	

Cascade Recycling	• **Textiles and plastics recycling companies** that want to integrate (mechanical) tarp recycling in their business portfolio
	• **(Chemical) industries** that are equipped to integrate (dissolution) recycling
	• **Associations, public authorities and NGOs** that want to stimulate recycling and circularity in EU and abroad
		• **EU deployed capacity for cascade recycling of tarp and other soft PVC-based products**:
		  • At least 3 RETAIN tarp recycling technologies further developed to TRL9 within 2 years after RETAIN
		  • RETAIN partners SF, EKOTREND and KKF strongly increase their tarp recycling activities after the project

Uptake of recyclates	• **Tarp manufacturers (also outside EU)** wanting to switch to more sustainable and circular tarps
	• **EU Plastics converters** that can apply recycled tarp components such as PVC, plasticizers and stabilizers in their products
	• **Standardisation & certification bodies**: develop circular economy standards, offer certification services
	• **Policy makers** at regional, national and international levels that regulate and target higher 'recycled content'
		• **Uptake of secondary raw materials from EOL tarp by EU manufacturers into soft PVC-based and other plastics products, eg**:
		  • rPVC from tarps will be applied by at least 5 PVC conversion companies
		  • rPET fibres (>50mm) from tarps will be applied by at least 3 PET nonwoven companies
		• **RETAIN tarp manufacturers produce only ecodesigned tarps with at least 50% recycled content** within 2 years after project end

Impact

- **Circular use of over 90% of EU manufactured tarps by 2036**
- **Economic/technological**:
  - **Sustainable and Circular Value chain for tarp and related products established**, comprising collection, screening and sorting, deconfectioning, remanufacturing, cascade recycling and uptake of recycled streams
  - **Market acceptance of high R-strategies** (repair, re-use, remanufacturing) accepted by the market, eg established remanufacturing market of **ca. €300mio**, assuming a share of 10% of EOL tarps at €15/m2)
  - **Large scale infrastructure for cascade recycling of EOL tarps available** (covering mechanical, dissolution and extraction) – estimated market of **ca. €100mio** (assuming €1000/ton and assuming a share of 80% of EOL tarps at an average of €1000/ton)
  - **Uptake of secondary raw materials from EOL tarps** (rPVC, rPET but eg also plasticisers & FR-additives) - especially in a wide range of soft PVC products (tarp, roofing, flooring)
  - **Competitiveness of EU plastics industry boosted by ability to offer truly circular soft PVC products**, opening also export opportunities.
- **Societal/environmental**:
  - **Reduced environmental impact**, eg incineration of all EOL tarps represents over 400kton CO2eq/kg, remanufacturing of 10% EOL tarps brings ca. 55ktonCO2eq reduction
  - **Generation of (novel types) of jobs** all along the circular value chain, estimated at ca. **800FTE**. An important share, eg for staffing deconfectioning workstations, is anticipated to be **at social enterprises** (ca. 550FTE).
  - **Better image for plastics industry at level of consumers and citizens** as circular manufacturing and use of tarp materials will be communicated as a flagship example of how circular economy can work in EU.

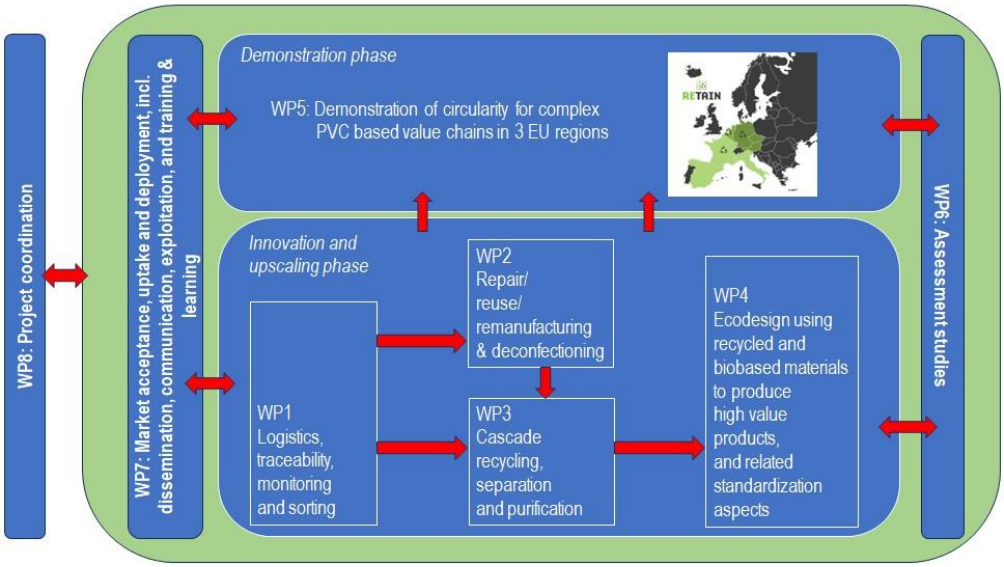
#§IMP-ACT-IA§

3. Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#

3.1 Work plan and resources

3.1.1. Overall structure of the work plan

The RETAIN workplan is composed of 8 work packages. WP1-4 focus on specific aspects of the RETAIN recycling chain, being logistics,



repair/reuse/remanufacture, deconfectioning, recycling, and conversion of recyclates into value added products. WP5 is the actual demonstration part of the project, whereas WP6-7 are supporting tasks. WP8 finally covers the project management and coordination of RETAIN.

WP1 (M1-M24) focusses on

logistics, traceability, monitoring *Figure 7: RETAIN work plan* and sorting. Lead partner IRIS, partner VEOLIA and several other partners will develop and implement a strict gate control procedure that distinguishes between various tarp qualities. The development and integration of a digital product passport for tarp will be an important part of the gate control procedure. **WP2** (M1-M30) deals with repair, reuse and remanufacturing, and deconfectioning of post-production and EOL tarps. Lead partner VWERVEN and partner EKOTREND will develop a (semi)automized deconfectioning system to remove eyelets, buckets, zippers and other functionalities from EOL tarps. Partners SIOEN, SF and several other partners will study repair/reuse/remanufacturing practices to increase the lifetime of tarps. In **WP3** (M1-M30) the actual recycling of tarps will be studied. Various mechanical recycling (EKOTREND, KKF, VWERVEN) and solvent based recycling (SF and CTB) methodologies will be investigated and prepared for demonstration in WP5. Furthermore, selective extraction of valuable additives from tarps (AIMPLAS), and strategies definition and development of a decision support tool for recycling processes (RINA-C) will be developed. Various other partners will be active in this WP under the leadership of partner CTB. **WP4** (M1-M39) covers the conversion of recycled and biobased materials into high value products. producing new tarps through extrusion coating/calendaring as executed by lead partner SIOEN and partner MEHLER. Various other partners such as RENO, BIG, SATTLER and SF will apply plasticizers, PVC and other recyclates into their products. Finally, partner CTB and some other partners will focus on standardization aspects related to tarp recycling. In **WP5** (M19-M42) the actual demonstration of circularity for complex PVC based value chains will be executed in three EU regions. Lead partner VEOLIA will coordinate this work. This WP covers 3 main activities: demonstration of the gate control process, demonstration of recycling processes and technologies, and uptake of recycled products. In the final stage of the project assessment of the RETAIN recycling chain and the demos will conclude on the technical and economic aspects of the RETAIN recycling chain. **WP6** (M19-M42), led by partner RINA-C, will focus on sustainability assessment studies. In **WP7** (M1-M42) lead partner EuPC and all other partners will execute all activities related to market acceptance, uptake and deployment. This WP includes dissemination, communication, exploitation and training & learning activities. Finally, in **WP8** (M1-M42) all project management and coordination activities will take place.

3.1.2. Timing of the Work Packages and main tasks - Gantt Chart

	GANTT chart RETAIN project			Year 1												Year 2												Year 3												Year 4					
	Work package / Task descriptions	**WP/task leader**	**Total cap.**	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
WP1	**Logistics, traceability, monitoring and sorting**	**IRIS**	**81**																																										
T1.1	Definition of logistic value chain and gate control for complex, flexible PVC plastic waste	VEOLIA										D1.1																																	
T1.2	Development of interoperable data platform (IDP) and Digital Product Passport (DPP) prototypes	IRIS																									D1.2																		
T1.3	Validation of spectroscopic photonics based monitoring technologies for the identification of legacy additives	IRIS																									D1.3																		
T1.4	Definition of best sorting strategies for circularity in logistic value chain	VWERV																																											
WP2	**Repair/reuse/remanufacturing, and deconfectioning**	**VWERV**	**47**																																										
T2.1	Repair solutions for confectioned tarps	SIOEN										D2.1																																	
T2.2	Remanufacture of post-production and EOL tarps	SIOEN																									D2.2																		
T2.3	(Semi)automized deconfectioning of EOL tarps	VWERV																									D2.3																		
WP3	**Cascade recycling, separation and purification**	**CTB**	**172**																																										
T3.1	Mechanical recycling of PVC tarps	KKF																									D3.1																		
T3.2	Solvent based recycling	SF																									D3.2																		
T3.3	Selective extraction of valuable additives from tarps	AIMPLAS																																											
T3.4	Strategies definition and development of decision support tool for recycling processes	RINA																			D3.3												D3.4												
WP4	**Ecodesign using recycled and biobased materials to produce high value products & standardization**	**SIOEN**	**67**																																										
T4.1	Setting the ecodesign framework for circular RETAIN value chains	SIOEN										D4.1																																	
T4.2	Introducing recycled tarp components into high value products	RENO																																											
T4.3	Development of sustainable PVC tarps	SIOEN																															D4.2												
T4.4	Standardisation scenario development	CTB																																								D4.3			
WP5	**Demonstration of circularity for complex PVC based value chains in three EU regions**	**VEOLIA**	**117**																																										
T5.1	Planning of demonstration phase	VEOLIA																															D5.1												
T5.2	Demonstration of gate control process and deconfectioning	IRIS																																					D5.2						
T5.3	Demonstrations at scale of repair/reuse/remanufacturing, and recycling	VEOLIA																																								D5.3			
T5.4	Demonstration of products with recycled content	RENO																																								D5.4			
T5.5	Assessing the RETAIN recycling chain, and demos	VEOLIA																																											D5.5
WP6	**Assessment studies**	**RINA**	**55**																																										
T6.1	Experimental LCC and LCA of proposed processes	RINA																									D6.1																D6.2		
T6.2	Experimental SLCA of proposed processes	RINA																																											
WP7	**Market acceptance, uptake and deployment, incl. dissemination, communication, exploitation and training & learning**	**EuPC**	**76**																																										
T7.1	Development of dissemination, exploitation and communication (DEC) plan	EuPC							D7.2																		D7.4																		
T7.2	Exploitation strategies and intellectual property (IPR) management	EuPC																																											D7.7
T7.3	Circular business model supporting RETAIN exploitation	RINA																																					D7.5						
T7.4	Communication and Dissemination Activities	EUPC				D7.1																	D7.3																						D7.8
T7.5	Training & learning material creation	EuPC																																											D7.6
WP8	**Project management & coordination**	**SIOEN**	**45**																																										
T8.1	Project coordination and impact management	SIOEN																																											
T8.2	Technical coordination and project quality assessment	SIOEN				D8.1																																							
T8.3	Day-to-day management, reporting and finances	SIOEN																																											
T8.4	Risk management	CTB							D8.2																																	D8.5			
T8.5	Data management, and ethics	EuPC							D8.3																		D8.4															D8.6			
						MS1						MS2									MS3						MS4						MS5									MS6		MS7

Milestones MS1 - GO for digital platform development and start of solvent based recycling, website online

MS2 - GO for remanufacturing & deconfectioning trials and GO for development of ecodesigned products

MS3 - Start of planning for demo & of assessment studies

MS4 - GO for screening demo activities

MS5 - GO for recycling demo activities, relying on the developed options for remanufacturing, deconfectioning, recycling and ecodesign

MS6 - End of experimental demo activities, switch focus on evaluation and finalisation of documentation

3.2 Capacity of participants and consortium as a whole

3.2.1. Description of the consortium

RETAIN is a strongly industry driven Innovation Action, with a consortium composition focused on developing and industrial deployment of circular tarpaulins. The consortium consists of 12 companies (9 large enterprises (LE) and 3 SMEs), 2 industry associations, and 2 RTD partners. All of them are considered renowned organizations within their respective sectors. Together, these 16 complementary partners form a multidisciplinary team that covers all parts needed for circularity (i.e. from tarp producers over re-use to applicators of recycled and biobased materials). This well-balanced set of competences and organizations is brought together from 8 EU countries (Belgium, Netherlands, Germany, Austria, France, Spain, Italy and Czech Republic).

Figure 8: contribution of participants

The idea for this project was initiated in 2022 by the 4 largest European tarp producers, SIOEN (BE), SF (FR), MEHLER (DE) and SATTLER (AU), representing over 90% of the EU tarp production. At that time these large enterprises defined the need for a European industrial value chain that enhances the circularity of the 200,000 tons per year of post-production and end-of-life tarps. Together they decided to join forces to make, what is by now called the RETAIN project, a success. They are the core drivers of the RETAIN project and their roles are quite diverse: development of a digital product passport for tarps, application of recyclates into novel tarps, re-use and reconfectioning of post-production and end-of-life tarps to enable a valuable, second life for these materials, ecodesign for uptake of recycled and biobased materials, market acceptance, uptake and deployment, and demonstration of processes.

These 4 tarp producers carefully selected further companies for completion of the industrial value chain needed for circularization, covering the aspects of waste collection, recycling and plastic convertors for the uptake of recycled streams as well as for developing the digital logistics platform.

For waste collection VEOLIA (FR) one of Europe's largest waste management companies is included. With subsidiaries all across the EU, VEOLIA is the ideal RETAIN partner to tackle all logistical challenges that come with the collection and sorting of post-production and end-of-life tarps. Veolia will work in close collaboration with most other partners as they are also coordinating the demonstration work package WP5.

The **recycling industry is strongly involved in RETAIN via 3 complementary recycling partners**: **VWERVEN** (NL), the largest recycler of hard plastics in Europe, including rigid PVC; **EKOTREND** (CZ), the largest EU recycler of soft PVC plastics; and **KKF** (DE) a key recycler of rigid and soft PVC. Their complementarity is demonstrated by their respective RETAIN tasks: VWERVEN will focus on the gate control process, more specifically on (semi)automized deconfectioning and washing of tarps, EKOTREND on grinding/shredding, slicing/shaving, granulation and analytical techniques, and KKF on mechanical separation techniques. Together they will also execute tasks related to the fast characterization of legacy additives.

From the **converters side RETAIN has two strong large industry partners**: **BIG (BE) world leader in the use of plasticized PVC for flooring** applications and **RENO (DE) world leader for roofing and pool applications**. They will test and demonstrate the use of recycled and biobased products in these applications. This work will be executed in WP4 (Ecodesign using recycled and biobased materials to produce high value products, and related standardization aspects).

For setting up the necessary **digital infrastructure and software development the companies RINA-C (LE; IT) and IRIS (SME; ES)** will work together, bringing together their own specific expertise: **RINA-C is a leading engineering consulting company, with ample experience in decision support tools**, **IRIS has unique expertise on optical technologies**, related monitoring and data processing.
The RETAIN project is **strengthened by 2 industry associations: EUPC and VINYLPLUS**. Their main role is related to the activities in WP7: Market acceptance, uptake and deployment, incl. dissemination, communication, exploitation and training & learning. Both associations are strongly integrated in the EU plastics industry and as such are the ideal partners for creating awareness about the RETAIN project via their members and for ensuring industrial leveraging and uptake of specific RETAIN results. Moreover, their participation ensures access of RETAIN to a large number of external experts.
Although RETAIN is a strongly industry driven project, there is **R&D support from CTB (BE) and AIMPLAS (ES). Both research and technology centres are strongly involved in plastics and textiles recycling and have a strong track record in EU projects** for many years. Within RETAIN, CTB will lead WP3 on recycling technologies. Furthermore, CTB will lead the standardization activities and participate in various other parts of the RETAIN project. Partner AIMPLAS will be focused on the separation and purification of recyclates, participate in the standardization activities, and contribute to the assessment studies. **Dr. Ir. Bert Groenendaal will coordinate the RETAIN project on behalf of SIOEN**. Bert obtained his PhD in polymer chemistry at the Eindhoven University of Technology (Netherlands). After a postdoctoral fellowship at the University of California at Berkeley he moved to industry (Bayer AG and Agfa-Gevaert NV) Since 2008 Bert is active at SIOEN as R&D project coordinator. As such **he coordinated various (inter)national R&D&I projects over the past 26 years, always with a focus on tangible results**. A **Project Steering Committee (PSC)** will be established as responsible for the overall management of the project. **Its members will consist of SIOEN as coordinator** (and WP2 and WP8 leader) **and the following members: Quality Manager** – Mr Guy Buyle (CTB) – wide experience with EU project management & EU plastics communities (internal quality procedures, including organising internal deliverables review and reporting, liaison activities), **Demonstration Manager** – Ms Elsie Haertjens (VEOLIA) – experienced project manager and WP5 leader (end-users view, ensure RETAIN demonstration is driving market uptake by endusers), **Innovation Manager** – Mr Albert Torres (IRIS) – seasoned project manager and WP1 leader (tech-provider & SME view, focus on marketability of RETAIN horizontal (digital) technologies), **Uptake Manager** – Ms Valeria Masci (EUPC) – expert reaching impact with research and innovation projects (supervise and coordinate the dissemination, communication & exploitation strategy, association view).
Transition Support Group (TSG) members. Next to the actual partners in the consortium, we also put together a group of entities, including European associations, public actors, clusters/hubs, that are linked to RETAIN objectives and expressed their interest to collaborate, see cover page. The support letters have been collected and available for reference here.
Resources. The total eligible costs of the RETAIN project is 6,538,156 EUR. The requested EU contribution to eligible costs is 4,999,379 EUR. **From the EU contribution 19.3% goes to SMEs, and 52.5% to LEs**.

3.2.2. Industrial/commercial involvement in the project

RETAIN has a strong industrial involvement: 12 out of 16 consortium members are companies, further supported by 2 industry associations. The companies consist of 9 large ones and 3 SMEs.
The industrial and commercial partners cover the necessary domains and expertise, see also their description in section 3.2.1. The industrial partners participate to bring RETAIN to a success and to bring the anticipated solutions to the market. As such, they are investing a considerable number of resources: **the RETAIN company partners have together almost 75% of the EU funding** and **the 'own funding' brought by the companies is over €1.5mio**.
We also aimed at including 'exploitation leverage' in RETAIN: each partner has a unique role in the project but in daily business some have overlapping activities. For example, partners VWERVEN, EKOTREND and KKF are PVC recyclers and all three have high interest in deploying the sorting, deconfectioning and recycling technologies after the project. Such an interest within the partnership will be beneficial for realizing the exploitation plans (from section 2.2). Within the TSG we searched for further such 'leverage'.

3.2.3. Other countries. Not applicable.

4. Ethics self assesment

SELF-ASSESSMENT

Ethical dimension of the objectives, methodology and likely impact

2. Humans and 4. Personal data. Within RETAIN we foresee the involvement of human volunteers for workshops (WP7). In this case, subjects will be adults participating voluntary according to informed consents and privacy regulations. Personal data (name, address, phone number, email) will be collected and handled according to GDPR (anonymized and kept in partners' safe places) and deleted at the end of the project or upon request.

7. Health & Safety. For the materials and processes being developed in WP2 to WP4 and demonstrated in WP5, we intend to use chemicals that should be handled with care. By using the necessary precautions, none of this handling raises significant concerns about individual security or safety or the protection of public health.

8. AI. Within WP1 and WP4 the software development involves using AI. However, we do not foresee any ethical concerns in relation to human rights or values as AI will be applied for pure technical aspects, ie processing of spectroscopic data for materials identification (WP1) and optimizing PVC waste channeling to the most appropriate recycling method (WP4).

Compliance with ethical principles and relevant legislations 2. Humans.

For surveys and workshops conducted with partners and external stakeholders, informed consent will be prepared in a language understandable by the volunteers and will include information about the project activities, the objectives and the anticipated outcome. Information on data processing methods, data privacy protection, handling of personal data and other rights of the participant will also be informed. The participants will be informed on their right to participate, decline to participate or withdraw from the project at any time. All partners that are involved in processing volunteers' data will be accountable for the compliance with the applicable legislative requirements. The data management plan of RETAIN, including the personal data protection protocols, will be described in the DMP related deliverables (part of Task 8.5).

All personal details and identification obtained from RETAIN activities will be handled according to the legislative requirements and the EU GDPR principles. Such data will be kept confidential and not used in any public documents that may be produced using the data.

7. Health & Safety.

For the implementation of RETAIN, the appropriate health and safety procedures conforming to relevant local/national guidelines/legislation (17/1996 "Measures for safety and health improvement of employees in the workplace, in accordance with 89/391/EU and 91/383/EU", 16/1996 "Minimum safety and health specifications in the workplace, in accordance with 89/654/EU) will be followed by the staff involved in the RETAIN project. All European health and safety conformities (eg Directive 89/391 - OSH "Framework Directive",) will be strictly adhered. Local laws and standards will be applied to as part of the grant. Staff members are/will be aware of local and European standards in the areas of engineering practice, health and safety conformance and in personal health and safety aspects. Specifically, we will ensure that all the RETAIN partners comply with the following:

1. All staff members involved in RETAIN handling potentially hazardous substances will be informed
2. All laboratory hazards will be properly identified and marked (compliance with health and safety legislation and codes of practice)

3. Provisions will be taken for appropriate resources, information, supervision, instruction and training
4. Ensuring that managers and staff are equipped with the knowledge, competence, confidence and capacity to deal effectively with health and safety issues, and have access to competent specialist advice
5. All staff members will be made aware of proper laboratory equipment handling, use of necessary protective equipment, first aid and emergency procedures
6. All chemicals involved in RETAIN research will be handled and used according to Safety Datasheets (SDS)
7. All laboratory equipment will be accompanied with user manuals and service logbooks

8. **AI**.

In RETAIN, we will comply with the EU AI Act (June 2023). Further, we will strive to inform stakeholders that we use AI based solutions and what they are used for. In addition, we will strive to develop fair and unbiased AI solutions.

DECLARATION:

The Consortium confirms that compliance with ethical principles and applicable international, EU and national law in the implementation of research activities not originally envisaged (or not described in detail) in the DoA will be ensured. The Consortium also confirms that any ethical concerns raised by those activities will be handled following rigorously the recommendations provided in the European Commission Ethics Self-Assessment Guidelines.

ANNEX 2